WEAK-TYPE ENDPOINT BOUNDS FOR BOCHNER–RIESZ MEANS IN EVERY DIMENSION

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ABSTRACT. *At the critical index, the Bochner–Riesz means of every dimension $n \geq 2$ are of weak type (p,p) for $1 < p < 2n/(n+1)$, uniformly in the radius, and converge in the weak L^p norm. They also satisfy sparse bounds that are sharp except at one endpoint, and weighted weak-type bounds for every A_1 weight with a range of p governed by its reverse Hölder exponent. The proof reproduces the recent proof of the local smoothing conjecture by OpenAI and the ε-removal of Tao in the form of Beltran, Roos and Seeger.*

1. INTRODUCTION

For $n \geq 2$, $a > 0$, $\lambda \geq 0$ and $t > 0$ let

$$R^{\lambda}_{a,t} f(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \Big(1 - \frac{|\xi|^a}{t^a}\Big)^{\lambda}_{+} \hat{f}(\xi)\, e^{i\langle x,\xi\rangle}\, d\xi,$$

where $\hat{f}(\xi) = \int_{\mathbb{R}^n} f(y) e^{-i\langle y,\xi\rangle}\, dy$. For $a = 2$ these are the Bochner–Riesz means of the Fourier inversion integral, and for $a = 1$ the Riesz means. For $1 \leq p < 2n/(n+1)$ the number

$$\lambda(p) = n\Big(\frac{1}{p} - \frac{1}{2}\Big) - \frac{1}{2} \tag{1.1}$$

is the critical index: $R^{\lambda}_{a,t}$ is unbounded on $L^p(\mathbb{R}^n)$ for $\lambda \leq \lambda(p)$, because its kernel decays like $|x|^{-(n+1)/2-\lambda}$ and fails to lie in L^p when $\lambda \leq \lambda(p)$ [Ste93]. The Bochner–Riesz conjecture asserts boundedness on L^p for every $\lambda > \lambda(p)$ when $1 \leq p < 2n/(n+1)$. At the critical index itself the expected statement is the *weak-type endpoint Bochner–Riesz conjecture*: for $1 < p < 2n/(n+1)$, the operators $R^{\lambda(p)}_{a,t}$ map $L^p(\mathbb{R}^n)$ to the Lorentz space $L^{p,\infty}(\mathbb{R}^n)$, uniformly in t. This is the title problem of Tao's paper [Tao98].

The case $p = 1$, with index $\lambda(1) = (n-1)/2$, is due to Christ [Chr88], and was extended by Vargas to $L^1(w) \to L^{1,\infty}(w)$ for every A_1 weight w [Var96]. For $p > 1$, Christ proved the weak-type endpoint in the range $1 < p < 2(n+1)/(n+3)$ [Chr87], Tao proved it at $p = 2(n+1)/(n+3)$ [Tao98], and Seeger settled the plane [See96]. Tao then showed that for $1 < p < 2n/(n+1)$ the weak-type endpoint follows from the strong-type Bochner–Riesz bounds for all $\lambda > \lambda(p)$, by an ε-removal argument [Tao98]. In higher dimensions only partial ranges were known, since the Bochner–Riesz conjecture itself was open for $n \geq 3$.

On 25 October 2026 OpenAI released a proof of Sogge's local smoothing conjecture for the Euclidean wave equation in every dimension [OAI26], and deduced from it the Bochner–Riesz conjecture for every $n \geq 2$ and every $1 \leq p \leq \infty$. Combined with the conditional theorems of Tao [Tao98] and of Beltran, Roos and Seeger [BRS25, BRS23], this settles the endpoint questions below. The deductions are short; the length of this paper comes from reproducing in full the two inputs that have not been refereed, so that every step is checked here. Consequences of

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another OpenAI release, for weighted counting and exact optimisation, are derived in the same manner in the paper archived as [Bha26b].

1.1. **Results.** Write $\mathrm{Sp}(p,q)$ for the class of operators with a (p,q) sparse bound (Definition 9.1); A_1 and RH_σ are the Muckenhoupt and reverse Hölder classes (Section 9.4).

Theorem A (Weak-type endpoint)**.** *Let $n \geq 2$, $a > 0$ and $1 < p < 2n/(n+1)$. Then*

$$\sup_{t>0} \|R_{a,t}^{\lambda(p)} f\|_{L^{p,\infty}(\mathbb{R}^n)} \leq C_{n,a,p} \|f\|_{L^p(\mathbb{R}^n)},$$

and $\|R_{a,t}^{\lambda(p)} f - f\|_{L^{p,\infty}} \to 0$ as $t \to \infty$ for every $f \in L^p(\mathbb{R}^n)$.

Theorem B (Sparse bounds)**.** *Let $n \geq 2$, $a > 0$, $1 \leq p < 2n/(n+1)$ and*

$$q > q_{\mathrm{opt}}(p) = \frac{(n-1)p}{n+1-2p}.$$

Then $R_{a,t}^{\lambda(p)} \in \mathrm{Sp}(p,q)$, with norm independent of t.

For $\lambda = \lambda(p)$ the pairs $(1/p, 1/q)$ of Theorem B fill the critical vertical side P_1P_2 of the trapezoid of exponents described in [BRS25, (1.5)], except for its upper corner P_2, where $q = q_{\mathrm{opt}}(p)$. Sparse bounds fail outside the closed trapezoid [BBL17], so Theorem B is sharp except at the single point P_2, which remains open.

Theorem C (Weights)**.** *Let $n \geq 2$, $a > 0$, and let $w \in A_1$ belong to RH_σ for some $\sigma > 1$. Put*

$$p_1(w) = 1 + \frac{n-1}{n+1}\Big(1 - \frac{1}{\sigma}\Big).$$

Then for $1 < p < p_1(w)$ the operators $R_{a,t}^{\lambda(p)}$ map $L^p(w)$ to $L^{p,\infty}(w)$ uniformly in t, and $R_{a,t}^{\lambda(p)} f \to f$ in $L^{p,\infty}(w)$ for every $f \in L^p(w)$.

Every A_1 weight lies in some RH_σ with $\sigma > 1$, so Theorem C gives a range $p > 1$ for every A_1 weight, as in [BRS25, Theorem 1.1]. The difference lies in the size of the range. In [BRS25, §9] the reverse Hölder exponent had to be capped at $(n+3)/2$, which limits the range to $p < 2(n+1)/(n+3)$; here no cap is needed, and as $\sigma \to \infty$ the range fills the whole interval $1 < p < 2n/(n+1)$. The constant weight, which lies in RH_σ for every σ, recovers Theorem A.

For $n = 2$ the Bochner–Riesz conjecture is due to Carleson and Sjölin [CS72], and Theorems A–C were proved in [See96, BRS25]; the new content is $n \geq 3$.

1.2. **What is borrowed and what is reproduced.** Theorems A–C are obtained from three ingredients.

(i) The Bochner–Riesz conjecture, proved by OpenAI [OAI26] through local smoothing.
(ii) The off-diagonal ε-removal of Tao [Tao98], in the vector-valued form of Beltran, Roos and Seeger [BRS23]: non-endpoint bounds imply the multi-scale inequalities $\mathrm{VBR}(p,r)$ of Definition 8.1.
(iii) The sparse domination theorem of Beltran, Roos and Seeger [BRS25, Theorem 2.3], which turns $\mathrm{VBR}(p,r)$ into sparse bounds, and the weighted theorem of Frey and Nieraeth [FN19, Theorem 1.4], which turns sparse bounds into weighted weak-type bounds.

Items (iii) are refereed and are cited. Items (i) and (ii) are a preprint and an Oberwolfach preprint, and are reproduced here with complete proofs: Part I gives OpenAI's proof of the critical annular estimate, of local smoothing and of the Bochner–Riesz conjecture, with the Dirichlet problem it uses in Appendix A; Part II gives the ε-removal. The arguments are those

of [OAI26] and [BRS23]; we follow them closely, specialise them to the round sphere, which is all that is needed, and supply some intermediate computations. Part III proves Theorems A–C.

We searched arXiv, the OpenAI release repository and Zenodo up to 27 October 2026 and found no statement of Theorems A–C for $n \geq 3$. They are corollaries of [OAI26] and the conditional theorems above, which is how we present them.

1.3. **Organisation.** Sections 2–7 form Part I. Section 2 builds a positive conserved current from a wave; Section 3 develops the quaternionic Hessian and its cofactor; Section 4 defines a determinant energy on spatial fibres; Section 5 estimates it; Section 6 compares two circles in the auxiliary time plane; Section 7 removes the period and proves local smoothing and the Bochner–Riesz conjecture. Section 8 is Part II. Section 9 proves Theorems A–C, and Section 10 lists what remains open. Exact-arithmetic checks of the algebraic identities and of the exponent computations are available with the sources [Bha26c]; no statement depends on them.

1.4. **Notation.** In Part I the spatial dimension is $n \geq 2$ and $d = n - 1$; in Parts II and III the dimension is again n, and d is not used. We write $A \lesssim B$ when $A \leq CB$ with a constant depending only on fixed parameters, and $A \asymp B$ when both $A \lesssim B$ and $B \lesssim A$. The Fourier multiplier with symbol m is $m(D)$.

Part I. Local smoothing and the Bochner–Riesz conjecture

2. The statements of Part I and a conserved current

Part I reproduces the proof by OpenAI [OAI26] of Sogge's local smoothing conjecture and of its consequence for Bochner–Riesz means. Throughout Part I the spatial dimension is $n \geq 2$, we put $d = n - 1$, and

$$U(T) = e^{-iT|D|}, \qquad I = [1, 2], \qquad J = (1 - \Delta)^{1/2},$$
$$s_+(p) = \max\Big\{0, \frac{n-1}{2} - \frac{n}{p}\Big\}, \qquad p_{\mathrm{c}} = \frac{2n}{n-1}.$$

Theorem 2.1 (Local smoothing [OAI26])**.** *For $n \geq 2$, $2 \leq p < \infty$ and $\varepsilon > 0$,*

$$\|U(T)f\|_{L^p(\mathbb{R}^n \times I)} \leq C_{n,p,\varepsilon} \|J^{s_+(p)+\varepsilon} f\|_{L^p(\mathbb{R}^n)}.$$

The whole proof rests on one estimate at the exponent p_{c}, on a single frequency annulus.

Theorem 2.2 (Critical annular estimate [OAI26])**.** *Let $n \geq 2$, $\varphi \in C_c^\infty(\mathbb{R}^n \setminus \{0\})$ and $P_N = \varphi(D/N)$. For $N \geq 2$,*

$$\|U(T)P_N f\|_{L^{p_{\mathrm{c}}}(\mathbb{R}^n \times I)} \leq C_{n,\varphi} \, [\log(2 + N)]^{2/p_{\mathrm{c}}} \, \|f\|_{L^{p_{\mathrm{c}}}(\mathbb{R}^n)}.$$

2.1. **The plan.** Put $h = N^{-1}$ and $a = Kh$ with a large fixed K. Besides the time T there will be two auxiliary real time variables $\tau \in \mathbb{R}^2$, which come from an imaginary displacement of the wave along light rays. For data f on a fixed torus whose frequencies lie in one small cap of the annulus $|\xi| \asymp h^{-1}$ and with $\|f\|_2 = 1$, we build a nonnegative energy $\mathcal{F}(T, \tau)$ and prove, with $\mathcal{H}(T, s)$ the mean of $\mathcal{F}$ over $|\tau| = s$,

$$\int_1^2 \|U(T)f\|_{p_{\mathrm{c}}}^{p_{\mathrm{c}}} \, dT \lesssim a^{-2} \int_1^2 \mathcal{H}(T, 2a) \, dT \lesssim [\log(2/h)]^2 a^{-2} \mathcal{H}(0, 4a) \lesssim [\log(2/h)]^2 \|f\|_{p_{\mathrm{c}}}^{p_{\mathrm{c}}}.$$

The outer inequalities compare the energy with the wave (Section 5); the middle one uses only the differential inequality $\Delta_{T,\tau} \mathcal{F} \leq C s^{-2} e^{-cs/h} \mathcal{F}$ on $2a \leq |\tau| \leq 3$ (Section 6).

The energy is built as follows. A finite sum of plane waves whose directions lie in one cap is continued to complex spacetime. Translation by $-i\ell(1, \omega)$ along a light ray multiplies a mode of

frequency ξ by $\exp(-\ell|\xi||\omega-\xi/|\xi||^2/2)$, a Gaussian of angular width $(N\ell)^{-1/2}$. The fluxes of the symmetric first-order form of the wave equation produce such displacements, and two flows built from the continued wave give a positive semidefinite tensor with divergence-free columns whose time component still sees the wave on the cap (this section). On each spatial fibre, the time component μ of a regularised version of this tensor is used as the density of a quaternionic Monge–Ampère equation $\det H_{\mathrm{fib}}(\lambda|\Phi|^2+u)=\lambda\mu$ with zero boundary values, where Φ is a complex scalar constant along the current and λ is large. The background $\lambda|\Phi|^2$ has a Hessian of rank one; adding αI to it changes its determinant to $(\beta+\alpha)\alpha^{n-1}$ with $\beta\asymp\lambda$. Hence a density $\mu\ge m$ on a ball forces the depth of u at its centre to be at least a multiple of the squared radius times $m^{1/(n-1)}$, and $\int(-u)\mu$ detects $m^{n/(n-1)}$. Since μ is a square of the wave, the exponent is $2n/(n-1)=p_{\mathrm{c}}$. A correction of Aubin–Mabuchi type makes the full determinant cancel in $\Delta_{T,\tau}$ of the energy, and the only positive contribution left is the wave flux through the side of the fibre, which carries a Gaussian factor $e^{-cs/h}$.

2.2. **Null coordinates and modes.** Choose an orthonormal frame $X=(x,b)\in\mathbb{R}^d\times\mathbb{R}$ of $\mathbb{R}^n$ and put

$$t=(T+b)/\sqrt{2},\qquad s_0=(b-T)/\sqrt{2}.$$

In these coordinates the wave operator $\partial_T^2-\Delta_X$ becomes, up to sign, $2\partial_t\partial_{s_0}+\Delta_x$. A plane wave with phase $X\cdot\xi-T|\xi|$ and $\xi_b+|\xi|>0$ is

$$u_{\kappa,v}=c_{\kappa,v}\exp\Big(i\kappa\big(s_0+v\cdot x-\tfrac12|v|^2t\big)\Big),\qquad \kappa=\frac{\xi_b+|\xi|}{\sqrt2},\qquad v=\frac{\xi_x}{\kappa}.\tag{2.1}$$

Indeed, the phase is $x\cdot\xi_x+s_0(\xi_b+|\xi|)/\sqrt2+t(\xi_b-|\xi|)/\sqrt2$, and $(\xi_b-|\xi|)/\sqrt2=-|\xi_x|^2/(2\kappa)=-\kappa|v|^2/2$. Throughout the construction $u_{\mathrm{w}}=\sum_{\kappa,v}u_{\kappa,v}$ is a finite sum with $\kappa>0$ and complex coefficients.

Index the states by $0,\dots,d$, let E_{ab} be the elementary $n\times n$ matrices, and in the real spacetime coordinates $\mathsf{X}=(t,x,s_0)$ put

$$C^t=2E_{00},\qquad C^{x_j}=E_{0j}+E_{j0},\qquad C^{s_0}=-\sum_{j=1}^{d}E_{jj},\qquad \mathscr{P}=\sum_{\mu}C^\mu\partial_{\mathsf{X}_\mu}.$$

For a scalar ψ the vector $(\partial_{s_0}\psi,\nabla_x\psi)$ satisfies $\mathscr{P}(\partial_{s_0}\psi,\nabla_x\psi)^{\mathrm{t}}=((2\partial_t\partial_{s_0}+\Delta_x)\psi,0,\dots,0)^{\mathrm{t}}$: the first entry is $2\partial_t\partial_{s_0}\psi+\sum_j\partial_{x_j}\partial_{x_j}\psi$, and the jth entry is $\partial_{x_j}\partial_{s_0}\psi-\partial_{s_0}\partial_{x_j}\psi=0$. At the covector $p=\kappa(-|v|^2/2,v,1)$ of (2.1) the symbol $C(p)=\sum_\mu p_\mu C^\mu$ satisfies

$$-C(p)=\kappa\begin{pmatrix}|v|^2 & -v^{\mathrm{t}}\\ -v & I_d\end{pmatrix},\qquad -w^{\mathrm{t}}C(p)w=\kappa|w'-vw_0|^2\quad(w=(w_0,w')\in\mathbb{R}^n).\tag{2.2}$$

Its kernel is spanned by $E(v)=(1,v)^{\mathrm{t}}$, and $f_{\mathrm{sys}}=\sum_{\kappa,v}\kappa^{d/2}E(v)u_{\kappa,v}$ solves $\mathscr{P}f_{\mathrm{sys}}=0$.

2.3. **The Gaussian from an imaginary displacement.** Translating (T,X) by $-i\varrho(1,\omega)$ with $\varrho>0$ and $|\omega|=1$ multiplies a mode of frequency ξ by

$$e^{\varrho(\xi\cdot\omega-|\xi|)}=e^{-\varrho|\xi||\hat\xi-\omega|^2/2},\qquad \hat\xi=\xi/|\xi|,$$

since $|\hat\xi-\omega|^2=2-2\hat\xi\cdot\omega$. The system produces such light rays as quadratic fluxes. For a real state $w=(w_0,w')$ let $q(w)^\mu=w^{\mathrm{t}}C^\mu w/2$; in null coordinates $q(w)=(w_0^2,w_0w',-|w'|^2/2)$. Since $\partial_t=(\partial_T+\partial_b)/\sqrt2$ and $\partial_{s_0}=(\partial_b-\partial_T)/\sqrt2$, its physical components satisfy $q(w)^T=|q(w)^X|=$

$(w_0^2+|w'|^2/2)/\sqrt{2}$, so $q(w)$ is a light ray; at $w=E(v)$ it is $(|\xi|,\xi)/\kappa$. We use the sum of the fluxes of two real states,

$$q^\mu = q(w_1)^\mu + q(w_2)^\mu = \frac{1}{2}\sum_{\ell=1}^{2} w_\ell^{\mathrm{t}} C^\mu w_\ell, \qquad w_1, w_2 \in \mathbb{R}^n. \tag{2.3}$$

At $\mathsf{X}-iq$ a mode acquires the factor $e^{p\cdot q}$. Introduce a second imaginary unit $\mathbf{k}$, with $\mathbf{k}^2=-1$, put $\mathbb{C}_{\mathbf{k}}=\{\alpha+\mathbf{k}\beta : \alpha,\beta\in\mathbb{R}\}$, and pair the two states into

$$z = w_1 + \mathbf{k} w_2 = (R,\zeta) \in \mathbb{C}_{\mathbf{k}} \times \mathbb{C}_{\mathbf{k}}^d.$$

Then $q^\mu = z^* C^\mu z/2$, where * is the conjugate transpose, and adding the two real squares of (2.2) gives

$$e^{p\cdot q} = \exp\Big(-\frac{\kappa}{2}|\zeta - vR|^2\Big). \tag{2.4}$$

For each R the square of $\kappa^{d/2}e^{p\cdot q}$ has integral π^d in ζ, which explains the factor $\kappa^{d/2}$ in f_{sys}.

2.4. **Two auxiliary flows.** For constant real symmetric matrices C^μ the bilinear flux of two states is $j^\mu(a,b)=a^{\mathrm{t}}C^\mu b$, and its divergence for two fields is $(\mathscr{P}a)^{\mathrm{t}}b + a^{\mathrm{t}}\mathscr{P}b$; this is the conservation law of symmetric hyperbolic systems [Fri54]. The derivative of q^μ in the direction δw_ℓ is $j^\mu(w_\ell,\delta w_\ell)$, and this suggests the following vector fields.

Lemma 2.3 ([OAI26])**.** *Let $\mathscr{P}=\sum_\mu C^\mu\partial_{\mathsf{X}_\mu}$ have constant real symmetric coefficients, let f_{sys} be an entire solution of $\mathscr{P}f_{\mathrm{sys}}=0$, and let q be the displacement (2.3). In the real coordinates (X,w_1,w_2) put*

$$G=\overline{f_{\mathrm{sys}}(\mathsf{X}-iq)}, \quad V_{\ell a}=\sum_\mu (C^\mu w_\ell)_a\,\partial_{\mathsf{X}_\mu}, \quad D_{\ell a}=\partial_{w_{\ell a}}+iV_{\ell a}, \quad \mathcal{Z}_\ell=\sum_a G_a D_{\ell a}.$$

Then the real tensor

$$B=\mathrm{Re}\sum_{\ell=1}^{2}\mathcal{Z}_\ell\otimes\overline{\mathcal{Z}_\ell} \tag{2.5}$$

is symmetric and positive semidefinite, and each of its columns has zero divergence.

Proof. The field $D_{\ell a}$ annihilates every component of $\mathsf{X}-iq$: it sends $\mathsf{X}_\mu-iq^\mu$ to $-i(C^\mu w_\ell)_a+i(C^\mu w_\ell)_a=0$. Since f_{sys} is holomorphic, $D_{\ell a}\overline{G}=0$, that is

$$\partial_{w_{\ell a}}G = iV_{\ell a}G.$$

The coefficients of $V_{\ell a}$ depend only on w, so $D_{\ell a}$ has zero divergence, and

$$\operatorname{div}\mathcal{Z}_\ell=\sum_a D_{\ell a}G_a=\sum_a(\partial_{w_{\ell a}}G_a+iV_{\ell a}G_a)=2i\sum_a V_{\ell a}G_a=2i\,w_\ell^{\mathrm{t}}\,\overline{(\mathscr{P}f_{\mathrm{sys}})(\mathsf{X}-iq)}=0.$$

For the flat connection the identity $D_{\ell a}\overline{G}=0$ shows that, in $\nabla_{\mathcal{Z}_\ell}\overline{\mathcal{Z}_m}=\sum_{a,b}G_aD_{\ell a}(\overline{G_b})\,\overline{D_{mb}}+\sum_{a,b}G_a\overline{G_b}\,D_{\ell a}(\overline{D_{mb}})$, only the derivatives of the coefficients of $\overline{D_{mb}}=\partial_{w_{mb}}-iV_{mb}$ remain. These are $-i\,\delta_{\ell m}\sum_\mu(C^\mu)_{ba}\partial_{\mathsf{X}_\mu}$, so

$$\nabla_{\mathcal{Z}_\ell}\overline{\mathcal{Z}_m}=-i\,\delta_{\ell m}\sum_\mu(G^{\mathrm{t}}C^\mu\overline{G})\partial_{\mathsf{X}_\mu}.$$

The vector on the right is real since the C^μ are real symmetric, so its product with $-i$ is purely imaginary. The column divergence of $\mathcal{Z}_\ell\otimes\overline{\mathcal{Z}_\ell}$ is $(\operatorname{div}\mathcal{Z}_\ell)\overline{\mathcal{Z}_\ell}+\nabla_{\mathcal{Z}_\ell}\overline{\mathcal{Z}_\ell}$, whose real part vanishes. Finally, if $\mathcal{Z}_\ell=A_\ell+iB_\ell$ with real fields, its contribution to (2.5) is $A_\ell\otimes A_\ell+B_\ell\otimes B_\ell\ge 0$. □

The imaginary part of the last identity at $m=\ell$ gives $[A_\ell,B_\ell]=\sum_\mu(G^{\mathrm{t}}C^\mu\overline{G})\partial_{\mathsf{X}_\mu}$, which identifies the physical current inside the lifted variables. We now express the four real squares of B in coordinates attached to spacetime.

2.5. **Quaternionic coordinates adapted to the wave.** The quaternions $\mathbb{H}$ have units $\mathbf{i},\mathbf{j},\mathbf{k}$ with $\mathbf{i}^2=\mathbf{j}^2=\mathbf{k}^2=-1$, $\mathbf{ij}=\mathbf{k}$ and $\mathbf{ki}=\mathbf{j}$. The Fourier complex plane is identified with $\mathbb{C}_{\mathbf{i}}=\mathbb{R}+\mathbb{R}\mathbf{i}$; every quaternion is uniquely $\alpha+\mathbf{k}\beta$ with $\alpha,\beta\in\mathbb{C}_{\mathbf{i}}$, and the planes $\mathbb{C}_{\mathbf{i}}$ and $\mathbf{k}\mathbb{C}_{\mathbf{i}}$ are orthogonal for $\langle u,v\rangle=\mathrm{Re}(u\bar{v})$. Columns carry scalars acting on the right, and * denotes the quaternionic conjugate transpose. We look for one auxiliary pair for each of the $n+1$ real spacetime coordinates and write each triple as $\mathsf{X}_\mu+\omega_\mu\mathbf{i}$ with $\omega_\mu\in\mathbb{C}_{\mathbf{k}}$; these triples lie in $\mathcal{D}=\mathrm{span}_{\mathbb{R}}(1,\mathbf{i},\mathbf{j})$.

For a quaternionic Hermitian matrix A on spacetime indices put

$$A^{\mathbb{R}}_{\mu\alpha,\nu\beta}=\mathrm{Re}(\overline{e_\alpha}A_{\mu\nu}e_\beta),\qquad (e_0,e_1,e_2)=(1,\mathbf{i},\mathbf{j}). \tag{2.6}$$

If $A=\chi\chi^*$ this real tensor is the Gram matrix of the rows $\overline{e_\alpha}\chi_\mu$, so the three rows over each μ are $Q,-\mathbf{i}Q,-\mathbf{j}Q$.

In source coordinates write the four squares of (2.5) as a Gram matrix by setting $Q_I=\mathcal{Z}_1^I+\mathbf{k}\mathcal{Z}_2^I$ for each coordinate I. Orthogonality of $\mathbb{C}_{\mathbf{i}}$ and $\mathbf{k}\mathbb{C}_{\mathbf{i}}$ gives $B^{IJ}=\mathrm{Re}(Q_I\overline{Q_J})$. With $M_{\mu a}=(C^\mu z)_a$ the rows in the real spacetime and in the state directions are

$$Q_{\mathsf{X}_\mu}=\sum_a M_{\mu a}\,\mathbf{i}\,G_a,\qquad (Q_{w_{1a}},Q_{w_{2a}})=(G_a,\mathbf{k}G_a).$$

For a holomorphic coordinate $\omega_\mu(z)$ let $L_{\mu a}=\partial_{z_a}\omega_\mu$. Multiplication by $c=c_1+\mathbf{k}c_2$ on an auxiliary pair has real rows $(c_1,-c_2)$ and (c_2,c_1), so it sends $(G_a,\mathbf{k}G_a)$ to $(\bar{c}G_a,\mathbf{k}\bar{c}G_a)$; the new pair over μ therefore starts with $\sum_a\overline{L_{\mu a}}G_a$. The choice $L=M$ gives exactly $-\mathbf{i}Q_{\mathsf{X}_\mu}$, because $-\mathbf{i}c\,\mathbf{i}=\bar{c}$ for $c\in\mathbb{C}_{\mathbf{k}}$, and its partner is $\mathbf{k}(-\mathbf{i}Q_{\mathsf{X}_\mu})=-\mathbf{j}Q_{\mathsf{X}_\mu}$.

The holomorphic flux map. By symmetry, $L_{\mu a}=(C^\mu z)_a$ integrates to the quadratic $z^{\mathrm{t}}C^\mu z/2$: the Hermitian square $z^*C^\mu z/2$ weights the modes, and the bilinear square moves their Gram rows. There are n complex state coordinates and $n+1$ spacetime coordinates, so add a passive variable $\eta\in\mathbb{C}_{\mathbf{k}}$ in whose two real directions the flows have no component. For a constant real symmetric system as in Lemma 2.3 and a fixed $c\in\mathbb{R}^{n+1}$ set

$$\Theta:(z,\eta)\longmapsto\omega,\qquad \omega_\mu=\tfrac{1}{2}z^{\mathrm{t}}C^\mu z+c^\mu\eta.$$

It fixes the real spacetime coordinates, and on a local inverse chart its real volume Jacobian is $J=|\det_{\mathbb{C}}[M,c]|^2$. Push B forward as a density,

$$\widetilde{B}\circ\Theta=J^{-1}\,D\Theta\,B\,D\Theta^{\mathrm{t}}. \tag{2.7}$$

The differentials transform the Gram rows as described, and J^{-1} converts source volume to target volume, so

$$\widetilde{B}=(\widetilde{\chi}\widetilde{\chi}^*)^{\mathbb{R}},\qquad \widetilde{\chi}_\mu\circ\Theta=J^{-1/2}\sum_a M_{\mu a}\mathbf{i}G_a. \tag{2.8}$$

As a function of ω, the passive coordinate satisfies $(\partial_\omega\eta)M=0$, because varying z at fixed η produces the columns of M; hence $(\partial_\omega\eta)\widetilde{\chi}=0$.

The tensor $\widetilde{B}$ is again conserved. Test its Kth column against the gradient of a real φ with compact support in the target chart. Changing variables in (2.7), the weak divergence becomes $-\sum_{I,L}\int B^{IL}(\partial_L\Theta^K)\partial_I(\varphi\circ\Theta)$, and integration by parts in the source gives

$$J(\mathrm{div}\,\widetilde{B})^K\circ\Theta=\sum_L(\partial_L\Theta^K)(\mathrm{div}\,B)^L+\sum_{I,L}B^{IL}\partial_I\partial_L\Theta^K.$$

The first sum vanishes by Lemma 2.3. In the second, the two diagonal state blocks of B both equal $(\mathrm{Re}(G_a\overline{G_b}))_{ab}$ and the mixed state block vanishes, while each real component of the holomorphic map satisfies $\partial_{w_{1a}}\partial_{w_{1b}}\Theta^K+\partial_{w_{2a}}\partial_{w_{2b}}\Theta^K=0$. The other coordinates enter Θ affinely, so the second sum vanishes too.

The wave in these coordinates. For the wave matrices add η in the s_0 coordinate. In the null frame the flux map is

$$\Theta : (R, \zeta, \eta) \longmapsto (Y, P, S_1) = (R^2, R\zeta, \eta - \zeta \cdot \zeta/2). \tag{2.9}$$

If (τ, P, Z) are the target auxiliary coordinates in the physical frame (T, x, b), then $Y = (\tau + Z)/\sqrt{2}$ and $S_1 = (Z - \tau)/\sqrt{2}$. On $Y \neq 0$, inverting (2.9) gives $R = \sqrt{Y}$ on a local branch and $\zeta = P/R$, and the single-valued coordinates are

$$W = \frac{P}{Y} = \frac{\zeta}{R}, \qquad \Phi = S_1 + \frac{P \cdot P}{2Y} = \eta, \qquad y = |Y|. \tag{2.10}$$

All dots between complex vectors are bilinear, and $|\cdot|$ is the Euclidean norm. The complex Jacobian determinant of Θ is $2R^{d+1}$, so

$$J = 4|R|^{2d+2}. \tag{2.11}$$

Under M a polarisation vector $E(v)$ maps to

$$ME(v) = (2R, \zeta + Rv, -\zeta \cdot v)^{\mathrm{t}} = 2RV^v, \qquad V^v = \big(1, (v + W)/2, -W \cdot v/2\big)^{\mathrm{t}}. \tag{2.12}$$

Put $\ell(w) = (-w \cdot w/2, w, 1)$, so that $\kappa\ell(v)$ is the real phase covector; one checks $\ell(w)V^v = -(w - W) \cdot (w - v)/2$. The column V^v thus lies in the kernels of both $\ell(v)$ and $\partial\Phi = \ell(W)$, and differentiating $W = P/Y$ gives the second identity in

$$(\partial\Phi)V^v = 0, \qquad (\partial W)V^v = \frac{v - W}{2Y}. \tag{2.13}$$

These are complex differentials in (Y, P, S_1). The current stays on a level set of Φ, and each mode moves W in proportion to its angular mismatch $v - W$.

For $e = Y/y$ put $\iota_e(\alpha + i\beta) = \alpha + e\mathbf{i}\beta$. This embeds the Fourier complex numbers isometrically into $\mathbb{H}$, since $\mathbf{i}c = \bar{c}\mathbf{i}$ for $c \in \mathbb{C}_{\mathbf{k}}$ gives $(e\mathbf{i})^2 = -1$. Define the positive weight

$$m_{\kappa,v} = (\kappa/y)^{d/2} \exp\Big(-\frac{\kappa y}{2}|W - v|^2\Big);$$

the factor $y^{-d/2}$ comes from the change of volume.

Proposition 2.4 (The wave current [OAI26])**.** *For a finite superposition of modes* (2.1) *set*

$$\chi = \sum_{\kappa,v} V^v \, \iota_e\big(\overline{u_{\kappa,v}}\big)\, m_{\kappa,v}, \qquad A^0 = \chi\chi^*. \tag{2.14}$$

On $Y \neq 0$ the matrix A^0 is smooth and positive semidefinite, $(\partial\Phi)A^0 = 0$, and each column of $(A^0)^{\mathbb{R}}$ has zero real divergence. The same holds after the orthogonal change from null to physical coordinates, and

$$V^v_T = (1 + W \cdot v/2)/\sqrt{2}, \qquad A^0_{TT} = |\chi_T|^2.$$

Proof. We identify the column (2.8) on one inverse chart. By (2.4) and (2.12) the contribution of a single mode to $\widetilde{\chi}$ is $V^v g\, \mathbf{i}\, \overline{u_{\kappa,v}}\, m_{\kappa,v}$ with $g = R/|R|$, since $|\zeta - vR|^2 = y|W - v|^2$ and $2|R|/\sqrt{J} = y^{-d/2}$. Multiplying every row on the right by the unit quaternion $(g\mathbf{i})^{-1}$ preserves the Gram matrix; conjugation by g takes $\mathbf{i}$ to $g^2\mathbf{i} = e\mathbf{i}$, and $\mathbf{i}$ commutes with the Fourier amplitudes, so the column becomes (2.14). This formula involves Y itself, hence is smooth and independent of the branch of the square root. Positivity follows from the outer product form, tangency from (2.13), and conservation from the discussion of $\widetilde{B}$. The change to physical coordinates acts by the same real orthogonal matrix in each of the three coordinate slots, so it preserves column divergence and the pairing of rows and columns. Finally $T = (t - s_0)/\sqrt{2}$ gives V^v_T. □

For a single κ and real W the exponential in $m_{\kappa,v}$ is a Gaussian window in the transverse frequency κv centred at κW, and since $V_t^v = 1$ the null component A^0_{tt} is, up to normalisation, the Husimi density of the scalar wave [Hus40, KLO16], a phase-space density of the kind also used for quantum states on symplectic manifolds [Bha26a]. For the full wave, (2.14) sums the amplitudes over κ before taking the square, which keeps the interference that produces a focus.

2.6. **The time component on a frequency cap.** At a fixed auxiliary point write $V_T^v = b_0 + \mathbf{k} b_1$, so that

$$b_0 = \frac{1 + \mathrm{Re}(W)\cdot v/2}{\sqrt{2}}, \qquad b_1 = \frac{\mathrm{Im}_{\mathbf{k}}(W)\cdot v}{2\sqrt{2}}, \qquad \mathrm{Im}_{\mathbf{k}}(\alpha + \mathbf{k}\beta) = \beta.$$

Let S_ℓ be the scalar spatial multiplier sending each mode of (2.1) to $b_\ell m_{\kappa,v} u_{\kappa,v}$ at this auxiliary point. By (2.14),

$$\chi_T = \iota_e\big(\overline{S_0 u_{\mathrm{w}}}\big) + \mathbf{k}\,\iota_e\big(\overline{S_1 u_{\mathrm{w}}}\big), \qquad A^0_{TT} = |S_0 u_{\mathrm{w}}|^2 + |S_1 u_{\mathrm{w}}|^2, \tag{2.15}$$

because the two terms lie in the conjugates of the orthogonal planes $\mathbb{C}_{\mathbf{i}}$ and $\mathbf{k}\mathbb{C}_{\mathbf{i}}$ by a square root of e in $\mathbb{C}_{\mathbf{k}}$. At $W = 0$ the second multiplier vanishes and $2A^0_{TT}$ is exactly the squared modulus of the wave with weight $(\kappa/y)^{d/2}\exp(-\kappa y|v|^2/2)$ on each mode.

For the estimates take $u_{\mathrm{w}} = e^{-iT|D|} f$ on the fixed torus $\mathbb{T}^n = \mathbb{R}^n/(100\mathbb{Z})^n$, with the axis b along the centre of a frequency cap; the torus itself is not rotated. For $0 < h \le 1$ suppose $\hat f$ is supported in

$$c_0/h \le |\xi| \le C_0/h, \qquad |v| \le \gamma,$$

with fixed $0 < c_0 < C_0$ and a small aperture $\gamma > 0$ independent of h. The identity $|\xi| = \kappa(1 + |v|^2/2)/\sqrt{2}$ gives $\kappa \asymp h^{-1}$. At $y \asymp h$ and $|W|, |v| \le 1/10$ the Gaussian and b_0 are bounded below, so $h^d S_0$ has a symbol bounded above and below on the cap; for larger y/h the cost of inverting S_0 depends on the mismatch $|W - v|$. There are $O(h^{-n})$ frequencies in the annulus, uniformly in the frame, so Cauchy–Schwarz and Parseval give $\sum_{\kappa,v} |c_{\kappa,v}| \lesssim h^{-n/2}\|f\|_2$. The formulas in null coordinates may be used on the covering space; the resulting current is periodic in the original spatial variables. Nonzero cap data are normalised by $\|f\|_2 = 1$ in the energy estimates.

3. The quaternionic Hessian and its cofactor

The current of Proposition 2.4 is positive and its real columns are divergence free. It will be paired with a Hessian through the real contraction $A : H(v) = A^{\mathbb{R}} : D^2 v$, in which each spacetime index keeps its three real coordinates. Two features of this Hessian matter later. The squared modulus of a holomorphic function contributes at most one quaternionic direction: in n spatial variables, if its nonzero eigenvalue is β, adding εI gives the determinant $(\beta + \varepsilon)\varepsilon^{n-1}$, and this is where the denominator of $p_{\mathrm{c}} = 2n/(n-1)$ enters. Secondly, the cofactor of the Hessian has divergence-free columns, so the determinant equation can be integrated by parts up to the boundary.

3.1. **The Hessian in real and quaternionic coordinates.** Use the triples $z_a = \mathsf{X}_a + P_a \mathbf{i} \in \mathcal{D}$ of Section 2, with $P_a \in \mathbb{C}_{\mathbf{k}}$. Appendix A also uses a fourth real coordinate, which rotates the normal at the boundary. So for each of m coordinate indices write

$$z_a = \sum_{\alpha=0}^{k-1} e_\alpha z_{a\alpha}, \qquad k \in \{3, 4\}, \qquad (e_0, e_1, e_2, e_3) = (1, \mathbf{i}, \mathbf{j}, \mathbf{k}),$$

with $z_{a\alpha}$ real; we speak of k *slots*. For three slots, $P_a = p_a + \mathbf{k}q_a$ gives $P_a\mathbf{i} = p_a\mathbf{i} + q_a\mathbf{j}$, since $\mathbf{ki} = \mathbf{j}$. For a real C^2 function u put $\partial_{a\alpha} = \partial/\partial z_{a\alpha}$ and

$$G_a(u) = \sum_\alpha e_\alpha\, \partial_{a\alpha} u, \qquad H_{ab}(u) = \sum_{\alpha,\beta} e_\alpha \overline{e_\beta}\, \partial_{a\alpha}\partial_{b\beta} u. \tag{3.1}$$

Thus $G(u)$ is the Euclidean gradient written as a quaternionic column. The symbols ∇, Δ and ∂_n denote the Euclidean gradient, the Laplacian and the outward normal derivative.

Lemma 3.1 (Hessian calculus [OAI26])**.** *In three or four slots, $H(u)$ is Hermitian and $\operatorname{tr} H(u) = \Delta u$. The linear map from the real Hessian to $H(u)$ sends $\xi\xi^{\mathrm{t}}$ to gg^*, where $g_a = \sum_\alpha e_\alpha \xi_{a\alpha}$, and so sends nonnegative real forms to nonnegative quaternionic forms. The product rule is*

$$H(uv) = uH(v) + vH(u) + G(u)G(v)^* + G(v)G(u)^*. \tag{3.2}$$

In four slots H is covariant under quaternionic unitary changes of coordinates, and in three slots under real orthogonal changes of the coordinate index applied in all slots. If u depends only on $P \in \mathbb{C}^m_{\mathbf{k}}$, then

$$G_a(u) = 2(\partial_{\bar{P}_a} u)\,\mathbf{i}, \qquad H_{ab}(u) = 4\,\partial_{\bar{P}_a}\partial_{P_b} u. \tag{3.3}$$

In particular, a holomorphic map $F = (F_1, \dots, F_l)$ with complex Jacobian $J_{\nu a} = \partial_{P_a} F_\nu$ satisfies

$$H\Big(\sum_\nu |F_\nu|^2\Big) = 4J^*J, \tag{3.4}$$

and for every smooth real function ρ of F, the column $G(\rho\circ F)$ lies in the quaternionic image of J^*.

Proof. Interchanging the two real derivatives in H_{ba} and conjugating shows $H_{ba} = \overline{H_{ab}}$. In a diagonal entry the terms with $\alpha \neq \beta$ come in pairs with coefficients $e_\alpha\overline{e_\beta} + e_\beta\overline{e_\alpha} = 2\operatorname{Re}(e_\alpha\overline{e_\beta}) = 0$, and the terms with $\alpha = \beta$ have coefficient one; hence $\operatorname{tr} H(u) = \Delta u$. Formula (3.1) applied to the quadratic function $(\xi\cdot z)^2/2$, whose real Hessian is $\xi\xi^{\mathrm{t}}$, gives $H_{ab} = g_a\overline{g_b}$; every nonnegative real form is a sum of such rank-one forms. The same formula and the ordinary product rule give (3.2).

For covariance, note that $\xi\cdot z = \operatorname{Re}(g^*z)$, because $\operatorname{Re}(\overline{e_\alpha}e_\beta) = \delta_{\alpha\beta}$. In four slots a quaternionic unitary U acting by $z \mapsto Uz$ therefore sends the covector g to U^*g, since $\operatorname{Re}(g^*Uz) = \operatorname{Re}((U^*g)^*z)$. The rank-one case gives $H(u\circ U)(z) = U^*H(u)(Uz)U$ whenever the real Hessian of u at Uz is rank one, and the chain rule for real Hessians together with linearity extends it to every u. Real orthogonal matrices acting on the index a in all slots preserve the three-slot subspace and commute with the units, which gives the three-slot statement.

For $P_a = p_a + \mathbf{k}q_a$ we have $2\partial_{\bar{P}_a} = \partial_{p_a} + \mathbf{k}\partial_{q_a}$. Since u is real, $2(\partial_{\bar{P}_a}u)\mathbf{i} = \mathbf{i}\partial_{p_a}u + \mathbf{ki}\,\partial_{q_a}u = \mathbf{i}\partial_{p_a}u + \mathbf{j}\partial_{q_a}u$, which is $G_a(u)$. In H_{ab} the slots $\mathbf{i},\mathbf{j}$ give the coefficients $\mathbf{i}\bar{\mathbf{i}} = 1$, $\mathbf{i}\bar{\mathbf{j}} = -\mathbf{k}$, $\mathbf{j}\bar{\mathbf{i}} = \mathbf{k}$, $\mathbf{j}\bar{\mathbf{j}} = 1$ in the order $(p_ap_b, p_aq_b, q_ap_b, q_aq_b)$, while

$$4\partial_{\bar{P}_a}\partial_{P_b} = (\partial_{p_a} + \mathbf{k}\partial_{q_a})(\partial_{p_b} - \mathbf{k}\partial_{q_b}) = \partial_{p_a}\partial_{p_b} - \mathbf{k}\partial_{p_a}\partial_{q_b} + \mathbf{k}\partial_{q_a}\partial_{p_b} + \partial_{q_a}\partial_{q_b},$$

which is (3.3). For holomorphic F, $4\partial_{\bar{P}_a}\partial_{P_b}\sum_\nu F_\nu\overline{F_\nu} = 4\sum_\nu \overline{J_{\nu a}}J_{\nu b}$, which is (3.4). Finally the complex chain rule gives $2\partial_{\bar{P}_a}(\rho\circ F) = 2\sum_\nu \overline{J_{\nu a}}\,(\partial_{\bar{F}_\nu}\rho)(F)$, so $G(\rho\circ F) = 2J^*\big((\partial_{\bar{F}_\nu}\rho)(F)\big)_\nu\,\mathbf{i}$. □

3.2. **The Moore cofactor.** On Hermitian quaternionic $m\times m$ matrices use the real inner product $A : B = \operatorname{Re}\operatorname{tr}(AB)$. Since $\operatorname{Re}(xy) = \operatorname{Re}(yx)$ for quaternions, the real part of the trace is cyclic, and in particular $A : (vv^*) = v^*Av$; moreover $A : A = \sum_{a,b}|A_{ab}|^2$, so the inner product is positive definite. A Hermitian matrix has an orthonormal eigenbasis with real eigenvalues: maximise v^*Av on the unit sphere and repeat on the quaternionic orthogonal complement. The *Moore determinant* $\det A$ is the product of the m real eigenvalues counted with quaternionic multiplicity. These conventions agree with [Ale03a, §1.1, Theorems 1.1.8 and 1.1.9]. The

identities below also apply to the full Hessian in the energy argument, whose spatial block is positive definite while its time entry need not have a sign.

Identify $\mathbb{H}^m$ with $\mathbb{R}^{4m}$ with the inner product $\mathrm{Re}(v^*w)$. Then $v \mapsto Av$ is a symmetric real linear map $A_{\mathbb{R}}$, since $\mathrm{Re}(v^*Aw) = \mathrm{Re}((Av)^*w)$; if $Av = \lambda v$ then $A(vq) = \lambda vq$ for every quaternion q, so each eigenvalue of A occurs four times in $A_{\mathbb{R}}$. Hence

$$\det_{\mathbb{R}} A_{\mathbb{R}} = (\det A)^4, \qquad \mathrm{tr}_{\mathbb{R}}\, A_{\mathbb{R}} = 4\,\mathrm{Re}\,\mathrm{tr}\, A. \tag{3.5}$$

Lemma 3.2 (Matrix identities [OAI26])**.** *The Moore determinant is a homogeneous polynomial of degree m in the real entries of A. Its cofactor $C(A)$ is the Hermitian matrix defined by*

$$C(A) : B = \frac{d}{dt} \det(A + tB)\Big|_{t=0}. \tag{3.6}$$

It is a homogeneous polynomial of degree $m-1$, and

$$C(A) = (\det A)A^{-1} \qquad (A > 0). \tag{3.7}$$

If $A \geq 0$, then $C(A) \geq 0$, and if moreover A is singular, $C(A)$ vanishes on the image of A. On positive definite matrices $\log\det$ *is concave and increasing in the matrix order.*

For a unit vector $v \in \mathbb{H}^m$ let $A|_{v^\perp}$ denote the compression of the form to the quaternionic orthogonal complement of v. For every Hermitian A and real t,

$$v^*C(A)v = \det(A|_{v^\perp}), \qquad \det(A + tvv^*) = \det A + t\, v^*C(A)v, \tag{3.8}$$

where the determinant on the zero-dimensional space is one. Finally, for a Hermitian $M = \begin{pmatrix} \ell & c^* \\ c & D \end{pmatrix}$ with $D > 0$, put $\sigma = \ell - c^*D^{-1}c$ and let v_0 be the first coordinate vector. Then

$$\det M = (\det D)\,\sigma, \qquad M - \sigma v_0 v_0^* \geq 0. \tag{3.9}$$

Proof. If $A = UDU^*$ with U unitary and D real diagonal, cyclicity gives $\mathrm{Re}\,\mathrm{tr}(A^j) = \sum_i \lambda_i^j$. By Newton's identities the elementary symmetric function $\lambda_1 \cdots \lambda_m$ is a polynomial with rational coefficients in these power sums, $1 \leq j \leq m$, which are polynomials in the entries of A. This proves polynomiality and homogeneity; (3.6) then defines $C(A)$ as the gradient of det for the inner product :, a homogeneous polynomial of degree $m-1$.

For $A > 0$, (3.5) gives $\log\det A = \frac{1}{4}\log\det_{\mathbb{R}} A_{\mathbb{R}}$. The map $A \mapsto A_{\mathbb{R}}$ is real linear and $(A^{-1}B)_{\mathbb{R}} = A_{\mathbb{R}}^{-1}B_{\mathbb{R}}$, so

$$\frac{d}{dt}\log\det(A + tB)\Big|_{t=0} = \frac{1}{4}\,\mathrm{tr}_{\mathbb{R}}(A_{\mathbb{R}}^{-1}B_{\mathbb{R}}) = \mathrm{Re}\,\mathrm{tr}(A^{-1}B) = A^{-1} : B,$$

which is (3.7). Concavity of $\log\det$ follows from that of the real log-determinant composed with a linear map, and monotonicity from $A^{-1} : B = \mathrm{Re}\,\mathrm{tr}(A^{-1/2}BA^{-1/2}) \geq 0$ for $B \geq 0$.

Since $\det(U^*AU) = \det A$ and $U^* \cdot U$ preserves the inner product, $C(U^*AU) = U^*C(A)U$. For a positive diagonal matrix, (3.7) shows that $C(A)$ is diagonal with ith entry $\prod_{j\neq i}\lambda_j$; both sides are polynomials in the diagonal entries, so the same holds for every real diagonal matrix. Diagonalising, $C(A) \geq 0$ when $A \geq 0$; if $\lambda_1 = 0$, every eigenvector v_i with $\lambda_i \neq 0$ has $i \neq 1$, and its cofactor eigenvalue contains the factor $\lambda_1 = 0$. Thus $C(A)$ vanishes on the image of A.

For (3.9), the form of M at $(b, w) \in \mathbb{H} \times \mathbb{H}^{m-1}$ is $\bar{b}\ell b + 2\,\mathrm{Re}(\bar{b}c^*w) + w^*Dw$, and completing the square gives

$$(b, w)^*(M - \sigma v_0 v_0^*)(b, w) = (w + D^{-1}cb)^*D(w + D^{-1}cb) \geq 0.$$

The factorisation

$$M = \begin{pmatrix} 1 & c^*D^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} \sigma & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} 1 & 0 \\ D^{-1}c & I \end{pmatrix}$$

has unipotent outer factors, so (3.5) gives $(\det M)^4 = \sigma^4(\det D)^4$. For $\sigma > 0$ the matrix M is positive definite by the square just completed, so $\det M > 0$ and the positive fourth root gives $\det M = (\det D)\sigma$. Both sides are polynomials in ℓ, hence the identity holds for every σ.

For (3.8) choose a unitary basis with v first, so that A takes the form of M with $D = A|_{v^\perp}$, and adding tvv^* changes only ℓ. If $D > 0$, (3.9) gives $\det(A + tvv^*) = \det A + t\det(A|_{v^\perp})$, and differentiating at $t = 0$ gives $v^*C(A)v = \det(A|_{v^\perp})$ by (3.6). Both identities are polynomial in A and hold on the open set $\{A|_{v^\perp} > 0\}$, hence for every Hermitian A. For $m = 1$ they hold directly. □

3.3. **Column divergence and the boundary term.** In k slots the real tensor of a Hermitian field $A(z)$, as in (2.6), is

$$A^{\mathbb{R}}_{a\alpha,b\beta} = \mathrm{Re}(\overline{e_\alpha} A_{ab} e_\beta). \tag{3.10}$$

It is symmetric, because $\mathrm{Re}(\overline{e_\beta} A_{ba} e_\alpha) = \mathrm{Re}(\overline{e_\alpha A_{ab} e_\beta})$, and it represents the Hermitian form on encoded real vectors:

$$(\nabla v)^{\mathrm{t}} A^{\mathbb{R}} \nabla u = \sum \partial_{a\alpha} v \; \mathrm{Re}(\overline{e_\alpha} A_{ab} e_\beta)\, \partial_{b\beta} u = \mathrm{Re}\big(G(v)^* A G(u)\big).$$

Likewise $\mathrm{Re}(A_{ab}H_{ba}) = \sum \mathrm{Re}(\overline{e_\beta} A_{ab} e_\alpha)\partial_{a\beta}\partial_{b\alpha} u$, so $A : H(u) = A^{\mathbb{R}} : D^2 u$ with the ordinary real matrix pairing on the right. Consequently, whenever the real columns of $A^{\mathbb{R}}$ have zero divergence,

$$A : H(u) = \mathrm{div}(A^{\mathbb{R}} \nabla u). \tag{3.11}$$

A sufficient condition is the quaternionic row identity

$$\sum_{a,\alpha} \overline{e_\alpha}\, \partial_{a\alpha} A_{ab} = 0 \qquad (1 \le b \le m)\,: \tag{3.12}$$

multiplying on the right by e_β and taking real parts gives $\sum_{a,\alpha} \partial_{a\alpha} A^{\mathbb{R}}_{a\alpha,b\beta} = 0$. Also, if $AG(w) = 0$ for a real function w, then wA inherits either form of column conservation from A: the product rule adds the row $\sum_{a,\alpha} \overline{e_\alpha}(\partial_{a\alpha} w) A_{ab} = (G(w)^* A)_b$, and $G(w)^* A = (AG(w))^* = 0$.

Alesker proved that the real tensor of the Moore cofactor of a positive definite quaternionic Hessian has divergence-free columns, which puts the linearised Monge–Ampère operator in divergence form [Ale13, Lemma 4.6]. The following polynomial argument of [OAI26] also covers the three slots used by the current, and allows an indefinite or singular Hessian.

Lemma 3.3 (Cofactor divergence [OAI26])**.** *For $k \in \{3, 4\}$ and a smooth real u,*

$$\sum_{a,\alpha} \overline{e_\alpha}\, \partial_{a\alpha} C(H(u))_{ab} = 0 \qquad (1 \le b \le m). \tag{3.13}$$

Proof. For $m = 1$ the cofactor is the constant 1. Otherwise put $j_0 = m - 1$ and polarise: for Hermitian $M_1, \dots, M_{j_0}$ let $Q(M_1, \dots, M_{j_0})$ be the coefficient of $t_1 \cdots t_{j_0}$ in $C(\sum_j t_j M_j)$. It is symmetric and multilinear, and since C is homogeneous of degree j_0, $Q(M, \dots, M) = j_0!\, C(M)$.

Take $M_j = g_j g_j^*$ with $g_j \in \mathbb{H}^m$. For $t_j > 0$ the matrix $\sum_j t_j g_j g_j^*$ is nonnegative with image the quaternionic span of the $j_0 < m$ columns g_j, so it is singular, and by Lemma 3.2 its cofactor vanishes on every g_i. This is a polynomial identity in t on an open orthant, so it holds coefficient by coefficient: $Q_0 = Q(g_1 g_1^*, \dots, g_{j_0} g_{j_0}^*)$ satisfies $Q_0 g_i = 0$, and $g_i^* Q_0 = 0$ since Q_0 is Hermitian.

Now take $u_j(z) = \exp(\xi_j \cdot z)$ for real covectors ξ_j, and $(g_j)_a = \sum_\alpha e_\alpha (\xi_j)_{a\alpha}$. The real Hessian of u_j is $u_j \xi_j \xi_j^{\mathrm{t}}$, so by Lemma 3.1, $H(u_j) = u_j g_j g_j^*$ and $Q(H(u_1), \dots, H(u_{j0})) = E\, Q_0$ with $E = \exp((\sum_j \xi_j) \cdot z)$. Its row divergence is

$$\sum_{a,\alpha} \overline{e_\alpha}\, \partial_{a\alpha} (E Q_0)_{ab} = E \sum_a \overline{\Big(\sum_j g_j\Big)_a} (Q_0)_{ab} = E\Big((\textstyle\sum_j g_j)^* Q_0\Big)_b = 0.$$

For general inputs, the row divergence $\mathcal{R}(u_1,\dots,u_{j_0})$ of $Q(H(u_1),\dots,H(u_{j_0}))$ is a multilinear differential expression with constant quaternionic coefficients: it is a sum of terms $c\,\partial^{\beta_1}u_1\cdots\partial^{\beta_{j_0}}u_{j_0}$ in which one multi-index has order three and the others order two. On the exponentials it equals E times $\sum c\,\xi_1^{\beta_1}\cdots\xi_{j_0}^{\beta_{j_0}}$. Distinct tuples $(\beta_1,\dots,\beta_{j_0})$ give distinct monomials in the independent variables $\xi_1,\dots,\xi_{j_0}$, so the vanishing of this polynomial forces the total coefficient of each tuple to vanish, and $\mathcal{R}\equiv 0$. Taking every $u_j=u$ gives $\mathcal{R}(u,\dots,u)=j_0!$ times the left side of (3.13). □

The boundary convention. Let $\Omega=\{r<0\}$ be a smooth domain with $\nabla r\neq 0$ on $\partial\Omega$. Write $n=\nabla r/|\nabla r|$ for the real outward unit normal and $\nu=G(r)/|\nabla r|$ for the same vector in quaternionic coordinates. For a Hermitian field A whose real columns are divergence free, (3.11) and the divergence theorem give Green's identity

$$\int_\Omega \big(v\,A:H(u)-u\,A:H(v)\big)=\int_{\partial\Omega}\mathrm{Re}\big(v\,\nu^*AG(u)-u\,\nu^*AG(v)\big)\,dS, \tag{3.14}$$

since the terms $\nabla v\cdot A^{\mathbb{R}}\nabla u$ cancel by symmetry and $n\cdot A^{\mathbb{R}}\nabla u=\mathrm{Re}(\nu^*AG(u))$. Volume and surface measure are Euclidean. The same formula holds in periodic coordinates, for test functions with compact support in any unbounded direction.

If u vanishes on $\partial\Omega$, then $G(u)=\nu\,\partial_n u$ there. Moreover $u=\rho r$ near $\partial\Omega$ for a smooth ρ: use r as one local coordinate and integrate the derivative of u in that coordinate from 0 to r. By (3.2), $H(u)=\rho H(r)+rH(\rho)+G(\rho)G(r)^*+G(r)G(\rho)^*$. On $\partial\Omega$ the second term vanishes, $G(r)=|\nabla r|\nu$, and the last two terms vanish on compression to $\nu^\perp$. Hence

$$H(u)|_{\nu^\perp}=\rho\,H(r)|_{\nu^\perp}\qquad\text{on }\partial\Omega. \tag{3.15}$$

In four slots this is the invariance of the quaternionic Levi form under a change of defining function [Ale03b]; functions independent of the fourth slot give the same formula in three slots. By (3.8) the normal flux coefficient of the cofactor, $\nu^*C(H(u))\nu=\det(H(u)|_{\nu^\perp})=\rho^{m-1}\det(H(r)|_{\nu^\perp})$, depends only on this compressed form. This is why the boundary term of the determinant can be evaluated without the second derivative of u in the real normal direction.

4. The determinant energy and its boundary flux

The time component of the current is a sum of squares. After regularisation it serves as the density of a Dirichlet potential on each spatial fibre, and a lower bound for the density on a ball forces a definite depth at its centre. To compare these depths at different times we need a scalar energy of the three real coordinates $z=(T,\tau)\in\mathbb{R}\times\mathbb{C}_{\mathbf{k}}$ whose Laplacian is bounded above by the flux of the wave through the side of the fibre.

Keep the annular wave on the torus from Section 2.6, normalised by $\|f\|_2=1$. Put $w_0=1/10$ and assume that the aperture satisfies $\gamma<w_0/4$. Write $s=|\tau|$ and $a=Kh$, where $K\geq 1$ is fixed and h is so small that $4a\leq 1$. All constructions of this section take place on $a<s<4$.

4.1. The geometry of the fibres.

At fixed z let X run over the spatial torus, and let the imaginary spatial variables $(P,Z)\in\mathbb{C}^d_{\mathbf{k}}\times\mathbb{C}_{\mathbf{k}}$ vary in the domain

$$\Omega_z:\quad r=\frac{|W|^2}{w_0^2}+100|Q|^2-1<0,\qquad Q=\Phi/a. \tag{4.1}$$

The shape of the fibre reflects two properties of the current of Proposition 2.4: it is tangent to the level sets of Φ, and the Gaussian of a mode of slope v is centred at $W=v$. Only the W part of the wall normal can therefore carry wave flux. In the coordinates (W,Q) all fibres are the

same ellipsoid. The definitions $Y = (\tau + Z)/\sqrt{2}$, $W = P/Y$ and $\Phi = (Z - \tau)/\sqrt{2} + P \cdot P/(2Y)$ give the inverse map

$$Y = \frac{\sqrt{2}(\tau + aQ)}{1 + W \cdot W/2}, \qquad P = WY, \qquad Z = \sqrt{2}\, Y - \tau. \tag{4.2}$$

Indeed, substituting $P = WY$ and $Z = \sqrt{2}Y - \tau$ into Φ gives $\Phi = Y(1 + W \cdot W/2)/\sqrt{2} - \sqrt{2}\tau$, so $aQ = \Phi$ is solved by (4.2). On the closed fibre, $|W| \le w_0$ keeps the denominator within $1/200$ of one, and $|Q| \le 1/10$ gives $|aQ| \le a/10 \le s/10$; hence $y = |Y| \asymp s$. The chart and its inverse are holomorphic on neighbourhoods of the closed fibres and depend smoothly on τ. On the total domain

$$\Omega = \{(z, X, P, Z) : a < s < 4,\ (X, P, Z) \in \Omega_z\}$$

H and G denote the full Hessian and gradient of Lemma 3.1, with the time index T carrying the triple (T, τ) and the n spatial indices carrying (X, P, Z); their spatial parts at fixed z are H_{fib} and G_{fib}. Fibre integrals use the volume $d\zeta = dX\, dP\, dZ$ in the original coordinates.

Lemma 4.1 (Fibre geometry [OAI26])**.** *The spatial Hessian $H_{\mathrm{fib}}(r)$ is positive definite. The full $H(r)$ is positive semidefinite, with a one-dimensional quaternionic kernel contained both in $G(r)^{\perp}$ and in $\ker H(|\Phi|^2)$. On the closed fibres, uniformly for $a < s < 4$,*

$$H_{\mathrm{fib}}(r) \ge cs^{-2} I, \qquad G_{\mathrm{fib}}(r) G_{\mathrm{fib}}(r)^* \le H_{\mathrm{fib}}(r), \tag{4.3}$$

$$dP\, dZ \asymp a^2 s^{2d}\, dW\, dQ, \qquad \det H_{\mathrm{fib}}(r) \asymp a^{-2} s^{-2d}. \tag{4.4}$$

The spatial Hessian $H_{\mathrm{fib}}(|\Phi|^2)$ has rank one, with nonzero eigenvalue comparable to one, and $|G(|W|^2)| \lesssim s^{-1}$.

Proof. Set $F = (W/w_0, 10Q) \in \mathbb{C}^n_{\mathbf{k}}$, a holomorphic function of (τ, P, Z), and let J be its complex Jacobian in these $n + 1$ variables, with spatial $n \times n$ block J_{fib}. Since $r = |F|^2 - 1$ does not depend on the real parts (T, X), Lemma 3.1 gives

$$H(r) = 4J^* J, \qquad G(r) = 2J^* F\, \mathbf{i}.$$

At fixed τ the map $(P, Z) \mapsto (W, Q)$ is the inverse of the chart (4.2), so J_{fib} is invertible and $H_{\mathrm{fib}}(r) = 4J^*_{\mathrm{fib}} J_{\mathrm{fib}} > 0$. In the full space we may vary τ while keeping (W, Q) fixed; this direction spans $\ker J$ over $\mathbb{C}_{\mathbf{k}}$. Writing a quaternionic column as $\alpha + \beta\mathbf{j}$ with $\alpha, \beta \in \mathbb{C}^{n+1}_{\mathbf{k}}$ we get $J(\alpha + \beta\mathbf{j}) = J\alpha + (J\beta)\mathbf{j}$, so the quaternionic kernel of $H(r)$, which equals that of J because $v^* H(r) v = 4|Jv|^2$, is the quaternionic span of this line. It is orthogonal to $G(r)$, which lies in the image of J^*, and it lies in $\ker H(|\Phi|^2)$, since the last row of J is $10a^{-1}\partial\Phi$ and $H(|\Phi|^2) = 4(\partial\Phi)^* \partial\Phi$.

By (4.2),

$$\partial_W Y = -\frac{\sqrt{2}(\tau + aQ) W}{(1 + W \cdot W/2)^2}, \quad \partial_Q Y = \frac{\sqrt{2}a}{1 + W \cdot W/2}, \quad \partial_W P = Y\, I + W \otimes \partial_W Y, \quad \partial_Q P = W \partial_Q Y,$$

and $dZ = \sqrt{2}\, dY$; so the derivatives of (P, Z) in W are $O(s)$ and those in Q are $O(a)$. Hence $\|J^{-1}_{\mathrm{fib}}\| \lesssim s$ and $H_{\mathrm{fib}}(r) \ge 4\|J^{-1}_{\mathrm{fib}}\|^{-2} I \ge cs^{-2} I$. Moreover $G_{\mathrm{fib}}(r) G_{\mathrm{fib}}(r)^* = 4J^*_{\mathrm{fib}} F F^* J_{\mathrm{fib}} \le 4|F|^2 J^*_{\mathrm{fib}} J_{\mathrm{fib}} \le H_{\mathrm{fib}}(r)$, because $|F| \le 1$ on the closed fibre. This proves (4.3).

For (4.4) take the exterior product of the differentials at fixed τ: $dP_i = Y\, dW_i + W_i\, dY$ and $dZ = \sqrt{2}\, dY$ give

$$dP_1 \wedge \cdots \wedge dP_d \wedge dZ = \sqrt{2}\, Y^d\, dW_1 \wedge \cdots \wedge dW_d \wedge dY = \frac{2aY^d}{1 + W \cdot W/2}\, dW_1 \wedge \cdots \wedge dW_d \wedge dQ.$$

The real volume Jacobian of a holomorphic map is the squared modulus of its complex determinant, which gives the volume formula. For the determinant, $4J^*_{\mathrm{fib}} J_{\mathrm{fib}}$ is a complex Hermitian

matrix, whose Moore determinant is its ordinary determinant $4^n|\det_{\mathbb{C}} J_{\mathrm{fib}}|^2$; and $\det_{\mathbb{C}} J_{\mathrm{fib}}$ is a constant multiple of $(1+W\cdot W/2)/(2aY^d)$. Finally $\partial_P\Phi = P/Y = W$ is bounded and $\partial_Z\Phi = (1-W\cdot W/2)/\sqrt{2}$ is bounded away from zero, so $H_{\mathrm{fib}}(|\Phi|^2) = 4(\partial_{\mathrm{fib}}\Phi)^*\partial_{\mathrm{fib}}\Phi$ has rank one with eigenvalue $4|\partial_{\mathrm{fib}}\Phi|^2 \asymp 1$; and the first derivatives $\partial_P W = Y^{-1}$, $\partial_Y W = -W/Y$ of $W = P/Y$ are $O(s^{-1})$, which bounds $G(|W|^2)$. □

4.2. **The Dirichlet potential.** The time density can vanish. We first smooth the current in X and cut it off in the passive coordinate Q. To make the time density strictly positive we then add a conserved current whose image is the null line of $H(r)$; it has no flux through the wall. For $\ell > 0$ let J_ℓ be the periodisation on the spatial torus of the probability kernel $c_n\ell^{-n}(1+|X/\ell|^2)^{-n-2}$. Choose a smooth $0 \le g \le 1$ on $\mathbb{C}_{\mathbf{k}}$, equal to one on $|Q| \le 1/100$ and zero on $|Q| \ge 1/20$, and define

$$A_{\mathrm{w}} = g(Q)\,(J_h *_X A^0), \qquad A = A_{\mathrm{w}} + h^{50n}C(H(r)), \qquad \mu = A_{TT}. \tag{4.5}$$

The convolution acts only in the real spatial variable X, and commutes with the derivatives, so $J_h *_X A^0$ is again positive semidefinite with conserved columns. Since $\partial\Phi$ is independent of X, the averaged current still satisfies $(\partial\Phi)(J_h *_X A^0) = 0$. By the last statement of Lemma 3.1, $G(g(Q)) = (\partial\Phi)^* q$ for a quaternion-valued q, so $(J_h *_X A^0)G(g(Q)) = 0$, and by the remark after (3.12) the cut-off current keeps conserved columns. By Lemmas 3.2 and 3.3 the cofactor $C(H(r))$ is positive semidefinite with conserved columns. Since $H(r) \ge 0$ has a one-dimensional kernel, $C(H(r))$ vanishes on the image of $H(r)$, so its image is the kernel line; this line is perpendicular to $G(r)$ and lies in $\ker H(|\Phi|^2)$. Its TT entry is, by (3.8), the determinant of the compression of $H(r)$ to the spatial coordinates, $\det H_{\mathrm{fib}}(r) > 0$. Therefore

$$A \ge 0, \qquad \mu > 0, \qquad A : H(|\Phi|^2) = 0, \tag{4.6}$$

the last because $A^0 : H(|\Phi|^2) = 4\,\mathrm{Re}\big((\partial\Phi)A^0(\partial\Phi)^*\big) = 0$ and the regulariser is supported on $\ker H(|\Phi|^2)$.

Put $B_0 = \lambda|\Phi|^2$, $B = H(B_0)$ and $B_{\mathrm{fib}} = H_{\mathrm{fib}}(B_0)$, with $\lambda = h^{-30n}$. By (4.6), $A : B = 0$. The spatial block B_{fib} has one nonzero eigenvalue $\beta \asymp \lambda$. The factor h^{50n} keeps the added density below the wave estimates of Section 5, and λ is large enough that

$$\mu^{1/d} \le \lambda \tag{4.7}$$

for all sufficiently small h. Indeed, on the fibres $y \asymp s \ge a \ge h$, so each Gaussian amplitude factor in (2.14) is at most $(\kappa/y)^{d/2} \lesssim h^{-d}$. The columns V^v are bounded and $\sum|c_{\kappa,v}| \lesssim h^{-n/2}$ by Section 2.6, so $\|A^0\| \lesssim h^{-n-2d}$; the regulariser adds only $O(h^{50n}a^{-2}s^{-2d})$ to μ by (4.4). Thus $\mu^{1/d} \lesssim h^{-(n+2d)/d} \le h^{-4}$, as $n \le 2d$.

On each spatial fibre there is a unique smooth real u with

$$D_u = H_{\mathrm{fib}}(B_0+u) > 0, \qquad \det D_u = \lambda\mu, \qquad u|_{\partial\Omega_z} = 0. \tag{4.8}$$

Indeed, $H_{\mathrm{fib}}(r)$ is positive definite, $B_{\mathrm{fib}} \ge 0$ has zero determinant because $n \ge 2$, and $\lambda\mu > 0$; these are the hypotheses of Theorem A.1 with $m = n$, the lattice $(100\mathbb{Z})^n$ in the real slots, and V the image of $\{r < 0\}$ in $\mathbb{C}^n_{\mathbf{k}}$. The chart (4.2) and the parameter statement of that theorem give smooth dependence on z up to the lateral boundary, and its barriers (A.2) give $u \le 0$. To see how the equation measures depth, add αI to the spatial background: then $\det(B_{\mathrm{fib}}+\alpha I) = (\beta+\alpha)\alpha^d$, so for $0 < \alpha \le \lambda$ a density $\lambda\mu$ corresponds to an isotropic comparison quadratic of curvature $\alpha \asymp \mu^{1/d}$. Section 5 uses such quadratics on balls.

4.3. **Cancelling the full determinant.** The quantity that measures the depth of the potential against its density is

$$X_0(z) = \int_{\Omega_z} (-u)\,\mu\,d\zeta.$$

Since the real columns of A are divergence free, Green's formula for $\Delta_z X_0$ has the interior term $-\int_{\Omega_z} A : H(u)\,d\zeta$. In a related estimate for real positive tensors, Serre uses the determinant of a Schur complement to isolate a diagonal entry of a second tensor [Ser23, Proposition 2.2]. Here only the spatial block of $H(B_0+u)$ is known to be positive, and we separate off its scalar Schur complement:

$$H(B_0+u) = \begin{pmatrix} \ell & c^* \\ c & D_u \end{pmatrix} = \sigma_0 e_T e_T^* + \Sigma, \quad \sigma_0 = \ell - c^* D_u^{-1} c, \quad \Sigma = \begin{pmatrix} c^* D_u^{-1} c & c^* \\ c & D_u \end{pmatrix} \geq 0, \quad (4.9)$$

where e_T is the time coordinate vector and $\Sigma \geq 0$ by (3.9). In the contraction with A the background $A : B$ vanishes; in the determinant, (3.9) and $\det D_u = \lambda\mu$ give $\det H(B_0+u) = \lambda\mu\sigma_0$. Therefore

$$A : H(u) = A : \Sigma + \mu\sigma_0 = A : \Sigma + \lambda^{-1} \det H(B_0+u). \quad (4.10)$$

The full determinant has no prescribed sign, so we look for a second fibre integral whose Laplacian cancels it. Write $D_v = H_{\mathrm{fib}}(B_0+v)$. Along the segment $B_0+\theta u$ the cofactor of the full Hessian has TT entry $\det D_{\theta u}$, and its contraction with $H(u)$ is the θ-derivative of the full determinant. This suggests integrating the spatial determinant along the segment. For a smooth real v on Ω_z with $D_v > 0$ and zero boundary values set

$$E_z(v) = \frac{1}{\lambda} \int_{\Omega_z} v \int_0^1 \det D_{\theta v}\,d\theta\,d\zeta, \qquad J_z(v) = E_z(v) - \int_{\Omega_z} v\,\mu\,d\zeta. \quad (4.11)$$

The segment integral is the quaternionic counterpart of the Aubin–Mabuchi energy: Mabuchi's functional integrates the variation of a Kähler potential against its complex Monge–Ampère measure [Mab86, (2.2.1)]. Here the determinant is a polynomial, so the integrand is smooth up to the singular endpoint $D_0 = B_{\mathrm{fib}}$. Define

$$\mathcal{F}(z) = J_z(u) = E_z(u) + X_0(z).$$

Proposition 4.2 (Energy balance [OAI26])**.** *The energy $\mathcal{F}$ is smooth on $a < |\tau| < 4$. Let ρ be the boundary value of u/r, which extends smoothly from the interior. In the three variables $z = (T, \tau)$,*

$$\Delta_z \mathcal{F}(z) + \int_{\Omega_z} A : \Sigma\,d\zeta = \int \rho\,G(r)^* A_{\mathrm{w}} G(r)\,\delta(r)\,d\zeta. \quad (4.12)$$

The normal seen by the current is that of the moving wall in the total space, so $G(r)$ is the full gradient. The integral on the right is over one fibre: $\delta(r)\,d\zeta = dS_{\mathrm{fib}}/|\nabla_{\mathrm{fib}} r|$, and the spatial gradient does not vanish by the ellipsoid chart. Since $A : \Sigma \geq 0$, the boundary integral bounds $\Delta_z \mathcal{F}$ from above.

Proof. Denote the right side of (4.12) by $\mathcal{B}(z)$. The balance is the sum of

$$\Delta_z X_0 = \mathcal{B}(z) - \int_{\Omega_z} A : \Sigma\,d\zeta - \lambda^{-1} \int_{\Omega_z} \det H(B_0+u)\,d\zeta, \quad (4.13)$$

$$\Delta_z[E_z(u)] = \lambda^{-1} \int_{\Omega_z} \det H(B_0+u)\,d\zeta. \quad (4.14)$$

For a fixed complex product domain, Åhag et al. identified the Laplacian of the slice energy, up to normalisation, with the integral of the full Monge–Ampère measure over the fibre [ÅCKPZ09, Theorem 3.1]. Here the fibre moves, so (4.14) requires a boundary calculation. Both identities follow by testing with a smooth $\eta = \eta(z)$ compactly supported in the parameter domain. The only

entry of $H(\eta)$ is $H_{TT}(\eta) = \Delta_z \eta$, by Lemma 3.1. For a Hermitian field M with divergence-free real columns, Green's formula (3.14) on the total domain, with $v = \eta$, gives

$$\int_\Omega u\, M_{TT}\, \Delta_z \eta = \int_\Omega \eta\, M : H(u) - \int_{\partial_{\mathrm{lat}}\Omega} \eta\, (\nu^* M \nu)\, \partial_n u\, dS. \tag{4.15}$$

Here $\partial_{\mathrm{lat}}\Omega$ is the side $r = 0$, volume on Ω is $dz\, d\zeta$, and the term containing $u\nabla\eta$ vanishes because $u = 0$ on the lateral boundary; there are no end terms because of the support of η. The outward unit normal $n = \nabla r/|\nabla r|$ has quaternionic column $\nu = G(r)/|\nabla r|$, and $G(u) = \nu \partial_n u$ on the wall. Since r is a smooth defining function, $u = r\rho$ near the wall with ρ smooth, and $\rho \geq 0$ there because $u \leq 0$; in particular $\partial_n u = \rho |\nabla r|$.

For X_0 take $M = A$ in (4.15) and change sign, since X_0 integrates $(-u)\mu$. The interior term is given by (4.10). On the wall the regulariser $C(H(r))$ has image perpendicular to ν, so only A_{w} contributes, and substituting $\partial_n u = \rho|\nabla r|$ and $dS = |\nabla r|\, \delta(r)\, dz\, d\zeta$ produces $\mathcal{B}(z)$. This proves (4.13).

For $E_z(u)$ take $M_\theta = C(H(B_0 + \theta u))$, whose columns are conserved by Lemma 3.3. Here the whole normal coefficient vanishes. Indeed (3.15), applied to $\theta u = \theta\rho r$, gives on $r = 0$

$$H(B_0 + \theta u)|_{\nu^\perp} = (B + \theta\rho H(r))|_{\nu^\perp}.$$

The null line of $H(r)$ lies in $\nu^\perp$ and in $\ker B$ by Lemma 4.1, so the compressed form has a kernel and is singular, and (3.8) yields $\nu^* M_\theta \nu = 0$ whatever the sign of the full Hessian. The same formula in the direction e_T gives $(M_\theta)_{TT} = \det D_{\theta u}$. Integrate (4.15) in $\theta \in [0, 1]$ and divide by λ. On the left we obtain $\int E_z(u) \Delta_z \eta\, dz$; on the right, $M_\theta : H(u) = \partial_\theta \det H(B_0 + \theta u)$ by (3.6), which integrates to the full determinant at $\theta = 1$, since $\det B = 0$ for the rank-one B and $n + 1 \geq 2$. This proves (4.14), hence the balance in the sense of distributions. The chart (4.2) maps every fibre onto a fixed product of a torus and an ellipsoid, and u depends smoothly on z up to the boundary; so the volume and surface integrals are smooth in z and the balance holds pointwise. □

4.4. **Size of the energy on a fibre.** Fix z. The segment integral in (4.11) is a primitive of the spatial determinant: for a smooth variation w with zero boundary values,

$$dJ_z(v)[w] = \int_{\Omega_z} w\, (\lambda^{-1} \det D_v - \mu)\, d\zeta, \tag{4.16}$$

$$d^2 J_z(v)[w, w] = -\lambda^{-1} \int_{\Omega_z} G_{\mathrm{fib}}(w)^* C(D_v) G_{\mathrm{fib}}(w)\, d\zeta. \tag{4.17}$$

For (4.16), differentiating under the integral in (4.11) produces, besides $\lambda^{-1} \int w \int_0^1 \det D_{\theta v}\, d\theta$, the term $\lambda^{-1} \int v \int_0^1 \theta\, C(D_{\theta v}) : H_{\mathrm{fib}}(w)\, d\theta$. The spatial cofactor $C(D_{\theta v})$ has conserved columns by Lemma 3.3, and v, w vanish on the boundary, so (3.14) lets us interchange v and w in this fibre integral. The integrand in θ becomes $w(\det D_{\theta v} + \theta C(D_{\theta v}) : H_{\mathrm{fib}}(v)) = w\, \partial_\theta(\theta \det D_{\theta v})$, which integrates to $w \det D_v$. Differentiating (4.16) once more gives $\lambda^{-1} \int w C(D_v) : H_{\mathrm{fib}}(w)$, and the integration by parts of (3.14) gives (4.17). It is strictly negative for $w \neq 0$, since $C(D_v) > 0$ by (3.7). Hence the Dirichlet solution is the unique maximiser of J_z on this convex class: it is the smooth determinant variational principle with a rank-one background. For zero background Wan proved the first variation and the variational characterisation in quaternionic finite energy classes [Wan20, §4].

Since $u \leq 0$ and $D_{\theta u} = (1 - \theta) B_{\mathrm{fib}} + \theta D_u \geq 0$, the correction $E_z(u)$ is nonpositive. At a solution the linear term of $\mathcal{F}$ uses the determinant at $B_0 + u$, so $\lambda\mathcal{F}$ has the form of Aubin's J-functional [Aub84, p. 145], with $B_0 + u$ as reference potential and B_0 as argument in the relative formula [BBGZ13, §2.1]. The rank-one background lets us compute how much of X_0 remains in $\mathcal{F} = X_0 + E_z(u)$.

Lemma 4.3 (Energy comparison [OAI26]). *On every spatial fibre,*

$$\mathcal{F} = \frac{n-1}{n+1} X_0 + \frac{1}{\lambda n(n+1)} \int_{\Omega_z} G_{\mathrm{fib}}(u)^* C(D_u) G_{\mathrm{fib}}(u)\, d\zeta. \tag{4.18}$$

In particular $\frac{n-1}{n+1} X_0 \le \mathcal{F} \le \frac{n}{n+1} X_0$.

Proof. Write $B_{\mathrm{fib}} = \beta b b^*$ with $|b| = 1$. By (3.8) applied to θD_u, whose cofactor is $\theta^{n-1} C(D_u)$, the determinant along the segment has two terms:

$$\det D_{\theta u} = \theta^n \det D_u + (1-\theta)\theta^{n-1} C(D_u) : B_{\mathrm{fib}}.$$

Using $\int_0^1 \theta^n d\theta = \frac{1}{n+1}$ and $\int_0^1 (1-\theta)\theta^{n-1} d\theta = \frac{1}{n(n+1)}$, together with $I = \int_{\Omega_z} (-u)\, C(D_u) : B_{\mathrm{fib}}\, d\zeta \ge 0$ and $\det D_u = \lambda\mu$, we get

$$E_z(u) = -\frac{X_0}{n+1} - \frac{I}{\lambda n(n+1)}, \qquad \mathcal{F} = \frac{n}{n+1} X_0 - \frac{I}{\lambda n(n+1)},$$

in particular the upper bound. By homogeneity $C(D_u) : D_u = n \det D_u$, so

$$n\lambda X_0 - I = \int_{\Omega_z} (-u)\, C(D_u) : H_{\mathrm{fib}}(u)\, d\zeta = \int_{\Omega_z} G_{\mathrm{fib}}(u)^* C(D_u) G_{\mathrm{fib}}(u)\, d\zeta;$$

the last integration by parts uses (3.14) and $u = 0$ on the boundary. Substituting for I proves (4.18), and the lower bound follows since $C(D_u) \ge 0$. □

5. Estimating the energy on a fixed fibre

On a fibre at fixed (T, τ) the only positive contribution to the energy balance comes from the wave current crossing the spatial wall $r = 0$. The Gaussian is small where the current reaches the wall, and comparison with a spatial average of the wave bounds the other factor, the boundary value of u/r. This bounds the flux by a norm of the wave. To bound it by the energy itself we recover the same norm from the interior: there we invert the Gaussian and compare u with a quadratic on each ball of radius h.

Proposition 5.1 ([OAI26]). *Fix $0 < c_0 < C_0$. There are constants $\gamma > 0$, $K \ge 1$ and $c, C > 0$, depending only on n, c_0 and C_0, with the following property. Let $a = Kh$ with $0 < h < 1/(4K)$, and let f be a trigonometric polynomial on $\mathbb{T}^n = \mathbb{R}^n/(100\mathbb{Z})^n$ with $\|f\|_2 = 1$ and frequencies in $c_0/h \le |\xi| \le C_0/h$ and in the cap $|v| \le \gamma$ of some orthonormal frame. Write $g_T = e^{-iT|D|} f$ and let $\mathcal{F}(T,\tau)$ be its energy from Section 4. Then*

$$\mathcal{F}(0,\tau) \le C a^2 \|f\|_{p_c}^{p_c} \qquad (|\tau| = 4a), \tag{5.1}$$

$$\mathcal{F}(T,\tau) \ge c a^2 \|g_T\|_{p_c}^{p_c} \qquad (|\tau| = 2a), \tag{5.2}$$

$$\Delta_z \mathcal{F}(T,\tau) \le V(s)\, \mathcal{F}(T,\tau), \quad V(s) = C s^{-2} e^{-cs/h} \qquad (2a \le s = |\tau| \le 3). \tag{5.3}$$

The last two inequalities hold for all $T \in \mathbb{R}$.

Proof. Write $d = n-1$, $q = n/d = p_c/2$ and $\sigma = s/h$; throughout, $2a \le s \le 3$. On the fibre (4.1) we use two regions. On $r = 0$ the cut-off $g(Q)$ in the current vanishes unless $|Q| \le 1/20$, and then $|W|^2/w_0^2 = 1 - 100|Q|^2 \ge 3/4$, so $|W| \ge (\sqrt{3}/2) w_0$. Near the centre, where $|W| \le \gamma$ and $|Q| \le 1/400$, the Gaussian multiplier can be inverted. Figure 1 shows the separation between the two regions, which lets us compare two exponentials with different constants in their exponents. For now take $0 < \gamma < w_0/8$; we estimate uniformly in s, and allow the prefactor in the interior lower bound to depend on γ.

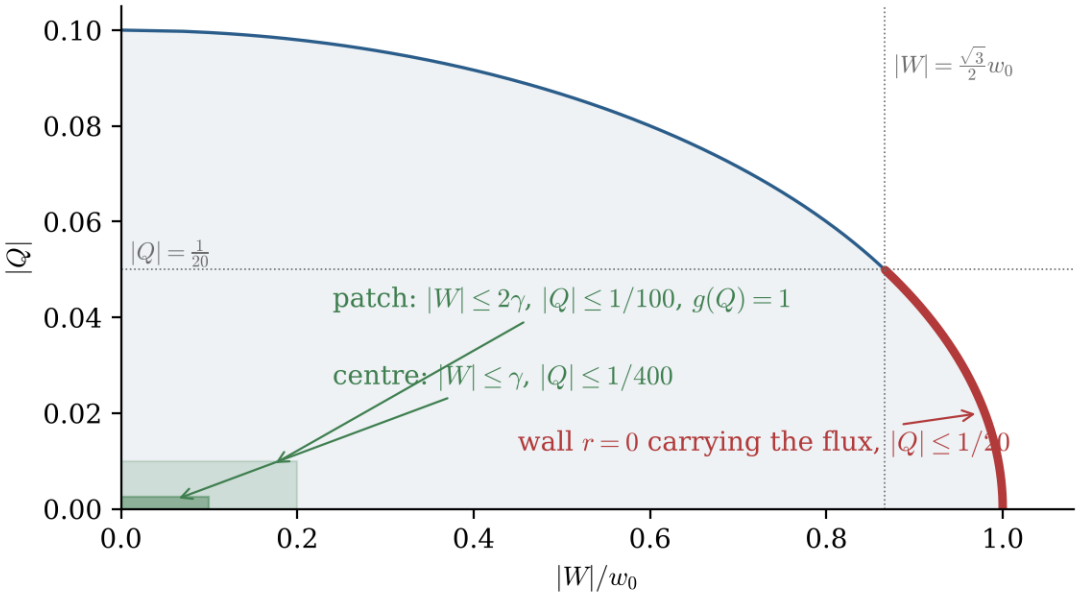


FIGURE 1. The fixed fibre in the modulus coordinates $(|W|, |Q|)$. The wave flux through the wall $r = 0$ is carried by the heavy arc, where $|Q| \le 1/20$ and hence $|W| \ge \sqrt{3}\, w_0/2$; this separation from the frequency cap gives the Gaussian decay $e^{-c\,s/h}$. Near the centre, balls of radius h about points with $|W| \le \gamma$, $|Q| \le 1/400$ stay in the patch $|W| \le 2\gamma$, $|Q| \le 1/100$, where $g(Q) = 1$ and the Gaussian can be inverted at the cost $e^{C\gamma^2 s/h}$.

A spatial average for the upper bound. For a dimensional constant $D \ge 1$, fixed below, set

$$P_*(X) = (J_{Ds} * |g_T|^2)(X)^{1/d}, \qquad M = 2n + 4,$$

where convolution is in the real spatial variables and J_ℓ is the periodisation of $c_n \ell^{-n}(1 + |X/\ell|^2)^{-M/2}$, of integral one, as in (4.5). Each derivative of this kernel costs at most a factor $C\ell^{-1}$ pointwise, $|\nabla^j J_\ell| \le C\ell^{-j} J_\ell$; hence $|\nabla^j (J_{Ds} * |g_T|^2)| \le C(Ds)^{-j} J_{Ds} * |g_T|^2$, and taking the dth root gives the first half of

$$|\nabla_X^j P_*| \le C(Ds)^{-j} P_* \quad (j = 1, 2), \qquad \int P_*^{d+1}\, dX \le \|g_T\|_{p_c}^{p_c}. \tag{5.4}$$

The second half is Young's inequality with exponent $q = (d+1)/d$: $\int P_*^{d+1} = \|J_{Ds} * |g_T|^2\|_q^q \le \||g_T|^2\|_q^q = \|g_T\|_{2q}^{2q}$. The positive kernel gives $P_* > 0$, since $\|g_T\|_2 = 1$.

We next bound the current A_{w} and its density μ. By Proposition 2.4, each real component of χ is the real or imaginary part of a Fourier multiplier applied to g_T, whose symbol is a bounded smooth factor times

$$(\kappa/y)^{d/2} e^{-\Psi}, \qquad \Psi = \tfrac{1}{2}\kappa y |W - v|^2, \qquad y = |Y| \asymp s.$$

Multiply the symbol by a fixed smooth annular cut-off equal to one on the data and supported in $|v| < w_0/3$. In the variable $\eta = h\xi$ its derivatives through order M are bounded by $C(hs)^{-d/2}\sigma^m$ for some fixed m: on this cut-off $h\kappa$ and v have bounded derivatives, $h\kappa$ is bounded away from zero, $(\kappa/y)^{d/2} \asymp (hs)^{-d/2}$, and each derivative of $\Psi = \frac{1}{2}(h\kappa)(y/h)|W - v|^2$ costs at most $C\sigma$. On the part of the wall where $g(Q) \ne 0$ we have $|W - v| \ge (\sqrt{3}/2 - 1/3)w_0$, so the same bound holds with an extra factor $e^{-c_*\sigma}$, where $c_* > 0$ does not depend on γ.

If B bounds the derivatives through order M of a symbol $a(h\xi)$ supported in a fixed compact set, then applying $(1-\Delta_\eta)^{n+2}$ inside the inverse Fourier transform bounds the absolute value of its kernel by $C\mathsf{B}h^{-n}(1+|X/h|^2)^{-M/2}$. On the torus the kernel of the multiplier is the periodisation of the Euclidean kernel, so it is majorised by $C\mathsf{B}J_h$, also for a rotated frame. By Cauchy–Schwarz the squared modulus of the output is at most $C\mathsf{B}^2 J_h * |g_T|^2$. There is one more convolution with J_h in A_{w}, and $J_h * J_h \le CJ_h$: in the Euclidean convolution one of $|Y|$, $|X-Y|$ is at least $|X|/2$, and the inequality survives periodisation. Moreover $J_h \le (Ds/h)^n J_{Ds}$, since $h^{-n}(1+|X/h|^2)^{-M/2} \le (L/h)^n L^{-n}(1+|X/L|^2)^{-M/2}$ for $L = Ds \ge h$. Using $(hs)^{-d} = s^{-2d}\sigma^d$ and enlarging m, we obtain, for the matrix operator norm,

$$\|A_{\mathrm{w}}\| \le Cs^{-2d}\sigma^m P_*^d \quad (r \le 0), \qquad \|A_{\mathrm{w}}\| \le Cs^{-2d}\sigma^m e^{-c_*\sigma} P_*^d \quad (r = 0). \tag{5.5}$$

Only the first bound is needed for μ, whose remaining term is $h^{50n}\det H_{\mathrm{fib}}(r) \le Ch^{50n}a^{-2}s^{-2d}$ by Lemma 4.1. The nearest lattice term of the kernel gives $\inf_X J_{Ds}(X) \ge c_D h^{M-n}$, since the torus has fixed diameter and $h \le s \le 3$; consequently $P_*^d \ge c_D h^{M-n}\|g_T\|_2^2 = c_D h^{M-n}$, which absorbs $h^{50n}a^{-2} \le h^{50n-2}$. In particular

$$\mu \le Cs^{-2d}\sigma^m P_*^d \qquad (r \le 0). \tag{5.6}$$

A comparison function vanishing on the wall. Put $R = H_{\mathrm{fib}}(r)$. By Lemma 4.1, $R \ge cs^{-2}I$ and $G_{\mathrm{fib}}(r)G_{\mathrm{fib}}(r)^* \le R$. We claim

$$H_{\mathrm{fib}}(rP_*) \ge (1 - C/D)P_* R. \tag{5.7}$$

The product rule (3.2) gives P_*R, $rH_{\mathrm{fib}}(P_*)$ and two terms pairing the gradients of r and P_*. For a quaternionic vector e, $|e^*G_{\mathrm{fib}}(r)| \le (e^*Re)^{1/2}$, while $|e^*G_{\mathrm{fib}}(P_*)| \le |\nabla P_*||e| \le C(Ds)^{-1}P_* \cdot Cs(e^*Re)^{1/2}$ by (5.4) and $R \ge cs^{-2}I$; this bounds the mixed terms by $CD^{-1}P_* e^*Re$. The term with $H_{\mathrm{fib}}(P_*)$ is at most $C(Ds)^{-2}P_*|e|^2 \le CD^{-2}P_* e^*Re$ by (5.4) and $|r| \le 1$.

Since $r < 0$, an upper bound for u/r calls for a comparison function below u whose determinant is at least $\lambda\mu$. Fix D so that the right side of (5.7) is at least $P_*R/2$. For $v_* = C_1\sigma^{m_1} r P_*$ this gives $H_{\mathrm{fib}}(v_*) \ge \alpha I$ with $\alpha = cC_1\sigma^{m_1}P_* s^{-2}$. The background $B_{\mathrm{fib}} = \lambda H_{\mathrm{fib}}(|\Phi|^2)$ has rank one and positive eigenvalue $\beta \asymp \lambda$ by Lemma 4.1, so by the monotonicity of det in Lemma 3.2 and (3.8),

$$\det\bigl(B_{\mathrm{fib}} + H_{\mathrm{fib}}(v_*)\bigr) \ge \det(B_{\mathrm{fib}} + \alpha I) = (\beta+\alpha)\alpha^d \ge c\lambda\bigl(C_1\sigma^{m_1}P_* s^{-2}\bigr)^d \ge \lambda\mu,$$

by (5.6), once $m_1 d \ge m$ and C_1 is large. The potentials v_* and u have the same zero boundary values, so the comparison principle in the proof of Theorem A.1 yields $v_* \le u \le 0$. Dividing by $r < 0$ and passing to the boundary,

$$0 \le u/r \le C_1\sigma^{m_1}P_* \qquad (r \le 0). \tag{5.8}$$

By Lemma 4.3, $\mathcal{F} \le X_0 = \int(-u)\mu\, d\zeta$, and $-u = |r|\,(u/r) \le C_1\sigma^{m_1}P_*$. The volume element in (P, Z) is comparable to $a^2 s^{2d}\, dW\, dQ$ on a fixed bounded set, by Lemma 4.1, and P_* depends only on X. Combining (5.4), (5.6) and (5.8) therefore gives, after enlarging m,

$$\mathcal{F}(T,\tau) \le Ca^2\sigma^m \|g_T\|_{p_{\mathrm{c}}}^{p_{\mathrm{c}}}. \tag{5.9}$$

For the flux we need the full gradient of r. Tangency in Proposition 2.4 gives $A_{\mathrm{w}}G(|\Phi|^2) = 0$, because $G(|\Phi|^2) = 2(\partial\Phi)^*\Phi\,\mathbf{i}$; the remaining part $w_0^{-2}G(|W|^2)$ has size $O(s^{-1})$ by Lemma 4.1. Thus $G(r)^* A_{\mathrm{w}} G(r) \le Cs^{-2}\|A_{\mathrm{w}}\|$. In the coordinates (W, Q) the function r is fixed with nonvanishing gradient on the wall, so the coarea formula and Lemma 4.1 give $\int \delta(r)\, dP\, dZ \le Ca^2 s^{2d}$. Inserting the wall bound (5.5) and (5.8) into the flux integral of Proposition 4.2,

$$\Delta_z \mathcal{F}(T,\tau) \le C(a/s)^2 \sigma^m e^{-c_*\sigma}\|g_T\|_{p_{\mathrm{c}}}^{p_{\mathrm{c}}}. \tag{5.10}$$

Recovering the wave in the interior. Consider a centre (X, P, Z) with $|W| \le \gamma$ and $|Q| \le 1/400$. In the ball of radius h about it in the $3n$ real spatial coordinates, $W = P/Y$ changes by at most $Ch/s \le C/K$ and Φ by at most Ch, so Q by at most C/K. If $K\gamma$ and K are large enough, the ball lies in $|W| \le 2\gamma$, $|Q| \le 1/100$, inside the fibre, and the cut-off in A_{w} equals one throughout it.

The first scalar multiplier S_0 of the time component in (2.15) has symbol $b_0(\kappa/y)^{d/2}e^{-\Psi}$ with $b_0 = (1 + \mathrm{Re}(W) \cdot v/2)/\sqrt{2}$ bounded away from zero, and the other component is orthogonal, so $|S_0 g_T|^2 \le A^0_{TT}$. Invert S_0 on the frequencies of g_T by multiplying its reciprocal symbol by a smooth annular cut-off equal to one on the data and supported in $|v| \le 2\gamma$. On its support $|W - v| \le 4\gamma$, so $\Psi \le C\gamma^2\sigma$, and the derivatives of the reciprocal symbol in $\eta = h\xi$ through order M are bounded by

$$\mathsf{B}_{\mathrm{inv}} = C_\gamma (hs)^{d/2} \sigma^m e^{C\gamma^2\sigma};$$

the derivatives of the cut-off enter only the prefactor C_γ, and the constant in the exponential does not depend on γ. The kernel estimate above yields $|g_T(X)|^2 \le C\mathsf{B}^2_{\mathrm{inv}}(J_h * A^0_{TT})(X)$ for each choice of the imaginary coordinates in the ball, with A^0_{TT} evaluated at those coordinates. Since $J_h(X - Y)$ and $J_h(X' - Y)$ are comparable for $|X' - X| \le h$, the density at every point of the ball is at least

$$m_X = c_\gamma (hs)^{-d} \sigma^{-m} e^{-C_2\gamma^2\sigma} |g_T(X)|^2. \tag{5.11}$$

For the interior lower bound on $-u$ we need a comparison function above u. Let l be the Euclidean distance from the centre. The quadratic $v_b = \varepsilon m_X^{1/d}(l^2 - h^2)$ vanishes on the boundary of the ball and has fibre Hessian $6\varepsilon m_X^{1/d} I$, since there are three real slots per spatial coordinate. If $m_X > 0$, its determinant after adding B_{fib} is

$$(\beta + 6\varepsilon m_X^{1/d})(6\varepsilon m_X^{1/d})^d \le C\big((\varepsilon m_X^{1/d})^n + \lambda(\varepsilon m_X^{1/d})^d\big) \le C(\varepsilon^n + \varepsilon^d)\lambda m_X \le \lambda m_X,$$

for a fixed small ε, because the density ceiling (4.7) and (5.11) give $m_X^{1/d} \le \mu^{1/d} \le \lambda$. Now $u \le 0 = v_b$ on the boundary of the ball and the determinant of $B_{\mathrm{fib}} + H_{\mathrm{fib}}(u)$ is $\lambda\mu \ge \lambda m_X$, so the comparison principle on the ball (Theorem A.1 and its proof) gives $u \le v_b$ there. At the centre the depth is therefore at least $\varepsilon h^2 m_X^{1/d}$, and multiplying by $\mu \ge m_X$ gives $(-u)\mu \ge \varepsilon h^2 m_X^{1+1/d}$, also when $m_X = 0$. Here $1 + 1/d = q$, and m_X^q contains $|g_T(X)|^{2q} = |g_T(X)|^{p_{\mathrm{c}}}$.

The imaginary volume of the region of centres is comparable to $a^2 s^{2d}\gamma^{2d}$. Integrate the centre inequality over it and over X, use the lower bound of Lemma 4.3, and note that $h^2 s^{2d}(hs)^{-dq} = h^{1-d}s^{d-1} = \sigma^{n-2}$ since $dq = d + 1$. Enlarging m and absorbing γ^{2d} into the prefactor,

$$\mathcal{F}(T, \tau) \ge c_\gamma a^2 \sigma^{-m} e^{-C_3\gamma^2\sigma} \|g_T\|^{p_{\mathrm{c}}}_{p_{\mathrm{c}}}. \tag{5.12}$$

Choose γ so small that $C_3\gamma^2 < c_*/2$, then fix K to meet the ball inclusions and so that $h < 1/(4K)$ lies in the range of (4.7). For $f \ne 0$ both sides below are positive, and dividing (5.10) by (5.12) gives

$$\frac{\Delta_z \mathcal{F}}{\mathcal{F}} \le C_\gamma s^{-2} \sigma^{2m} e^{-(c_* - C_3\gamma^2)\sigma}.$$

The exponential absorbs the power of σ, which proves (5.3). On the two circles in the τ-plane the scaled radii $4a/h = 4K$ and $2a/h = 2K$ are fixed. The upper bound (5.9) at $T = 0$ on the first circle gives (5.1), and the lower bound (5.12) on the second gives (5.2) for the wave at time T. □

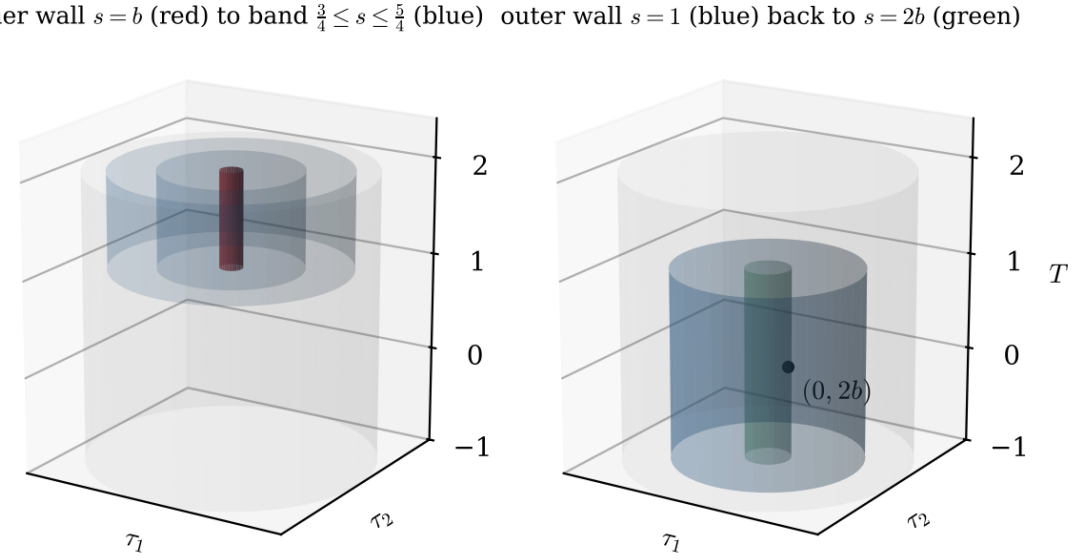


FIGURE 2. The two comparisons in the (T,s) half-plane, $s=|\tau|$, drawn schematically. Left: the mass of the circular mean on the inner wall $s=b$ for $1\le T\le 2$ bounds the mass in the band $3/4\le s\le 5/4$ from below, at the cost $1/\log(3/b)$. Right: a mean-value inequality on a fixed annulus gives a lower bound on the whole outer wall $s=1$, and the comparison solution $\cos(\pi T/2)u(s)$ carries it back to $s=2b$ at the cost $\log 2/\log(1/b)$. In the wave estimate $b=2Kh$, the inner wall controls the wave on $1\le T\le 2$, and the dot at $(0,2b)$ is controlled by the initial data.

6. Propagation across a thin cylinder

The energy of Proposition 5.1 controls the wave on the circle $|\tau|=2a$ for $1\le T\le 2$, and is controlled by the data on $|\tau|=4a$ at $T=0$. To compare the two sets we first carry the energy out to $|\tau|=1$, where a fixed neighbourhood connects the two time intervals, and then carry it back towards the inner wall. Each passage costs one logarithm, as Figure 2 illustrates. The radial harmonic functions used for comparison are the probabilities that planar Brownian motion leaves an annulus through one circle or the other [MP10, Theorem 3.18]; the maximum principle keeps this comparison in the presence of a potential.

Lemma 6.1 ([OAI26])**.** *Let $0<b<1/4$ and $D=\log(3/b)$. Suppose a continuous potential $V:[b,3]\to[0,\infty)$ satisfies*

$$\int_b^3 s\log(s/b)\,V(s)\,ds\le M,\qquad \sup_{1/2\le s\le 3/2}V(s)\le M. \tag{6.1}$$

Let $E\ge 0$ be smooth on $\mathbb{R}_T\times\{\tau\in\mathbb{R}^2:b\le|\tau|\le 3\}$ with $\Delta_{T,\tau}E\le V(|\tau|)E$ in the interior. Its circular mean

$$H(T,s)=\frac{1}{2\pi}\int_0^{2\pi}E\big(T,s(\cos\theta,\sin\theta)\big)\,d\theta$$

satisfies, for $b\le r\le 1$,

$$H(0,r)\ge c(M)\,\frac{\log(r/b)}{D^2}\int_1^2 H(T,b)\,dT. \tag{6.2}$$

In particular the factor at $r=2b$ is a constant multiple of D^{-2}.

Proof. Averaging over rotations preserves the differential inequality, so $H(T,|\tau|)$ is again a supersolution: $(-\Delta_{T,\tau}+V)H \geq 0$. All maximum principles below are the weak maximum principle for operators $\Delta - c$ with bounded $c \geq 0$ [GT01, Corollary 3.2].

Radial comparison functions. For $S \in \{1,3\}$ and a fixed $\kappa > 0$ consider

$$u'' + s^{-1}u' = (V+\kappa^2)u \qquad (b < s < S),$$

with endpoint values $0, 1$ in either order. We claim that u is at least a constant $c(M,\kappa)$ times the function affine in $\log s$ with the same endpoint values. The substitution $x = \log(s/b)$ gives $u'' + s^{-1}u' = s^{-2}u_{xx}$, so the equation becomes $u_{xx} = P(x)u$ on $[0,A]$ with $A = \log(S/b)$ and $P(x) = s^2(V(s)+\kappa^2) \geq 0$. Let ψ solve it with $\psi(0) = 0$, $\psi'(0) = 1$. Then $\psi'' = P\psi \geq 0$ while $\psi \geq 0$, so $\psi' \geq 1$ and $\psi(x) \geq x$. Integrating twice, $\psi(x) = x + \int_0^x (x-z)P(z)\psi(z)\,dz$, that is,

$$\frac{\psi(x)}{x} = 1 + \int_0^x \frac{z(x-z)}{x}\,P(z)\,\frac{\psi(z)}{z}\,dz.$$

Since $z(x-z)/x \leq \min(z, A-z)$, Gronwall's inequality gives $\psi(A)/A \leq \exp\bigl(\int_0^A \min(z, A-z)P(z)\,dz\bigr)$. In this integral, $dz = ds/s$ and $\min(z, A-z) \leq \log(s/b)$, so the contribution of V is at most $\int_b^3 s\log(s/b)V(s)\,ds \leq M$; with $t = A - z$, $s = Se^{-t}$, the contribution of κ^2 is at most $\kappa^2 S^2 \int_0^\infty te^{-2t}dt$. Hence $\psi(A) \leq C(M,\kappa)A$, and the solution $u = \psi/\psi(A)$ with $u(0) = 0$, $u(A) = 1$ satisfies $u(x) \geq x/(C(M,\kappa)A)$. The integral bound is unchanged under $x \mapsto A - x$, which gives the other order of the endpoint values.

The outward passage. Put $m(s) = \int_0^3 \sin(\pi T/3)H(T,s)\,dT$. The cut-off is positive on the observed times and its second derivative is a multiple of itself; two integrations by parts give

$$\int_0^3 \sin(\pi T/3)\,\partial_T^2 H(T,s)\,dT = -(\pi/3)^2 m(s) + (\pi/3)\bigl(H(0,s) + H(3,s)\bigr).$$

Since $H \geq 0$, multiplying $\partial_T^2 H + \partial_s^2 H + s^{-1}\partial_s H \leq VH$ by $\sin(\pi T/3)$ and integrating gives $m'' + s^{-1}m' \leq (V + \pi^2/9)m$. Let u be the radial solution with $S = 3$, $\kappa = \pi/3$, $u(b) = 1$ and $u(3) = 0$. Then $m - m(b)u$ is a supersolution vanishing at $s = b$ and nonnegative at $s = 3$, so it is nonnegative, and the radial estimate gives $m(s) \geq c\,m(b)\log(3/s)/D$. In particular

$$\int_0^3 \int_{3/4 \leq |\tau| \leq 5/4} H(T,|\tau|)\,d\tau\,dT \geq c\,m(b)/D. \tag{6.3}$$

A Harnack chain. On the fixed region $-2 < T < 4$, $1/2 < |\tau| < 3/2$ we have $\Delta H \leq MH$. The compact weak Harnack inequality gives $H(T,1) \geq c(M)m(b)/D$ for $|T| \leq 1$; we give the argument. Add a real variable v. The function $w = H(T,|\tau|)\cos(\sqrt{M+1}\,v)$ satisfies $\Delta w \leq (MH - (M+1)H)\cos(\sqrt{M+1}\,v) \leq 0$, so it is nonnegative and superharmonic on a fixed slab $|v| < v_0$. Its mean-value inequality on the four-dimensional ball of radius 2ρ about $(x,0)$, $x = (T,\tau)$, restricted to $|v| < \rho$, where each slice of the ball contains the three-dimensional ball $B_\rho(x)$ and the cosine is at least $1/2$, gives

$$H(x) \geq c\rho^{-3} \int_{B_\rho(x)} H$$

for one small radius ρ, throughout a neighbourhood of the compact connected band $[-1,3] \times \{3/4 \leq |\tau| \leq 5/4\}$; here $H(x) = H(T,|\tau|)$. Cover the band by finitely many balls of radius $\rho/4$. If two of them, B and B', overlap, then $B' \subset B_\rho(x)$ for every $x \in B$, so $\inf_B H \geq c\rho^{-3}\int_{B'} H$. By (6.3) one ball of the cover carries at least a fixed fraction of $c\,m(b)/D$, and a finite chain of overlapping balls gives $H \geq c(M)m(b)/D$ on the whole band.

The return passage. Take the radial solution u with $S = 1$, $\kappa = \pi/2$, $u(b) = 0$ and $u(1) = 1$. On $|T| < 1$, $b < |\tau| < 1$ the function $w = c(M)m(b)D^{-1}\cos(\pi T/2)u(|\tau|)$ satisfies $(-\Delta_{T,\tau} + V)w = 0$,

vanishes on $|T| = 1$ and on $|\tau| = b$, and lies below H on $|\tau| = 1$ by the Harnack chain. The maximum principle gives $H \geq w$, and the radial estimate gives

$$H(0, r) \geq \frac{c(M)m(b)}{D} \cdot \frac{\log(r/b)}{\log(1/b)} \qquad (b \leq r \leq 1).$$

Since $\log(1/b) \leq D$ and $\sin(\pi T/3) \geq \sqrt{3}/2$ on $[1, 2]$, this proves (6.2). □

Remark 6.2. The dependence on r and b in Lemma 6.1 is sharp for scalar supersolutions, even when $V = 0$. Let $p(s) = \log(3/s)/D$, the radial harmonic function with $p(b) = 1$, $p(3) = 0$, and $g(s) = s^2(1 + \log(3/s))/(4D)$. A direct computation gives $g'' + s^{-1}g' = p$. The radial harmonic interpolant $L(s) = p(s)g(b) + (1 - p(s))g(3)$ satisfies $\Delta_\tau(L - g) = -p \leq 0$ and $L - g = 0$ at $s = b, 3$, so $L \geq g$. Thus $E(T, s) = (1 - \cos T)p(s) + L(s) - g(s)$ is bounded and nonnegative, with $\Delta_{T,\tau}E = (\cos T - 1)p \leq 0$, and its trace at $s = b$ is $1 - \cos T$. Since g is increasing,

$$E(0, r) \leq (1 - p(r))g(3) = \frac{9}{4} \frac{\log(r/b)}{D^2}.$$

A fixed positive r would cost one logarithm only; the data control $r = 2b$, where a sharper estimate would need more information about the wave.

Proposition 6.3 ([OAI26])**.** *On $\mathbb{T}^n = \mathbb{R}^n/(100\mathbb{Z})^n$ let f have Fourier support in $c_0/h \leq |\xi| \leq C_0/h$, $0 < h \leq 1$, and in one cap of the aperture γ of Proposition 5.1. Then*

$$\Big(\int_1^2 \|e^{-iT|D|} f\|_{L^{p_c}(\mathbb{T}^n)}^{p_c} \, dT \Big)^{1/p_c} \leq C \, [\log(2/h)]^{2/p_c} \, \|f\|_{L^{p_c}(\mathbb{T}^n)}, \tag{6.4}$$

with C independent of h and of the direction of the cap.

Proof. Both sides are homogeneous, so for $f \neq 0$ we may normalise $\|f\|_{L^2(\mathbb{T}^n)} = 1$. Fix the constant K of Proposition 5.1, put $a = Kh$ and $b = 2a$, and first take h so small that $b < 1/4$. Apply Lemma 6.1 to the energy $E = \mathcal{F} \geq 0$ of that proposition, with $V(s) = Cs^{-2}e^{-cs/h}$; the energy is defined for $a < |\tau| < 4$, and (5.3) holds on $b \leq |\tau| \leq 3$. The first bound in (6.1) follows from the substitution $s = bt$,

$$\int_b^3 s \log(s/b) V(s) \, ds \leq C \int_1^\infty \frac{\log t}{t} \, e^{-2cKt} \, dt,$$

and $V \leq 4C$ on $1/2 \leq s \leq 3/2$. The bounds (5.1) and (5.2) are uniform on their circles, so the circular mean satisfies

$$H(0, 2b) \leq Ca^2 \|f\|_{p_c}^{p_c}, \qquad H(T, b) \geq ca^2 \|e^{-iT|D|} f\|_{p_c}^{p_c}.$$

Take $r = 2b$ in (6.2), insert both bounds, cancel a^2 and take p_c-th roots; since $D = \log(3/(2Kh)) \lesssim \log(2/h)$, this gives (6.4). For the remaining h, bounded below, the frequencies lie in a fixed finite set, and (6.4) holds because $\|e^{-iT|D|} f\|_\infty \leq \sum |\hat{f}| \leq C\|f\|_1 \leq C\|f\|_{p_c}$ on such functions. □

7. Euclidean space, local smoothing and Bochner–Riesz means

The aperture in Proposition 6.3 is fixed, so a finite partition of the sphere gives the critical estimate on a whole annulus with the same power of the logarithm. We remove the spatial period, interpolate, sum the dyadic annuli, and then deduce the Bochner–Riesz conjecture. Throughout, $I = [1, 2]$.

7.1. **Removing the period.**

Proof of Theorem 2.2. Write $h = N^{-1}$ and $P_h = \varphi(hD)$, with φ supported in a fixed compact annulus $c_0 \le |\eta| \le C_0$. Choose a smooth finite partition $\varphi = \sum_\nu \varphi_\nu$ subordinate to caps of aperture γ about finitely many directions. On the torus $\mathbb{T}^n = \mathbb{R}^n/(100\mathbb{Z})^n$ the multiplier $\varphi_\nu(hD)$ has as kernel the periodisation of the Euclidean kernel $h^{-n}\check{\varphi}_\nu(\cdot/h)$, and periodisation does not increase the L^1 norm, so $\|\varphi_\nu(hD)f\|_{L^{p_c}(\mathbb{T}^n)} \le C\|f\|_{L^{p_c}(\mathbb{T}^n)}$. Applying Proposition 6.3 to each $\varphi_\nu(hD)f$, in the frame of its cap, and summing over ν gives

$$\|U(T)P_hf\|_{L^{p_c}(I\times\mathbb{T}^n)} \lesssim [\log(2/h)]^{2/p_c}\|f\|_{L^{p_c}(\mathbb{T}^n)}, \qquad 0 < h \le 1/2. \tag{7.1}$$

The Euclidean kernel of $U(T)P_h$ is

$$K_h(T,X) = (2\pi h)^{-n}\int e^{i(X\cdot\eta - T|\eta|)/h}\varphi(\eta)\,d\eta.$$

For $T \in I$ and $|X| \ge 3$ the gradient of the phase before division by h, $X - T\eta/|\eta|$, has length at least $|X| - 2 \ge |X|/3$. Integrating by parts along this gradient gains a factor $h/|X|$ each time, since the amplitude is supported away from $\eta = 0$ and all phase derivatives of order at least two are bounded there. With $M + n$ integrations,

$$|K_h(T,X)| \le C_M h^M(1+|X|)^{-M} \qquad (T \in I,\ |X| \ge 3). \tag{7.2}$$

Tile $\mathbb{R}^n$ by unit cubes Q and let $10Q$ be the concentric cube of side ten. If $X \in Q$ and $Y \notin 10Q$ then $|X - Y| \ge 9/2$, so on Q the contribution of $f1_{(10Q)^c}$ is bounded pointwise by $C_M h^M((1+|\cdot|)^{-M} * |f|)(X)$. This single function bounds the far contributions on every cube, and for $M > n$ Young's inequality bounds its $L^{p_c}(I\times\mathbb{R}^n)$ norm by $O(h^M)\|f\|_{p_c}$.

Translate the centre of Q to the origin and periodise $f_Q = f1_{10Q}$ on the torus of side 100; its torus norm is $\|f\|_{L^{p_c}(10Q)}$. The periodic wave kernel is the periodisation of K_h. On Q its nonzero translates see only arguments $X - Y + 100k$ with $Y \in 10Q$ and $k \in \mathbb{Z}^n \setminus \{0\}$; since $|X_j - Y_j| \le 11/2$, each coordinate satisfies $|X_j - Y_j + 100k_j| \ge \frac{189}{2}|k_j|$, so these arguments have length at least $\frac{189}{2}|k|$. Summing (7.2) over k bounds the difference between the periodic and Euclidean outputs on Q by $C_M h^M\|f_Q\|_1 \le C_M h^M\|f\|_{L^{p_c}(10Q)}$. Hence (7.1) gives

$$\|U(T)P_hf_Q\|_{L^{p_c}(I\times Q)} \lesssim [\log(2/h)]^{2/p_c}\|f\|_{L^{p_c}(10Q)}.$$

Take p_c-th powers and sum over Q. The cubes $10Q$ have bounded overlap, so together with the far contributions this proves Theorem 2.2, since $\log(2/h) \asymp \log(2+N)$ for $N \ge 2$. □

7.2. **Interpolation and summation.** By Plancherel the norm of $U(T)P_h$ from $L^2(\mathbb{R}^n)$ to $L^2(I\times\mathbb{R}^n)$ is at most $\|\varphi\|_\infty$, since $|I| = 1$. From L^∞ to L^∞ its norm is $O(h^{-d/2})$. To see this, partition the annulus into $O(h^{-d/2})$ caps of aperture $c\sqrt{h}$ with bounded overlap. In coordinates $\eta = (\sqrt{h}\,z, \rho)$, $z \in \mathbb{R}^d$, about the axis of a cap, its cut-off is supported in a fixed compact set with ρ bounded below and has uniformly bounded derivatives; such cut-offs are obtained by dividing rescaled angular bumps by their sum, and the overlapping centres have bounded offsets in these coordinates. The linear phase $T\rho/h$ corresponds to a translation in the axis direction, and what remains of the phase is

$$\frac{T}{h}\Big(\sqrt{\rho^2 + h|z|^2} - \rho\Big) = \frac{T|z|^2}{\sqrt{\rho^2 + h|z|^2} + \rho},$$

which has uniformly bounded derivatives on the support. In these coordinates the kernel of a cap equals $(2\pi)^{-n}h^{-n+d/2}\check{b}(X'/\sqrt{h}, X_n/h)$, up to that translation, for a smooth b with bounded support and derivatives, so its L^1 norm is $(2\pi)^{-n}h^{-n+d/2+d/2+1}\|\check{b}\|_1 = O(1)$. Summing over the caps gives the bound.

Set $\beta(p) = \min\{d(1-2/p), 2/p\}$. Riesz–Thorin interpolation [Gra14, Ch. 1] of Theorem 2.2 with these two bounds gives

$$\|U(T)P_h f\|_{L^p(I\times\mathbb{R}^n)} \lesssim_p h^{-s_+(p)}[\log(2/h)]^{\beta(p)}\|f\|_{L^p(\mathbb{R}^n)}, \qquad 2 \le p \le \infty. \tag{7.3}$$

Indeed, for $2 \le p \le p_c$ write $\frac12 - \frac1p = \theta(\frac12 - \frac{1}{p_c}) = \frac{\theta}{2n}$; the power of the logarithm is $2\theta/p_c = n(1-\frac2p)\frac{d}{n} = d(1-\frac2p)$. For $p_c \le p \le \infty$ write $\frac1p = \frac{1-\theta}{p_c}$; the power of the logarithm is $2(1-\theta)/p_c = 2/p$ and that of h^{-1} is $\frac{d}{2}(1-\frac{p_c}{p}) = \frac{d}{2} - \frac{n}{p}$. The two expressions for β agree at $p = p_c$, and the minimum selects the correct one on each side.

Proof of Theorem 2.1. Let φ define a smooth dyadic partition $\chi_0(\xi) + \sum_{j\ge1}\varphi(2^{-j}\xi) = 1$, and set $g = J^{s_+(p)+\varepsilon}f$. With $h = 2^{-j}$, (7.3) bounds the jth piece by $C2^{-j\varepsilon}(1+j)^{\beta(p)}\|g\|_p$: a smooth enlargement of $\varphi(2^{-j}\cdot)$ times $2^{j(s_+(p)+\varepsilon)}J^{-s_+(p)-\varepsilon}$ is a multiplier whose kernel is uniformly integrable. These bounds are summable in j. On the low frequencies the symbol $\chi_0(\xi)e^{-iT|\xi|}J(\xi)^{-s_+(p)-\varepsilon}$ satisfies the Mihlin conditions uniformly in $T \in I$, so it is bounded on L^p for $1 < p < \infty$ [Gra14, Ch. 6]. At $p = 2$ Plancherel gives the estimate without loss. □

A finer summation gives the estimate at the exact exponent $s_+(p)$ with logarithmic factors [OAI26, Corollary 7.1]; it is not needed here.

7.3. **Bochner–Riesz means.** The Bochner–Riesz mean B_r^α has multiplier $(1-|\xi|^2/r^2)_+^\alpha$; write $B_*^\alpha f = \sup_{r>0}|B_r^\alpha f|$. A smooth cut-off to a shell of relative width δ is an average of waves over times of order δ^{-1}, which is how local smoothing controls these means. Sogge observed the implication for a single radius [Sog91, (4.5)–(4.6)], and Beltran, Hickman and Sogge give the fixed-radius and maximal reductions [BHS21, Propositions 39 and 40].

Theorem 7.1 (Bochner–Riesz means [OAI26])**.** *Let $n \ge 2$, $1 \le p \le \infty$, and $\delta(p) = \max\{0, n|1/p - 1/2| - 1/2\}$. If $\alpha > \delta(p)$, then*

$$\sup_{r>0}\|B_r^\alpha f\|_{L^p(\mathbb{R}^n)} \le C_{n,p,\alpha}\|f\|_{L^p(\mathbb{R}^n)}, \tag{7.4}$$

and $B_r^\alpha f \to f$ in L^p as $r \to \infty$ when $p < \infty$. If $2 \le p < \infty$ and $\alpha > s_+(p)$, then $\|B_*^\alpha f\|_p \le C_{n,p,\alpha}\|f\|_p$.

For $1 \le p < 2n/(n+1)$ we have $\delta(p) = \lambda(p)$ of (1.1). We first rescale Theorem 2.1 to a unit frequency annulus and a long time interval: for $2 \le p < \infty$, $\sigma > s_+(p)$ and $\hat g$ supported in a fixed annulus $c \le |\xi| \le C$,

$$\|U(t)g\|_{L^p(\mathbb{R}^n\times\{R\le|t|\le2R\})} \lesssim R^{1/p+\sigma}\|g\|_p, \qquad R \ge 1. \tag{7.5}$$

Indeed $g(R\,\cdot)$ has frequencies of order R and $U(t)[g(R\,\cdot)](x) = (U(Rt)g)(Rx)$. Changing variables in space and time contributes $R^{(n+1)/p}$, Theorem 2.1 with the Sobolev multiplier on that annulus contributes $O(R^\sigma)$, and the input norm contributes $R^{-n/p}$. Complex conjugation gives the negative times, since $\overline{U(t)g} = U(-t)\bar g$. For $|t| \le 1$ the unit annular wave multiplier has a uniformly integrable kernel.

Lemma 7.2 (Maximal estimate for a thin shell [OAI26])**.** *For $0 < \delta \le 1/8$ and $\psi \in C_c^\infty((1/2,2))$ let $A_{\delta,r}f = \psi((1-|D|/r)/\delta)f$. For $2 \le p < \infty$ and $\sigma > s_+(p)$,*

$$\Big\|\sup_{r>0}|A_{\delta,r}f|\Big\|_p \lesssim \delta^{-\sigma}\|f\|_p,$$

uniformly for ψ in bounded subsets of $C_c^\infty((1/2,2))$.

Proof. For $1 \le r \le 2$ the shell lies in $1/2 < |\xi| < 2$. Choose a smooth annular symbol χ equal to one there, let $Q_k = \chi(2^{-k}D)$ and $g = Q_0 f$. Writing ψ by Fourier inversion and substituting $\tau = \delta r t$,

$$A_{\delta,r} f(x) = \frac{\delta r}{2\pi} \int_{\mathbb{R}} e^{itr}\, \hat{\psi}(\delta r t)\, U(t) g(x)\, dt.$$

Taking the supremum over $r \in [1,2]$ bounds this by $C_M \delta \int (1+\delta|t|)^{-M} |U(t)g(x)|\, dt$. On a dyadic interval $R \le |t| \le 2R$, Hölder's inequality in time and (7.5) bound the L^p_x norm of this contribution by $C_M \delta (1+\delta R)^{-M} R^{1-1/p} R^{1/p+\sigma} \|g\|_p$. For $M > 1+\sigma$ the sum over dyadic $R \ge 1$ is $O(\delta^{-\sigma}\|g\|_p)$, as one sees by splitting at $R = \delta^{-1}$; the interval $|t| \le 1$ contributes $O(\delta \|g\|_p)$.

For $r = 2^k s$ with $1 \le s \le 2$ set $g_k(y) = (Q_k f)(2^{-k} y)$. Then $A_{\delta,r} f(2^{-k}y) = A_{\delta,s} g_k(y)$, and the factor $2^{-kn/p}$ in the output norm cancels $2^{kn/p}$ in $\|g_k\|_p$. So the L^p norm of the supremum over this octave is at most $C\delta^{-\sigma}\|Q_k f\|_p$. The supremum over all octaves is at most their pointwise ℓ^p norm, and since $p \ge 2$, Littlewood–Paley theory gives $(\sum_k \|Q_k f\|_p^p)^{1/p} \le \|(\sum_k |Q_k f|^2)^{1/2}\|_p \lesssim \|f\|_p$. □

Proof of Theorem 7.1. Take $2 \le p < \infty$ and $s_+(p) < \sigma < \alpha$. Choose $\eta \in C_c^\infty((1/2,2))$ with $\sum_{j\in\mathbb{Z}} \eta(2^j v) = 1$ for $v > 0$, and apply this partition to $v = 1-\rho$ on $0 \le \rho < 1$. For $\delta = 2^{-j} \le 1/8$ the corresponding term is

$$\eta\big(2^j(1-\rho)\big)(1-\rho^2)^\alpha = \delta^\alpha \psi_\delta\big((1-\rho)/\delta\big), \qquad \psi_\delta(u) = \eta(u) u^\alpha (2-\delta u)^\alpha,$$

because $1-\rho^2 = (1-\rho)(1+\rho)$ and $1+\rho = 2-\delta u$. These profiles have uniformly bounded smooth seminorms. The remaining terms, $0 \le j \le 2$, add up to a smooth radial multiplier of compact support which vanishes near $\rho = 1$ and equals $(1-\rho^2)^\alpha$ near $\rho = 0$; the supremum of its dilations is bounded by a constant times the Hardy–Littlewood maximal function, since its kernel has a radially decreasing integrable majorant. Lemma 7.2 bounds the other terms of B^α_* by $C\sum_{j\ge 3} 2^{-j(\alpha-\sigma)}\|f\|_p$. This proves the maximal estimate.

For $2 \le p < \infty$, $\delta(p) = s_+(p)$, so the maximal estimate implies (7.4). Each B^α_r is self-adjoint and $\delta(p) = \delta(p')$, so duality gives (7.4) for $1 < p < 2$. At $p = 1$ and $p = \infty$ the hypothesis is $\alpha > (n-1)/2$; the unit-radius kernel is a constant multiple of $|x|^{-n/2-\alpha} J_{n/2+\alpha}(|x|)$ [Ste93, Ch. IX], bounded at the origin and $O(|x|^{-(n+1)/2-\alpha})$ at infinity, hence integrable, and its dilates $r^n K(rx)$ have a common L^1 norm. Finally, for Schwartz functions with compactly supported Fourier transform the multipliers converge to one in all derivatives on the support as $r \to \infty$, so the means converge in the Schwartz topology; density and (7.4) give convergence in L^p for $p < \infty$. □

Part II. Removal of the ε-loss

8. Removal of the ε-loss

Theorem 7.1 bounds the Bochner–Riesz means of every order $\lambda > \lambda(p)$, with constants that grow as $\lambda \downarrow \lambda(p)$. To reach the critical index we need a multi-scale form of these bounds without the loss, the inequalities $\mathrm{VBR}(p,r)$ below. That they follow from the non-endpoint bounds is Tao's ε-removal [Tao98]; the off-diagonal form, with $r > p$, is due to Beltran, Roos and Seeger [BRS23]. This section reproduces their argument for the round sphere $\rho(\xi) = |\xi|$, which is the only case needed here. We follow [BRS23] step by step and write out the computations it leaves to the reader. The letter ℓ indexes the dyadic layers that [BRS23] index by n, since n is the dimension here.

8.1. **Statement.** A dyadic lattice $\mathcal{D}$ is a family of half-open cubes as in [LN19, §2]: the subfamily $\mathcal{D}_j$ of cubes of side 2^j partitions $\mathbb{R}^n$ for every $j \in \mathbb{Z}$, and the cubes of $\mathcal{D}_{j-1}$ inside a cube of $\mathcal{D}_j$ are its 2^n halves. Then $\mathcal{D}_0 = \{a + z + [0,1)^n : z \in \mathbb{Z}^n\}$ for some $a \in \mathbb{R}^n$, and for $j \geq 0$ every cube of $\mathcal{D}_j$ is a union of cubes of $\mathcal{D}_0$. For $M \geq 1$ let $\mathcal{Y}_M$ be the set of C^M functions χ on $\mathbb{R}$ with support in a compact subset of $(\frac{1}{2}, 2)$ and $\sum_{\nu=0}^{M} \|\chi^{(\nu)}\|_\infty \leq 1$. Throughout this section we fix an integer $N \geq 3n+1$, put $M_0 = N + n + 1$, and write $\mathcal{Y} = \mathcal{Y}_{M_0}$.

Definition 8.1. Let $1 \leq p \leq r < \infty$. The statement $\mathrm{VBR}(p, r)$ is: there are M and C such that for every dyadic lattice $\mathcal{D}$, all $\chi_j \in \mathcal{Y}_M$ $(j \geq 1)$ and all families $\{f_Q\}_{Q \in \mathcal{D}}$ of functions $f_Q \in L^p(\mathbb{R}^n)$ with $\operatorname{supp} f_Q \subset Q$,

$$\Big\| \sum_{j \geq 1} 2^{j\frac{n+1}{2}} \chi_j(2^j(1 - |D|)) \sum_{Q \in \mathcal{D}_j} f_Q \Big\|_{L^r(\mathbb{R}^n)} \leq C \Big(\sum_{Q \in \mathcal{D}} |Q| \, \|f_Q\|_{L^p(\mathbb{R}^n)}^r \Big)^{1/r}. \tag{8.1}$$

This is [BRS25, Definition 2.2] and [BRS23, Definition 1.1] for the unit ball. It suffices to prove (8.1) for families with finitely many nonzero members: for a general family with finite right-hand side, the partial sums of the left-hand side over increasing finite subfamilies then form a Cauchy sequence in L^r, since the difference of two partial sums is itself such a sum, and (8.1) passes to the limit. Applied to a single cube, (8.1) gives $\|\chi(2^j(1 - |D|))\|_{L^p \to L^r} \lesssim 2^{j\lambda(r)}$, with $\lambda(r) = n(\frac{1}{r} - \frac{1}{2}) - \frac{1}{2}$ as in (1.1); summing a decomposition of $(1 - |\xi|)_+^\lambda$ into such bumps then bounds the Riesz means of every order $\lambda > \lambda(r)$, so $\mathrm{VBR}(p, r)$ is a multi-scale strengthening of these bounds.

Write $p_{\mathrm{ST}} = \frac{2(n+1)}{n+3}$ for the Stein–Tomas exponent. For $p_{\mathrm{ST}} < p_\circ < \frac{2n}{n+1}$ and $p_{\mathrm{ST}} \leq p \leq p_\circ$ define $r_*(p) = r_*(p, p_\circ)$ by

$$\frac{1}{r_*(p)} = \frac{\frac{1}{p_\circ}\big(\frac{1}{p_{\mathrm{ST}}} - \frac{1}{p}\big) + \frac{1}{2}\big(\frac{1}{p} - \frac{1}{p_\circ}\big)}{\frac{1}{p_{\mathrm{ST}}} - \frac{1}{p_\circ}}. \tag{8.2}$$

Thus $1/r_*$ is the affine function of $1/p$ equal to $\frac{1}{2}$ at $p = p_{\mathrm{ST}}$ and to $\frac{1}{p_\circ}$ at $p = p_\circ$: the pair $(\frac{1}{p}, \frac{1}{r_*(p)})$ runs along the segment from $(\frac{1}{p_{\mathrm{ST}}}, \frac{1}{2})$ to $(\frac{1}{p_\circ}, \frac{1}{p_\circ})$. Since both endpoints satisfy $\frac{1}{r} \leq \frac{1}{p}$ and $\frac{1}{r} \geq \frac{1}{2}$,

$$p \leq r_*(p) \leq 2 \qquad (p_{\mathrm{ST}} \leq p \leq p_\circ). \tag{8.3}$$

As $p_\circ \to \frac{2n}{n+1}$ the denominator of (8.2) tends to $\frac{n+3}{2(n+1)} - \frac{n+1}{2n} = \frac{n-1}{2n(n+1)}$ and the numerator to $\frac{n+1}{2n}\big(\frac{n+3}{2(n+1)} - \frac{1}{p}\big) + \frac{1}{2}\big(\frac{1}{p} - \frac{n+1}{2n}\big) = \frac{1}{2n}\big(1 - \frac{1}{p}\big)$, so

$$\lim_{p_\circ \to \frac{2n}{n+1}} \frac{1}{r_*(p, p_\circ)} = \frac{n+1}{n-1}\Big(1 - \frac{1}{p}\Big), \qquad \lim_{p_\circ \to \frac{2n}{n+1}} r_*(p, p_\circ) = \frac{n-1}{n+1}\, p'. \tag{8.4}$$

This is the line $r = \frac{n-1}{n+1}p'$ of the Knapp examples, which bounds the range of $\mathrm{VBR}(p, r)$ [BRS23, §1].

The multiplier $(1 - |\xi|)_+^\lambda$ is not smooth at the origin, which plays no role in the problem. We therefore work with a localised form. Fix $\tilde{w} \in C_c^\infty((\frac{1}{8}, 2))$ with $\tilde{w} = 1$ on $[\frac{1}{4}, \frac{3}{2}]$ and set

$$R^\lambda = \tilde{w}(|D|)\,(1 - |D|)_+^\lambda \qquad (\lambda > -1). \tag{8.5}$$

Theorem 8.2 ([BRS23, Theorem 1.2])**.** *Let $n \geq 2$ and $p_{\mathrm{ST}} < p_\circ < \frac{2n}{n+1}$, and suppose that R^λ is bounded on $L^{p_\circ}(\mathbb{R}^n)$ for every $\lambda > \lambda(p_\circ)$. Then $\mathrm{VBR}(p, r)$ holds for $p_{\mathrm{ST}} \leq p < p_\circ$ and $p \leq r < r_*(p, p_\circ)$, with $M = M_0$.*

In [BRS23] the hypothesis is an $L^{p_\circ} \to L^{r_\circ}$ bound with $r_\circ \geq p_\circ$; we need only $r_\circ = p_\circ$, since the limit (8.4) already reaches the Knapp line.

Lemma 8.3. (i) *If $1 \le p \le \infty$ and $(1-|D|^2)_+^\lambda$ is bounded on L^p, then so is R^λ. In particular, R^λ is bounded on L^p for $1 \le p < \frac{2n}{n+1}$ and $\lambda > \lambda(p)$.*

(ii) *For $\lambda > -\frac12$, R^λ maps $L^{p_{\mathrm{ST}}}(\mathbb{R}^n)$ to $L^2(\mathbb{R}^n)$.*

Proof. (i) On the support of $\tilde w(|\xi|)$ we have $(1-|\xi|)_+^\lambda = (1-|\xi|^2)_+^\lambda (1+|\xi|)^{-\lambda}$, so $R^\lambda = a(D)\,(1-|D|^2)_+^\lambda$ with $a(\xi) = \tilde w(|\xi|)(1+|\xi|)^{-\lambda}$. The function a is smooth, since $\tilde w(|\xi|)$ vanishes near the origin, and has compact support; so its kernel is a Schwartz function and $a(D)$ is bounded on every L^p. The second statement is Theorem 7.1 with $r=1$, since $\delta(p) = \lambda(p)$ for $p < \frac{2n}{n+1}$.

(ii) By Plancherel's theorem and polar coordinates,

$$\|R^\lambda f\|_2^2 = (2\pi)^{-n} \int_0^1 \tilde w(t)^2 (1-t)^{2\lambda} t^{n-1} \int_{S^{n-1}} |\hat f(t\omega)|^2 \, d\sigma(\omega)\, dt.$$

For $\frac18 \le t \le 1$, $\hat f(t\,\cdot)$ is the Fourier transform of $t^{-n} f(\cdot/t)$, whose $L^{p_{\mathrm{ST}}}$ norm is $t^{-n/p'_{\mathrm{ST}}}\|f\|_{p_{\mathrm{ST}}} \le 8^n \|f\|_{p_{\mathrm{ST}}}$. The Stein–Tomas restriction theorem [Ste93, Ch. IX, §2.1] bounds the inner integral by $C\|f\|_{p_{\mathrm{ST}}}^2$, and $\int_0^1 (1-t)^{2\lambda}\,dt < \infty$ for $\lambda > -\frac12$. □

Corollary 8.4. *Let $n \ge 2$. For every $p_\circ \in (p_{\mathrm{ST}}, \frac{2n}{n+1})$, $\mathrm{VBR}(p,r)$ holds for $p_{\mathrm{ST}} \le p < p_\circ$ and $p \le r < r_*(p, p_\circ)$. Consequently $\mathrm{VBR}(p,r)$ holds whenever $p_{\mathrm{ST}} \le p < \frac{2n}{n+1}$ and $p \le r < \frac{n-1}{n+1} p'$.*

Proof. By Lemma 8.3(i) the hypothesis of Theorem 8.2 holds for every $p_\circ < \frac{2n}{n+1}$, which gives the first statement. For the second, given $p < \frac{2n}{n+1}$ and $r < \frac{n-1}{n+1}p'$, the limit (8.4) provides $p_\circ \in (p, \frac{2n}{n+1})$ with $r < r_*(p, p_\circ)$. □

The rest of this section proves Theorem 8.2; its hypothesis is assumed from now on, and implicit constants may depend on n, $p_\circ$, N and the exponents at hand, but not on the lattice, the cubes, the functions or the bumps χ_j.

8.2. **Single-scale estimates.** For $j \ge 1$ and $\chi \in \mathcal{Y}$ let

$$m_j(\xi) = h_j(|\xi|), \qquad h_j(\varrho) = \chi\big(2^j(1-\varrho)\big), \tag{8.6}$$

so that m_j is supported in $1 - 2^{1-j} < |\xi| < 1 - 2^{-j-1}$. The two coarsest scales need no analysis.

Lemma 8.5. *Let $j \in \{1,2\}$, $1 \le p \le r < \infty$, and $\chi \in \mathcal{Y}_M$ with $M \ge n+1$. Then the kernel K_j of $m_j(D)$ satisfies $|K_j(x)| \le C(1+|x|)^{-n-1}$, and*

$$\Big\| m_j(D) \sum_{Q \in \mathcal{D}_j} f_Q \Big\|_r \le C \Big(\sum_{Q \in \mathcal{D}_j} \|f_Q\|_p^r \Big)^{1/r} \qquad (\operatorname{supp} f_Q \subset Q).$$

Proof. Write $g(t) = \chi(2^j(1-t))$. For $j=2$, g vanishes on $[0, \frac12]$, so m_2 is smooth and $|\partial^\alpha m_2| \le C$ for $|\alpha| \le M$. For $j=1$, extend χ by zero; it is then C^M on $\mathbb{R}$ and vanishes on $[2,\infty)$, so $g^{(i)}(0) = 0$ for $i \le M$ and Taylor's formula gives $|g^{(i)}(t)| \le 2^M t^{M-i}$ for $0 \le t \le 1$ and $i \le M$. For $\xi \ne 0$ and $1 \le |\alpha| \le M$ the chain rule gives

$$\partial^\alpha [g(|\xi|)] = \sum_{i=1}^{|\alpha|} g^{(i)}(|\xi|)\, P_{i,\alpha}(\xi), \tag{8.7}$$

where $P_{i,\alpha}$ is a sum of products of i derivatives of $|\xi|$ of total order $|\alpha|$; it is smooth on $\xi \ne 0$, homogeneous of degree $i - |\alpha|$, and independent of g. Hence $|\partial^\alpha m_1(\xi)| \le C|\xi|^{M-|\alpha|} \le C$ on $0 < |\xi| < 1$. So m_1 is C^{M-1} on $\mathbb{R}^n$, with derivatives of order $M-1$ that are Lipschitz, and its distributional derivatives of order M are bounded. In both cases $|x^\alpha K_j(x)| \le (2\pi)^{-n}\|\partial^\alpha m_j\|_{L^1} \le C$ for $|\alpha| \le M$, which gives the kernel bound since $M \ge n+1$.

For the second assertion,

$$|K_j * f_Q(x)| \le C(1+\operatorname{dist}(x,Q))^{-n-1}\|f_Q\|_1 \le C'(1+\operatorname{dist}(x,Q))^{-n-1}\|f_Q\|_p,$$

because $|Q| \le 4^n$. For x in a cube $Q' \in \mathcal{D}_j$ this gives $\big|\sum_Q K_j * f_Q(x)\big| \le C\sum_Q (1+|c_Q - c_{Q'}|)^{-n-1}\|f_Q\|_p$, with c_Q the centre of Q. The right-hand side is the convolution, on the lattice of centres, of $(\|f_Q\|_p)_Q$ with a summable sequence, so its ℓ^r norm over Q' is at most $C(\sum_Q \|f_Q\|_p^r)^{1/r}$. Integrating over each Q', of measure at most 4^n, gives the claim. □

From now on $j \ge 3$, so that m_j is supported in $\frac34 < |\xi| < 1$. By (8.7), for $1 \le |\alpha| \le M_0$,

$$\partial^\alpha m_j(\xi) = \sum_{k=1}^{|\alpha|} (-2^j)^k\, \chi^{(k)}(2^j(1-|\xi|))\, P_{k,\alpha}(\xi), \qquad |\partial^\alpha m_j(\xi)| \le C\, 2^{j|\alpha|}, \tag{8.8}$$

since $|P_{k,\alpha}| \le C$ on $\frac34 \le |\xi| \le 1$.

Subordination. For $c > 0$ and a continuous function g on $\mathbb{R}$ whose support is bounded above, with enough decay at $-\infty$, let

$$I^c g(\varrho) = \frac{1}{\Gamma(c)} \int_\varrho^\infty (s-\varrho)^{c-1} g(s)\, ds.$$

For compactly supported g, Fubini's theorem and the Beta integral give $I^a I^c g = I^{a+c} g$; and $I^1(-g') = g$ for $g \in C_c^1$, so $I^k((-\frac{d}{d\varrho})^k g) = g$ for $g \in C_c^k$.

Lemma 8.6 ([BRS23, (2.3) and Lemma 2.1])**.** *Let $\lambda > -1$, let $k = \lfloor \lambda \rfloor + 2$, so that $b = k - \lambda - 1 \in (0,1]$, and let $h \in C_c^k(\mathbb{R})$. Put $H = I^b((-\frac{d}{d\varrho})^k h)$. Then*

$$h(\varrho) = \frac{1}{\Gamma(\lambda+1)} \int_\varrho^\infty (s-\varrho)^\lambda H(s)\, ds \qquad (\varrho \in \mathbb{R}). \tag{8.9}$$

If $h = h_j$ with $j \ge 3$ and $\|\chi\|_{C^k} \le 1$, then $H(s) = 0$ for $s \ge 1 - 2^{-j-1}$,

$$|H(s)| \le C\, \frac{2^{j(\lambda+1)}}{\big(1+2^j(1-s)\big)^{\lambda+2}} \quad (s<1), \qquad \int_{5/8}^1 |H(s)|\, ds \le C\, 2^{j\lambda}. \tag{8.10}$$

Proof. The function $(-\frac{d}{d\varrho})^k h$ has compact support, so $I^{\lambda+1} H = I^{\lambda+1} I^b (-\frac{d}{d\varrho})^k h = I^k (-\frac{d}{d\varrho})^k h = h$, which is (8.9). Now let $h = h_j$, supported in $[1-2^{1-j}, 1-2^{-j-1}]$, with $|h_j^{(k)}| \le 2^{jk}$. If $b = 1$ then $H = (-1)^{k-1} h_j^{(k-1)}$, which is supported in the same interval and bounded by $2^{j(k-1)} = 2^{j(\lambda+1)}$; this gives (8.10). Let $0 < b < 1$. Clearly $H(s) = 0$ for $s \ge 1 - 2^{-j-1}$. If $1 - 2^{2-j} \le s < 1$, then

$$|H(s)| \le \frac{2^{jk}}{\Gamma(b)} \int_s^1 (u-s)^{b-1}\, du \le \frac{2^{jk}(1-s)^b}{\Gamma(b+1)} \le C\, 2^{j(k-b)} = C\, 2^{j(\lambda+1)}.$$

If $s < 1 - 2^{2-j}$, then $u - s \ge \frac12(1-s)$ on the support of h_j, and h_j vanishes near $u = s$, so k integrations by parts give

$$|H(s)| = \frac{1}{\Gamma(b)} \Big| \int (b-1)\cdots(b-k)\, (u-s)^{b-1-k} h_j(u)\, du \Big| \le C\, 2^{-j} \Big(\frac{1-s}{2}\Big)^{-\lambda-2} = C\, \frac{2^{j(\lambda+1)}}{(2^j(1-s))^{\lambda+2}},$$

since $b - 1 - k = -\lambda - 2$. This proves the pointwise bound. Integrating it over $[\frac58, 1]$, the part $1 - s \le 2^{2-j}$ contributes $C 2^{j(\lambda+1)} 2^{2-j}$ and the rest $C 2^{-j} \int_{2^{2-j}}^\infty t^{-\lambda-2}\, dt \le C 2^{j\lambda}$, because $\lambda + 1 > 0$. □

Spatial localisation. Fix a radial $\phi_0 \in C_c^\infty(\mathbb{R}^n)$ with $\phi_0(x) = 1$ for $|x| \le \frac{1}{2}$ and $\phi_0(x) = 0$ for $|x| \ge 1$, and set

$$\phi_{j,0}(x) = \phi_0(2^{-j}x), \qquad \phi_{j,\ell}(x) = \phi_0(2^{-j-\ell}x) - \phi_0(2^{-j-\ell+1}x) \quad (\ell \ge 1), \tag{8.11}$$

so that $\sum_{\ell \ge 0} \phi_{j,\ell} = 1$, $\phi_{j,\ell}$ is supported in $|x| < 2^{j+\ell}$, and for $\ell \ge 1$ it vanishes on $|x| \le 2^{j+\ell-2}$. Let

$$m_{j,\ell} = (2\pi)^{-n}\, m_j * \widehat{\phi_{j,\ell}}. \tag{8.12}$$

Then $m_{j,\ell}(D)$ has kernel $K_j \phi_{j,\ell}$, where K_j is the kernel of $m_j(D)$; so this kernel is supported in $|x| < 2^{j+\ell}$, $m_j = \sum_\ell m_{j,\ell}$, and $m_{j,\ell}$ is radial. Since $\phi_{j,\ell} = \phi_{0,1}(2^{1-j-\ell}\,\cdot)$ for $\ell \ge 1$,

$$|\widehat{\phi_{j,\ell}}(\eta)| \le C_K\, 2^{(j+\ell)n}(1 + 2^{j+\ell}|\eta|)^{-K}, \qquad \int |\eta|^N |\widehat{\phi_{j,\ell}}(\eta)|\, d\eta \le C\, 2^{-(j+\ell)N}, \tag{8.13}$$

for every K, and $\int \eta^\beta \widehat{\phi_{j,\ell}}(\eta)\, d\eta = (2\pi)^n (i\partial)^\beta \phi_{j,\ell}(0) = 0$ for every multi-index β when $\ell \ge 1$. Taylor's formula for $m_j(\xi - \eta)$ about ξ, to order N, therefore gives for $\ell \ge 1$

$$m_{j,\ell}(\xi) = (2\pi)^{-n} \sum_{|\alpha| = N} \frac{N}{\alpha!} \int\!\!\int_0^1 (1-s)^{N-1} (-\eta)^\alpha\, \partial^\alpha m_j(\xi - s\eta)\, ds\, \widehat{\phi_{j,\ell}}(\eta)\, d\eta. \tag{8.14}$$

Lemma 8.7 ([BRS23, (2.13)])**.** *For $j \ge 3$, $\ell \ge 0$ and every N_1,*

$$|m_{j,\ell}(\xi)| \le C_{N_1}\, 2^{-\ell N} \big(1 + 2^j \big||\xi| - 1\big|\big)^{-N_1}.$$

Proof. Let $t = \big||\xi| - 1\big|$. Let $\ell \ge 1$. By (8.8) and (8.13), (8.14) gives $|m_{j,\ell}(\xi)| \le C 2^{jN} 2^{-(j+\ell)N} = C 2^{-\ell N}$. If $t \ge 2^{2-j}$, the integrand of (8.14) vanishes unless $\big||\xi - s\eta| - 1\big| < 2^{1-j}$, which forces $|\eta| \ge s|\eta| \ge t - 2^{1-j} \ge t/2$; by (8.13) with $K = N + n + N_1$,

$$|m_{j,\ell}(\xi)| \le C 2^{jN} \int_{|\eta| \ge t/2} |\eta|^N 2^{(j+\ell)n} (1 + 2^{j+\ell}|\eta|)^{-K}\, d\eta \le C 2^{-\ell N} (1 + 2^{j+\ell} t)^{-N_1}.$$

For $\ell = 0$, $|m_{j,0}(\xi)| \le C \int_{||\xi-\eta|-1| < 2^{1-j}} 2^{jn}(1 + 2^j|\eta|)^{-K} d\eta$, which is bounded, and is at most $C(1 + 2^j t)^{-N_1}$ when $t \ge 2^{2-j}$ by the same argument. ☐

Lemma 8.8 ([BRS23, Lemma 2.2])**.** *Let $1 \le p \le r < \infty$ and $-1 < \lambda < n - 1$, and suppose that R^λ maps L^p to L^r. Let $j \ge 3$ and $\ell \ge 0$.*

(i) *If $\chi \in C^{n+1}$ is supported in $(\frac{1}{2}, 2)$ with $\|\chi\|_{C^{n+1}} \le 1$, then $\|m_j(D)\|_{L^p \to L^r} \le C 2^{j\lambda}$.*

(ii) *If $\chi \in \mathcal{Y}$, then $\|m_{j,\ell}(D)\|_{L^p \to L^r} \le C 2^{-\ell N} 2^{j\lambda}$.*

(iii) *If $\chi \in \mathcal{Y}$ and $p \le q \le r$, then $\|m_{j,\ell}(D)\|_{L^p \to L^q} \le C 2^{-\ell(N-n)} 2^{j(\lambda + n(\frac{1}{q} - \frac{1}{r}))}$ and $\|m_j(D)\|_{L^p \to L^q} \le C 2^{j(\lambda + n(\frac{1}{q} - \frac{1}{r}))}$.*

Proof. (i) Fix a radial $\theta_1 \in C_c^\infty$ supported in $\frac{5}{8} < |\xi| < \frac{5}{4}$ and equal to one on $\frac{3}{4} \le |\xi| \le 1$. By (8.9) with $\varrho = |\xi|$, and since H_j vanishes on $[1, \infty)$,

$$m_j(\xi) = \theta_1(\xi) m_j(\xi) = \frac{1}{\Gamma(\lambda + 1)} \int_0^1 s^\lambda H_j(s)\, \theta_1(\xi) \Big(1 - \frac{|\xi|}{s}\Big)_+^\lambda ds.$$

The integrand vanishes for $s < \frac{5}{8}$. For $\frac{5}{8} \le s < 1$, on the support of $\theta_1(\xi)(1 - |\xi|/s)_+$ we have $\frac{5}{8} < |\xi|/s < 1$, where $\tilde{w}(|\xi|/s) = 1$; so the multiplier there is $\theta_1(\xi)$ times the dilate by s of the symbol of R^λ, whose $L^p \to L^r$ norm is $s^{n(\frac{1}{p} - \frac{1}{r})} \|R^\lambda\|_{L^p \to L^r}$. By Minkowski's inequality and (8.10) (note $k \le n + 1$), $\|m_j(D)\|_{L^p \to L^r} \le C \int_{5/8}^1 |H_j(s)|\, ds \le C 2^{j\lambda}$.

(ii) For $\ell = 0$, (8.12) writes $m_{j,0}$ as an average of the translates $m_j(\cdot - \eta)$ against the weight $(2\pi)^{-n}|\widehat{\phi_{j,0}}|$, of bounded L^1 norm; translating a multiplier conjugates it by a modulation, which does not change its norm. For $\ell \ge 1$ use (8.14) and (8.8). Let $\theta \in C_c^\infty(\{\frac{1}{2} < |\xi| < \frac{3}{2}\})$ be

equal to one on $\frac34 \le |\xi| \le 1$. Then $\partial^\alpha m_j = \sum_k (-2^j)^k\,(\theta P_{k,\alpha})(\xi)\,\chi^{(k)}(2^j(1-|\xi|))$, where $\theta P_{k,\alpha}$ is a fixed smooth compactly supported multiplier, bounded on L^r, and $\chi^{(k)} \in C^{M_0-k}$ with $M_0 - k \ge n+1$. By (i), $\|\partial^\alpha m_j(D - s\eta)\|_{L^p\to L^r} \le C 2^{jN} 2^{j\lambda}$ for $|\alpha| = N$, and (8.13) gives $\|m_{j,\ell}(D)\|_{L^p\to L^r} \le C 2^{jN+j\lambda} 2^{-(j+\ell)N}$.

(iii) Let $\mathcal{G}$ be the grid of cubes of side $2^{j+\ell}$. For a cube $Q \in \mathcal{G}$ the function $m_{j,\ell}(D)[f1_Q]$ is supported in the concentric cube $3Q$, and these cubes overlap at most 3^n times. By Hölder's inequality on $3Q$ and (ii),

$$\begin{aligned}\|m_{j,\ell}(D)f\|_q &\le C\Big(\sum_{Q\in\mathcal{G}} \big(|3Q|^{\frac1q-\frac1r}\|m_{j,\ell}(D)[f1_Q]\|_r\big)^q\Big)^{1/q}\\ &\le C 2^{(j+\ell)n(\frac1q-\frac1r)} 2^{-\ell N} 2^{j\lambda}\Big(\sum_Q \|f1_Q\|_p^q\Big)^{1/q},\end{aligned}$$

and $(\sum_Q \|f1_Q\|_p^q)^{1/q} \le \|f\|_p$ since $q \ge p$. Summing over ℓ gives the bound for m_j, because $N > n$. □

Corollary 8.9 ([BRS23, Corollary 2.3])**.** *Let $p_{\mathrm{ST}} \le p_1 < p_\circ$ and $p_1 \le r \le r_*(p_1)$. For every $\varepsilon > 0$, $j \ge 3$, $\ell \ge 0$ and $\chi \in \mathcal{Y}$,*

$$\|m_{j,\ell}(D)\|_{L^{p_1}\to L^r} \le C_\varepsilon 2^{-\ell(N-n)} 2^{j(\lambda(r)+\varepsilon)}, \qquad \|m_j(D)\|_{L^{p_1}\to L^r} \le C_\varepsilon 2^{j(\lambda(r)+\varepsilon)},$$

and R^λ maps L^{p_1} to L^r for every $\lambda > \lambda(r)$.

Proof. By Lemma 8.8(ii), the hypothesis of Theorem 8.2 with $\lambda = \lambda(p_\circ) + \varepsilon$ (which is less than $n-1$), and Lemma 8.3(ii) with $\lambda = -\frac12 + \varepsilon$,

$$\|m_{j,\ell}(D)\|_{L^{p_\circ}\to L^{p_\circ}} \le C_\varepsilon 2^{-\ell N} 2^{j(\lambda(p_\circ)+\varepsilon)}, \qquad \|m_{j,\ell}(D)\|_{L^{p_{\mathrm{ST}}}\to L^2} \le C_\varepsilon 2^{-\ell N} 2^{j(\lambda(2)+\varepsilon)},$$

where $\lambda(2) = -\frac12$. Write $\frac{1}{p_1} = \frac{\vartheta}{p_\circ} + \frac{1-\vartheta}{p_{\mathrm{ST}}}$ with $\vartheta \in (0,1]$; then $\frac{1}{r_*(p_1)} = \frac{\vartheta}{p_\circ} + \frac{1-\vartheta}{2}$ by (8.2), and since $\lambda(r)$ is an affine function of $\frac1r$, $\vartheta\lambda(p_\circ) + (1-\vartheta)\lambda(2) = \lambda(r_*(p_1))$. The Riesz–Thorin theorem gives the first bound for $r = r_*(p_1)$, with $2^{-\ell N}$. For $p_1 \le r < r_*(p_1)$ apply the Hölder argument of Lemma 8.8(iii) to this bound, noting that $\lambda(r_*(p_1)) + n(\frac1r - \frac{1}{r_*(p_1)}) = \lambda(r)$; summing in ℓ bounds $m_j(D)$.

For the last statement choose $\chi_0 \in C_c^\infty((\frac12, 2))$ with $\sum_{j\in\mathbb{Z}} \chi_0(2^j s) = 1$ for $s > 0$. For $j \ge 3$, $(1-\varrho)^\lambda \chi_0(2^j(1-\varrho)) = 2^{-j\lambda}\tilde\chi(2^j(1-\varrho))$ with $\tilde\chi(s) = s^\lambda \chi_0(s)$, a fixed multiple of an element of $\mathcal{Y}$, and these terms live in $\frac34 < \varrho < 1$, where $\tilde w = 1$. By the first part their multiplier norms are at most $C 2^{-j(\lambda-\lambda(r)-\varepsilon)}$, which is summable for $\varepsilon < \lambda - \lambda(r)$. The remaining part of the symbol of R^λ, $\tilde w(|\xi|)(1-|\xi|)_+^\lambda\big(1 - \sum_{j\ge3}\chi_0(2^j(1-|\xi|))\big)$, is smooth and supported in $\frac18 < |\xi| < \frac78$, so its kernel is a Schwartz function and it maps L^{p_1} to L^r by Young's inequality. □

Lemma 8.10 ([BRS23, Lemma 2.5])**.** *For $j \ge 3$ and $\ell \ge 0$ let $\kappa_{j,\ell} = \mathcal{F}^{-1}[|m_{j,\ell}|^2]$. Then*

$$\sup_{x\in\mathbb{R}^n} (1+|x|)^{\frac{n-1}{2}} |\kappa_{j,\ell}(x)| \le C\, 2^{-2\ell N} 2^{-j}.$$

Proof. Since $m_{j,\ell}$ is radial, $|m_{j,\ell}(\xi)|^2 = G(|\xi|)$, and in polar coordinates

$$\kappa_{j,\ell}(x) = (2\pi)^{-n} \int_0^\infty G(t)\, t^{n-1}\, \hat\sigma(tx)\, dt, \qquad \hat\sigma(y) = \int_{S^{n-1}} e^{i\langle y,\omega\rangle}\, d\sigma(\omega).$$

The Fourier transform of the surface measure satisfies $|\hat\sigma(y)| \le C(1+|y|)^{-\frac{n-1}{2}}$ [Ste93, Ch. VIII, §3.1]; it is a constant multiple of $|y|^{-\frac{n-2}{2}} J_{\frac{n-2}{2}}(|y|)$. Hence

$$|\kappa_{j,\ell}(x)| \le C \min\Big\{\int_0^\infty |G(t)| t^{n-1}\, dt,\ |x|^{-\frac{n-1}{2}} \int_0^\infty |G(t)| t^{\frac{n-1}{2}}\, dt\Big\}.$$

By Lemma 8.7 with $N_1 = n$, $|G(t)| \le C2^{-2\ell N}(1+2^j|t-1|)^{-2n}$, and both integrals are at most $C2^{-2\ell N}2^{-j}$. □

Lemma 8.10 replaces the stationary phase argument of [BRS23], which is needed there for general convex domains; for the sphere the radial symmetry reduces it to the decay of $\hat\sigma$.

8.3. **Square functions over separated cubes.** The gain over the trivial L^2 bound comes from cubes that are far apart, as in Tao's work [Tao98]. Fix $\psi \in C_c^\infty(\mathbb{R}^n)$ with $\psi \ge 0$, $\psi \ne 0$ and support in $|\xi| \le \frac{1}{4\sqrt{n}}$, let $\check\psi = \mathcal{F}^{-1}\psi$, and put

$$\eta = \frac{|\check\psi|^2}{c_0}, \qquad c_0 = \big(\cos(\tfrac12)\,\check\psi(0)\big)^2.$$

Then $\eta \ge 0$ is a Schwartz function, $\hat\eta = c_0^{-1}(2\pi)^{-n}\psi * \psi(-\cdot)$ is supported in $|\xi| \le \frac{1}{2\sqrt{n}}$, and $\eta(x) \ge 1$ for $|x|_\infty \le 2$: for such x and $\xi \in \operatorname{supp}\psi$ we have $|\langle x,\xi\rangle| \le 2\sqrt{n}\cdot\frac{1}{4\sqrt{n}} = \frac12$, so $\operatorname{Re}\check\psi(x) = (2\pi)^{-n}\int\psi(\xi)\cos\langle x,\xi\rangle\,d\xi \ge \cos(\frac12)\check\psi(0) > 0$. For a cube B of side $R \ge 1$ and centre x_B let

$$\eta_B(x) = \eta\Big(\frac{x-x_B}{R}\Big), \qquad \widehat{\eta_B}(\xi) = R^n e^{-i\langle x_B,\xi\rangle}\hat\eta(R\xi), \tag{8.15}$$

so that $\widehat{\eta_B}$ is supported in $|\xi| \le \frac{1}{2R}$, $\eta_B \ge 1$ on the concentric cube of side $4R$, and $\|\eta_B\|_1 = R^n\|\eta\|_1$.

Definition 8.11. Let $R \ge 1$ and $S \ge 3R$. A family $\mathcal{B}$ of cubes of side R is *S-separated* if $|x_B - x_{B'}| \ge S$ for all $B \ne B'$ in $\mathcal{B}$.

With ϕ_0 as in (8.11) let

$$\eta_{B,0}(x) = \phi_0\Big(\frac{x-x_B}{R}\Big)\eta_B(x), \qquad \eta_{B,\ell}(x) = \Big[\phi_0\Big(\frac{x-x_B}{2^\ell R}\Big) - \phi_0\Big(\frac{x-x_B}{2^{\ell-1}R}\Big)\Big]\eta_B(x) \quad (\ell \ge 1). \tag{8.16}$$

Then $\eta_B = \sum_{\ell\ge0}\eta_{B,\ell}$, $\eta_{B,\ell}$ is supported in the ball $|x-x_B| \le 2^\ell R$, and since $\eta_{B,\ell}$ vanishes on $|x-x_B| \le 2^{\ell-2}R$ for $\ell \ge 1$ and η is a Schwartz function,

$$\|\eta_{B,\ell}\|_\infty \le C_K 2^{-\ell K} \qquad \text{for every } K. \tag{8.17}$$

We need one elementary bound on the measure of thin shells. Here and below $B(c,\rho)$ is the ball of centre c and radius ρ.

Lemma 8.12. *Let $K \ge n+1$ and $j \ge 0$.*

(i) *For $c \in \mathbb{R}^n$ and $\rho > 0$, $\displaystyle\int_{B(c,\rho)} (1+2^j||w|-1|)^{-K}dw \le C\,2^{-j}\min\{\rho,1\}^{n-1}$.*

(ii) *For $0 < \rho \le 1$ and $\xi \in \mathbb{R}^n$, $\displaystyle\int \rho^{-n}\Big(1+\frac{|v|}{\rho}\Big)^{-K}(1+2^j||\xi-v|-1|)^{-K}dv \le C\min\{1, 2^{-j}\rho^{-1}\}$.*

Proof. (i) In polar coordinates the integral over all of $\mathbb{R}^n$ is $|S^{n-1}|\int_0^\infty(1+2^j|s-1|)^{-K}s^{n-1}ds \le C2^{-j}$, which settles $\rho \ge 1$. Let $\rho < 1$ and $V(t) = |\{w \in B(c,\rho) : ||w|-1| \le t\}|$. If $t \ge \frac12$ then $V(t) \le C\rho^n \le 2Ct\rho^{n-1}$. If $t < \frac12$, then $V(t) = \int_{1-t}^{1+t}\sigma_s(B(c,\rho))\,ds$, where σ_s is the surface measure of the sphere of radius $s \in [\frac12,\frac32]$; a ball of radius ρ meets such a sphere in a set of diameter at most 2ρ, of measure at most $C\rho^{n-1}$, so $V(t) \le Ct\rho^{n-1}$. Splitting the integral according to $||w|-1| \le 2^{-j}$ and $2^{i-1-j} < ||w|-1| \le 2^{i-j}$, $i \ge 1$, it is at most $\sum_{i\ge0}2^{-(i-1)K}V(2^{i-j}) \le C\rho^{n-1}2^{-j}\sum_i 2^{i-(i-1)K}$.

(ii) The integral is at most $\int\rho^{-n}(1+|v|/\rho)^{-K}dv = C$. Apply (i), with centre ξ and $w = \xi - v$, on $|v| \le \rho$ and on the shells $2^{i-1}\rho < |v| \le 2^i\rho$, $i \ge 1$: the contribution of the i-th piece is at most $C\rho^{-n}2^{-(i-1)K}2^{-j}(2^i\rho)^{n-1}$, and the sum over i is $C2^{-j}\rho^{-1}$. □

Lemma 8.13 ([BRS23, Lemma 3.3])**.** *Let $j \geq 3$, $\ell_1, \ell_2 \geq 0$, $R \geq 1$, $S \geq 3R$, and let $\mathcal{B}$ be a finite S-separated family of cubes of side R. For every $N_2 > 0$,*

$$\Big\| m_{j,\ell_1}(D) \sum_{B \in \mathcal{B}} \eta_{B,\ell_2} f_B \Big\|_2 \leq C\, 2^{-\ell_1 N} 2^{-\ell_2 N_2} (2^{-j}R)^{1/2} \Big(1 + S^{-\frac{n-1}{2}} R^{n-1} \#\mathcal{B}\Big)^{1/2} \Big(\sum_{B \in \mathcal{B}} \|f_B\|_2^2 \Big)^{1/2}$$

for all $f_B \in L^2(\mathbb{R}^n)$, where C depends on N_2.

Proof. Let $m = m_{j,\ell_1}$ and $\kappa = \kappa_{j,\ell_1} = \mathcal{F}^{-1}[|m|^2]$. First assume the stronger separation

$$|x_B - x_{B'}| \geq S + 2^{\ell_2+1} R \qquad (B \neq B' \text{ in } \mathcal{B}). \tag{8.18}$$

Since η_{B,ℓ_2} is real, the square of the left-hand side is $\sum_{B,B'} \langle T_{B,B'} f_B, f_{B'} \rangle$, where $T_{B,B'} f = \eta_{B',\ell_2} \, |m|^2(D)[\eta_{B,\ell_2} f]$ has kernel $\eta_{B',\ell_2}(x) \kappa(x-y) \eta_{B,\ell_2}(y)$.

Diagonal terms. If $x, y \in \operatorname{supp} \eta_{B,\ell_2}$, then $|x-y| \leq 2^{\ell_2+1} R$, so $\phi_0((x-y)/(2^{\ell_2+2}R)) = 1$ and the kernel of $T_{B,B}$ is $\eta_{B,\ell_2}(x)\, \mathcal{F}^{-1}[a](x-y)\, \eta_{B,\ell_2}(y)$ with

$$a(\xi) = (2\pi)^{-n} \int |m(\xi - v)|^2 (2^{\ell_2+2}R)^n \hat{\phi}_0(2^{\ell_2+2} R v)\, dv.$$

Thus $\|T_{B,B}\|_{L^2 \to L^2} \leq \|\eta_{B,\ell_2}\|_\infty^2 \|a\|_\infty$. By Lemma 8.7 with $N_1 = n+1$, $|m(\xi - v)|^2 \leq C 2^{-2\ell_1 N} (1 + 2^j ||\xi - v| - 1|)^{-n-1}$, and $|\hat{\phi}_0(u)| \leq C(1+|u|)^{-n-1}$; so Lemma 8.12(ii) with $\rho = (2^{\ell_2+2}R)^{-1}$ gives $\|a\|_\infty \leq C 2^{-2\ell_1 N} \min\{1, 2^{\ell_2+2-j} R\}$. With (8.17),

$$\sum_B |\langle T_{B,B} f_B, f_B \rangle| \leq C_K 2^{-2\ell_1 N} 2^{-\ell_2(2K-1)} 2^{-j} R \sum_B \|f_B\|_2^2.$$

Off-diagonal terms. If $B \neq B'$, $x \in \operatorname{supp} \eta_{B',\ell_2}$ and $y \in \operatorname{supp} \eta_{B,\ell_2}$, then $|x-y| \geq |x_B - x_{B'}| - 2^{\ell_2+1} R \geq S$ by (8.18), so Lemma 8.10 and (8.17) bound the kernel of $T_{B,B'}$ by $C_K 2^{-2\ell_1 N} 2^{-j} S^{-\frac{n-1}{2}} 2^{-2\ell_2 K}$ times the indicator of a product of two balls of radius $2^{\ell_2} R$. By Schur's test $\|T_{B,B'}\| \leq C_K 2^{-2\ell_1 N} 2^{-j} S^{-\frac{n-1}{2}} 2^{-\ell_2(2K-n)} R^n$, and by the Cauchy–Schwarz inequality

$$\begin{aligned} \sum_{B \neq B'} |\langle T_{B,B'} f_B, f_{B'} \rangle| &\leq \max_{B \neq B'} \|T_{B,B'}\| \Big(\sum_B \|f_B\|_2 \Big)^2 \\ &\leq C_K 2^{-2\ell_1 N} 2^{-\ell_2(2K-n)} 2^{-j} R\, S^{-\frac{n-1}{2}} R^{n-1} \#\mathcal{B} \sum_B \|f_B\|_2^2. \end{aligned}$$

Adding the two bounds proves the lemma under (8.18), with $2^{-\ell_2(2K-n)}$ in place of $2^{-2\ell_2 N_2}$.

Colouring. In general, let $T' = S + 2^{\ell_2+1} R$ and partition $\mathbb{R}^n$ into the half-open cubes of a grid of side $a = S/\sqrt{n}$. Two points of one such cube are at distance less than S, so each cube contains at most one centre x_B. Colour the cubes of the grid by their integer coordinates modulo $k_0 = \lceil T'/a \rceil + 1$. Points of two different cubes of the same colour are at distance more than $(k_0 - 1)a \geq T'$, so each colour class of $\mathcal{B}$ satisfies (8.18); and since $R \leq S/3$, the number of colours is $k_0^n \leq (\sqrt{n}(1 + 2^{\ell_2}) + 2)^n \leq C 2^{\ell_2 n}$. Applying the bound already proved to each class, with $\#\mathcal{B}$ in place of the size of the class, and using the Cauchy–Schwarz inequality over the classes, we lose a factor $C 2^{\ell_2 n/2}$. Choosing K with $2K - n \geq 2N_2 + n$ proves the lemma. □

Proposition 8.14 ([BRS23, Proposition 3.2])**.** *Let $j \geq 3$ and $1 \leq R \leq 2^j$. For each $Q \in \mathcal{D}_j$ let $S_Q \geq 3R$ and let $\mathcal{B}_Q$ be a finite S_Q-separated family of cubes of side R meeting Q. Put*

$\Lambda = \sup_Q (1 + S_Q^{-\frac{n-1}{2}} R^{n-1} \#\mathcal{B}_Q)^{1/2}$. *Then*

$$\Big\| m_j(D) \sum_{Q\in\mathcal{D}_j} \sum_{B\in\mathcal{B}_Q} \eta_B f_{Q,B} \Big\|_2 \le C(2^{-j}R)^{1/2} \Lambda \Big(\sum_Q \sum_B \|f_{Q,B}\|_2^2 \Big)^{1/2}, \tag{8.19}$$

$$\Big\| m_j(D) \sum_{Q\in\mathcal{D}_j} \sum_{B\in\mathcal{B}_Q} \eta_B f_{Q,B} \Big\|_1 \le C 2^{j\frac{n-1}{2}} R^{1/2} \Lambda \sum_Q \Big(\sum_B \|f_{Q,B}\|_2^2 \Big)^{1/2} \tag{8.20}$$

for all families of functions $f_{Q,B} \in L^2(\mathbb{R}^n)$ with finitely many nonzero members.

Proof. Write the operator as $\sum_{\ell_1,\ell_2\ge 0} m_{j,\ell_1}(D) \sum_Q \sum_B \eta_{B,\ell_2} f_{Q,B}$ and fix ℓ_1, ℓ_2. Let $g_Q = m_{j,\ell_1}(D) \sum_{B\in\mathcal{B}_Q} \eta_{B,\ell_2} f_{Q,B}$. Every $B \in \mathcal{B}_Q$ meets Q and has side $R \le 2^j$, so its centre lies within $\frac{\sqrt{n}}{2} R \le \sqrt{n} 2^{j+\ell_1}$ of Q; the kernel of $m_{j,\ell_1}(D)$ is supported in $|x| < 2^{j+\ell_1}$. Hence g_Q is supported in the set of points at distance at most $U_0 = 2^{\ell_2} R + (1+\sqrt{n}) 2^{j+\ell_1}$ from Q, and $U_0 \le 3\sqrt{n} U$ with $U = \max\{2^{j+\ell_1}, 2^{\ell_2} R\}$.

The cubes of $\mathcal{D}_j$ are $Q_k = a_j + 2^j(k + [0,1)^n)$, $k \in \mathbb{Z}^n$, for some $a_j \in \mathbb{R}^n$. Colour Q_k by k modulo $k_0 = \lceil 2U_0 2^{-j} \rceil + 2$. Two different cubes of the same colour are at distance at least $(k_0 - 1)2^j > 2U_0$, so within a colour class the g_Q have disjoint supports and $\| \sum_Q g_Q \|_2^2 = \sum_Q \|g_Q\|_2^2$. Lemma 8.13 bounds each $\|g_Q\|_2$. The number of classes is $k_0^n \le C(2^{-j}U)^n \le C 2^{(\ell_1+\ell_2)n}$, because $R \le 2^j$; the Cauchy–Schwarz inequality over the classes gives

$$\Big\| \sum_Q g_Q \Big\|_2 \le C 2^{-\ell_1(N-\frac{n}{2})} 2^{-\ell_2(N_2-\frac{n}{2})} (2^{-j}R)^{1/2} \Lambda \Big(\sum_Q \sum_B \|f_{Q,B}\|_2^2 \Big)^{1/2}. \tag{8.21}$$

With $N_2 = 2n+1$ the sum over ℓ_1, ℓ_2 converges, since $N \ge 3n+1$; this proves (8.19).

For (8.20), partition $\mathbb{R}^n$ into the cubes $\square$ of a grid of side $s_0 = \sqrt{n} 2^j + 2U_0 \le CU$. The set of points within U_0 of a cube $Q \in \mathcal{D}_j$ has diameter at most s_0 and therefore meets at most 2^n cubes $\square$. Let $\mathcal{Q}(\square)$ be the set of Q for which it meets $\square$. By the Cauchy–Schwarz inequality on $\square$ and (8.21) applied to the subfamily $\mathcal{Q}(\square)$,

$$\begin{aligned} \Big\| \sum_Q g_Q \Big\|_1 &\le \sum_{\square} |\square|^{\frac12} \Big\| \sum_{Q\in\mathcal{Q}(\square)} g_Q \Big\|_2 \\ &\le C U^{\frac{n}{2}} 2^{-\ell_1(N-\frac{n}{2})} 2^{-\ell_2(N_2-\frac{n}{2})} (2^{-j}R)^{\frac12} \Lambda \sum_{\square} \sum_{Q\in\mathcal{Q}(\square)} \Big(\sum_B \|f_{Q,B}\|_2^2 \Big)^{\frac12}. \end{aligned}$$

Each Q occurs for at most 2^n cubes $\square$, and $U^{n/2} \le 2^{jn/2} 2^{(\ell_1+\ell_2)n/2}$. Since $2^{jn/2}(2^{-j}R)^{1/2} = 2^{j\frac{n-1}{2}} R^{1/2}$, summing over ℓ_1, ℓ_2 proves (8.20). ☐

8.4. **Discretisation.** For $z \in \mathbb{Z}^n$ let q_z be the cube of $\mathcal{D}_0$ containing z, and $2q_z$ the concentric cube of side two. A cube $Q \in \mathcal{D}_j$ with $j \ge 0$ is a union of cubes of $\mathcal{D}_0$, and $q_z \subset Q$ exactly when $z \in Q$. In this subsection the bumps $\chi_j \in \mathcal{Y}$ ($j \ge 3$) are fixed, and m_j is defined by (8.6) with $\chi = \chi_j$.

Theorem 8.15 ([BRS23, Theorem 4.1])**.** *Let $p_{\mathrm{ST}} \le p < p_\circ$ and $p \le r < r_*(p)$. Then*

$$\Big\| \sum_{j\ge 3} 2^{j\frac{n+1}{2}} m_j(D) \sum_{Q\in\mathcal{D}_j} \sum_{z\in Q\cap\mathbb{Z}^n} \gamma(Q,z) F_{Q,z} \Big\|_r \le C \Big(\sum_{Q\in\mathcal{D}} |Q| \Big(\sum_z |\gamma(Q,z)|^p \Big)^{r/p} \Big)^{1/r} \tag{8.22}$$

for all finitely supported $\gamma : \mathcal{D} \times \mathbb{Z}^n \to \mathbb{C}$ and all measurable $F_{Q,z}$ with $\operatorname{supp} F_{Q,z} \subset 2q_z$ and $|F_{Q,z}| \le 1$.

In [BRS23, (4.1)] the support condition reads $\operatorname{supp} F_{Q,z} \subset 2q_z \cap Q$; the functions built in the proof below are supported in $2q_z$ but not always in Q, and the argument uses only $\operatorname{supp} F_{Q,z} \subset 2q_z$.

Proof of Theorem 8.2, assuming Theorem 8.15. By the remark after Definition 8.1 we may assume that only finitely many f_Q are nonzero. The terms with $j = 1, 2$ are bounded by Lemma 8.5, since $|Q| \geq 1$. For $j \geq 3$ let u be a Schwartz function with $\hat{u}$ compactly supported and $\hat{u}(\xi) = 1$ for $|\xi| \leq 2$, so that $m_j = m_j \hat{u}$. Choose a real even $\varphi \in C_c^\infty(\mathbb{R}^n)$ with $\int \varphi = 1$, and $\epsilon > 0$ so small that $|\hat{\varphi}(\epsilon\xi) - 1| \leq \frac{1}{2}$ on $\operatorname{supp} \hat{u}$ and $\operatorname{supp} \varphi(\cdot/\epsilon) \subset \{|x| < \frac{1}{2}\}$; then $\Phi = \epsilon^{-n}\varphi(\cdot/\epsilon)$ is supported in $|x| < \frac{1}{2}$ and $\hat{\Phi} \geq \frac{1}{2}$ on $\operatorname{supp} \hat{u}$. The function $v = \mathcal{F}^{-1}[\hat{u}/\hat{\Phi}]$ is a Schwartz function and $m_j(D)g = v * m_j(D)[\Phi * g]$, so it suffices to bound the left-hand side of (8.1), restricted to $j \geq 3$, with f_Q replaced by $\Phi * f_Q$.

For $Q \in \mathcal{D}_j$ and $z \in Q \cap \mathbb{Z}^n$ let $\gamma(Q,z) = \|f_Q 1_{q_z}\|_p$, and let $F_{Q,z} = \Phi * [f_Q 1_{q_z}]/(\|\Phi\|_{p'}\gamma(Q,z))$ if $\gamma(Q,z) \neq 0$ and $F_{Q,z} = 0$ otherwise. Then $\operatorname{supp} F_{Q,z} \subset q_z + B(0,\frac{1}{2}) \subset 2q_z$, $|F_{Q,z}| \leq 1$ by Hölder's inequality, and

$$\Phi * f_Q = \sum_{z \in Q \cap \mathbb{Z}^n} \Phi * [f_Q 1_{q_z}] = \|\Phi\|_{p'} \sum_{z \in Q \cap \mathbb{Z}^n} \gamma(Q,z) F_{Q,z}, \qquad \sum_z \gamma(Q,z)^p = \|f_Q\|_p^p.$$

Theorem 8.15 now gives (8.1). □

Theorem 8.15 is reduced to characteristic functions of finite sets.

Proposition 8.16 ([BRS23, Proposition 4.2])**.** *Let $p_{\mathrm{ST}} \leq p_1 < p_\circ$, $1 \leq r < r_*(p_1)$, $0 < \delta \leq 1$ and $M \geq 2$. Then*

$$\Big\| \sum_{j \geq 3} \sum_{\substack{Q \in \mathcal{D}_j \\ \#E_Q \leq M}} \sum_{z \in E_Q} 2^{j\frac{n+1}{2}} \beta_Q \, m_j(D) F_{Q,z} \Big\|_r \leq C_\delta M^\delta \Big(\sum_{Q \in \mathcal{D}} |Q| \, |\beta_Q|^r (\#E_Q)^{r/p_1} \Big)^{1/r} \tag{8.23}$$

for all finite sets $E_Q \subset Q \cap \mathbb{Z}^n$, empty for all but finitely many Q, all $\beta_Q \in \mathbb{C}$ and all $F_{Q,z}$ as in Theorem 8.15.

In [BRS23] the exponent r is tied to an exponent $p < p_1$; only $r < r_*(p_1)$ is used, and the statement above records this.

Proof of Theorem 8.15, assuming Proposition 8.16. The exponent $r_*(p)$ decreases continuously in p, since $\frac{1}{r_*}$ is an increasing affine function of $-\frac{1}{p}$. Choose $p_1 \in (p, p_\circ)$ with $r < r_*(p_1)$ and let $\delta = \frac{1}{2}(\frac{1}{p} - \frac{1}{p_1})$. Write $\gamma_Q = \gamma(Q,\cdot)$; we may assume $\gamma_Q \neq 0$ for the finitely many Q concerned. For $k \geq 0$ let

$$E_Q^k = \{z \in Q \cap \mathbb{Z}^n : 2^{-(k+1)/p}\|\gamma_Q\|_{\ell^p} < |\gamma(Q,z)| \leq 2^{-k/p}\|\gamma_Q\|_{\ell^p}\}, \qquad \beta_Q^k = 2^{-k/p}\|\gamma_Q\|_{\ell^p}.$$

These sets partition the support of γ_Q, and $\#E_Q^k \leq 2^{k+1}$ by Chebyshev's inequality. The functions $F_{Q,z}^k = \gamma(Q,z)F_{Q,z}/\beta_Q^k$, $z \in E_Q^k$, satisfy the hypotheses of Theorem 8.15. Proposition 8.16 with $M = 2^{k+1}$ and the triangle inequality bound the left-hand side of (8.22) by

$$C_\delta \sum_{k \geq 0} 2^{(k+1)\delta} \Big(\sum_Q |Q| \, 2^{-kr/p} \|\gamma_Q\|_{\ell^p}^r 2^{(k+1)r/p_1} \Big)^{1/r} \leq C \sum_{k \geq 0} 2^{-k\delta} \Big(\sum_Q |Q| \, \|\gamma_Q\|_{\ell^p}^r \Big)^{1/r},$$

because $\delta - \frac{1}{p} + \frac{1}{p_1} = -\delta$. □

8.5. **A stopping time.** The following is a discrete form of Tao's stopping time argument [Tao98, Lemma 4.3]. For $L \geq 0$ the cubes of $\mathcal{D}_L$ are $a_L + 2^L(k + [0,1)^n)$, $k \in \mathbb{Z}^n$, for some $a_L \in \mathbb{R}^n$: a dyadic lattice is a nested family, and two cubes of $\mathcal{D}_L$ lie in the standard subdivision of a common ancestor. For $\nu \in \{0,1\}^n$ let $\mathcal{D}_{L,\nu}$ consist of the cubes with $k \equiv \nu$ modulo 2. Two different cubes of $\mathcal{D}_{L,\nu}$ differ by at least 2 in some coordinate of k, so their distance is at least 2^L. For $z \in \mathbb{Z}^n$ let $B(z,L)$ denote the cube of $\mathcal{D}_L$ containing z.

Lemma 8.17 ([BRS23, Lemma 5.1])**.** *Let $E \subset \mathbb{Z}^n$ be finite and nonempty, $V \geq 1$ an integer, and $0 = L_0 < L_1 < \cdots < L_V$ integers. For $0 \leq k \leq V-1$ there are index sets $\mathcal{A}_k$ with $\#\mathcal{A}_k \leq 2^n (\#E)^{1/V}$ and families $\mathcal{B}_{k,\alpha}$, $\alpha \in \mathcal{A}_k$, of cubes of $\mathcal{D}_{L_k}$, such that two different cubes of $\mathcal{B}_{k,\alpha}$ are at distance at least $2^{L_{k+1}}$, and nonempty sets $E_{k,B} \subset E \cap B$ ($B \in \mathcal{B}_{k,\alpha}$) with*

$$E = \bigsqcup_{k=0}^{V-1} \bigsqcup_{\alpha \in \mathcal{A}_k} \bigsqcup_{B \in \mathcal{B}_{k,\alpha}} E_{k,B} \qquad \text{(a disjoint union).}$$

Proof. For $z \in E$ let $\kappa(z)$ be the least integer $\kappa \in [1,V]$ with $\#(E \cap B(z,L_\kappa)) \leq (\#E)^{\kappa/V}$; it exists because $\kappa = V$ qualifies. Let $E_k = \{z \in E : \kappa(z) = k+1\}$, so that $E = \bigsqcup_{k<V} E_k$. For $z \in E_k$,

$$\#(E \cap B(z,L_k)) \geq (\#E)^{k/V}, \qquad \#(E \cap B(z,L_{k+1})) \leq (\#E)^{(k+1)/V}: \tag{8.24}$$

the second is the definition of κ; the first follows from the minimality of $\kappa(z)$ if $k \geq 1$, and from $z \in E \cap B(z,0)$ if $k = 0$.

Let $\mathcal{D}_{L_k}(E_k)$ be the set of cubes of $\mathcal{D}_{L_k}$ that contain a point of E_k, and for $\nu \in \{0,1\}^n$ let $\mathcal{D}_{L_{k+1},\nu}(E_k)$ be the set of cubes of $\mathcal{D}_{L_{k+1},\nu}$ that contain a point of E_k. A cube $B' \in \mathcal{D}_{L_{k+1},\nu}(E_k)$ is $B(z,L_{k+1})$ for some $z \in E_k$; enumerate the cubes of $\mathcal{D}_{L_k}(E_k)$ inside it as $B_1(B'), \dots, B_{n(B')}(B')$. Each contains at least $(\#E)^{k/V}$ points of E by (8.24), they are disjoint, and B' contains at most $(\#E)^{(k+1)/V}$ points of E; so $n(B') \leq (\#E)^{1/V}$. For $1 \leq i \leq (\#E)^{1/V}$ let $\mathcal{B}_{k,(\nu,i)} = \{B_i(B') : B' \in \mathcal{D}_{L_{k+1},\nu}(E_k),\ n(B') \geq i\}$, and let $\mathcal{A}_k$ be the set of pairs (ν,i) for which it is nonempty. Different cubes of $\mathcal{B}_{k,(\nu,i)}$ lie in different cubes of $\mathcal{D}_{L_{k+1},\nu}$, so they are at distance at least $2^{L_{k+1}}$. Every cube of $\mathcal{D}_{L_k}(E_k)$ is $B_i(B')$ for exactly one ν, i and B', so $E_{k,B} = E_k \cap B$ gives the decomposition. □

8.6. **A restriction estimate.** At very large scales the proof uses a Fourier restriction bound with a small loss, which follows from the non-endpoint Bochner–Riesz bounds by an argument of Tao [Tao99, Theorem 1.1]; this is [BRS23, Proposition 6.1]. For the sphere the dual function ρ^* of [BRS23] is $|x|$ and $\nabla\rho^*(x) = x/|x|$. Write $\mu = \lambda + \frac{n+1}{2}$ and $e_n = (0,\dots,0,1)$.

Lemma 8.18. *Let $\lambda > -1$. Fix $\theta \in C^\infty(S^{n-1})$ supported in $|\omega - e_n| < \frac12$ with $\theta = 1$ on $|\omega - e_n| \leq \frac14$, and $\tilde w_2 \in C_c^\infty((\frac{1}{16},4))$ with $\tilde w_2 = 1$ on $[\frac18, 2]$. Let K be the kernel of $w(D)R^\lambda$, where $w(\xi) = \tilde w_2(|\xi|)\theta(\xi/|\xi|)$, and let $\Gamma = \{x \neq 0 : |x/|x| - e_n| \leq \frac18\}$. For every $J \geq 1$ there are smooth functions $c_0, \dots, c_{J-1}$ on $\Gamma \cap S^{n-1}$, with c_0 a nonzero constant, and a constant C_J, such that for $x \in \Gamma$ with $|x| \geq 16$*

$$K(x) = e^{i|x|}|x|^{-\mu}\Big(\sum_{k<J} c_k\big(\tfrac{x}{|x|}\big)|x|^{-k} + E_J(x)\Big), \qquad |E_J(x)| \leq C_J|x|^{-J}.$$

Proof. Since $\tilde w_2 \tilde w = \tilde w$, polar coordinates give

$$K(x) = (2\pi)^{-n}\int_0^1 \tilde w(t)(1-t)^\lambda t^{n-1}\,\hat\Theta(tx)\,dt, \qquad \hat\Theta(y) = \int_{S^{n-1}} \theta(\omega)e^{i\langle y,\omega\rangle}d\sigma(\omega).$$

The angular integral. Parametrise the cap $|\omega - e_n| < \frac12$ by $\omega = \Omega(u) = (u, s(u))$, $s(u) = (1-|u|^2)^{1/2}$, $|u| < \frac12$; then $d\sigma = s(u)^{-1}du$ and, for $y = |y|\vartheta$, $\hat\Theta(y) = \int a(u)e^{i|y|f_\vartheta(u)}du$ with

$a = (\theta \circ \Omega)/s$ and $f_\vartheta(u) = \langle \vartheta', u\rangle + \vartheta_n s(u)$, where $\vartheta = (\vartheta', \vartheta_n)$. Let $|\vartheta - e_n| \le \frac18$. The map $G(u) = u/s(u) = -\nabla s(u)$ has symmetric derivative $s^{-1}I + s^{-3}uu^{\mathsf T} \ge I$, so $|G(u)-G(v)| \ge |u-v|$. Since $\nabla f_\vartheta(u) = \vartheta_n(G(\vartheta') - G(u))$, the only critical point of f_ϑ is $u_\vartheta = \vartheta'$, where $\Omega(u_\vartheta) = \vartheta$ and $f_\vartheta = 1$, and $|\nabla f_\vartheta(u)| \ge \frac12 |u - u_\vartheta|$ on $|u| < \frac12$. At u_ϑ, $\nabla^2 f_\vartheta = -I - \vartheta_n^{-2}\vartheta'\vartheta'^{\mathsf T}$, which is negative definite with $|\det \nabla^2 f_\vartheta| = \vartheta_n^{-2}$. The method of stationary phase [Hör03, Theorem 7.7.5], whose constants are uniform for ϑ in this compact set, gives for $|y| \ge 1$

$$\hat\Theta(y) = e^{i|y|}|y|^{-\frac{n-1}{2}}\Big(\sum_{k<J} d_k(\vartheta)|y|^{-k} + O(|y|^{-J})\Big),$$
$$d_0 = (2\pi)^{\frac{n-1}{2}} e^{-i\pi\frac{n-1}{4}}\,\vartheta_n\, a(u_\vartheta) = (2\pi)^{\frac{n-1}{2}} e^{-i\pi\frac{n-1}{4}},$$

with d_k smooth in ϑ; here $\theta(\vartheta) = 1$ because $|\vartheta - e_n| \le \frac18$.

The radial integral. Let $x \in \Gamma$ with $|x| \ge 16$. For $t \in \operatorname{supp}\tilde w \cap (0,1)$ we have $t \ge \frac18$, so $|tx| \ge 2$ and $tx/|tx| = x/|x|$. Inserting the expansion, and bounding the remainder by $C|x|^{-\frac{n-1}{2}-J}\int_0^1 (1-t)^\lambda dt$,

$$K(x) = (2\pi)^{-n}\sum_{k<J} d_k\big(\tfrac{x}{|x|}\big)|x|^{-\frac{n-1}{2}-k}\int_0^1 (1-t)^\lambda h_k(t) e^{it|x|}dt + O(|x|^{-\frac{n-1}{2}-J}),$$

where $h_k(t) = \tilde w(t) t^{\frac{n-1}{2}-k}$ is smooth and vanishes for $t \le \frac18$. With $t = 1 - v$ the integral is $e^{i|x|}\int_0^\infty v^\lambda \varphi_k(v) e^{-iv|x|}dv$, where $\varphi_k(v) = h_k(1-v)$ for $v \le 1$ and $\varphi_k(v) = 0$ for $v \ge 1$ is smooth with compact support in $[0, \frac78]$. By Erdélyi's lemma [Erd56, §2.8],

$$\int_0^\infty v^\lambda \varphi_k(v) e^{-ivs}dv$$
$$= \sum_{m<J} \frac{\varphi_k^{(m)}(0)}{m!}\,\Gamma(\lambda+m+1)\, e^{-i\frac{\pi}{2}(\lambda+m+1)} s^{-\lambda-m-1} + O(s^{-\lambda-1-J}) \qquad (s\to\infty).$$

Collecting terms gives the expansion, with $c_0 = (2\pi)^{-n} d_0\,\Gamma(\lambda+1) e^{-i\frac{\pi}{2}(\lambda+1)}\varphi_0(0)$ and $\varphi_0(0) = \tilde w(1) = 1$. □

Proposition 8.19. *Let $1 \le p \le r < \infty$ and $\alpha > 0$, and suppose that $R^{\lambda(r)+\alpha}$ maps $L^p(\mathbb{R}^n)$ to $L^r(\mathbb{R}^n)$. Let $\eta \in C_c^\infty(\{\frac14 < |x| < 4\})$ and $T_A f(x) = \int e^{iA|x-y|}\eta(x-y) f(y)\,dy$. Then $\|T_A\|_{L^p\to L^r} \le C A^{\alpha - n/p'}$ for $A \ge 1$.*

Proof. For A in a bounded range the bound follows from Young's inequality, so let A be large. Cover $\operatorname{supp}\eta$ by finitely many balls $B(x_\nu, \frac{1}{64})$ with $\frac14 \le |x_\nu| \le 4$ and write $\eta = \sum_\nu \eta_\nu$ with $\eta_\nu \in C_c^\infty(B(x_\nu, \frac{1}{64}))$. For x in that ball, $|x/|x| - x_\nu/|x_\nu|| \le 2|x - x_\nu|/|x_\nu| \le \frac18$. Since the operator norms are invariant under rotations, we may assume $x_\nu/|x_\nu| = e_n$, so that $\operatorname{supp}\eta_\nu \subset \Gamma \cap \{\frac14 < |x| < 4\}$. Let K be as in Lemma 8.18 with $\lambda = \lambda(r) + \alpha$; then $\mu = \frac nr + \alpha$. Take $J = n$ and write $B_J(z) = \sum_{k<J} c_k(z/|z|)|z|^{-k}$. On $\operatorname{supp}\eta_\nu$ the functions $x \mapsto B_J(Ax)$, $A \ge 64$, have derivatives of order at most $n+1$ bounded uniformly in A, and $|B_J(Ax) - c_0| \le C/A$; choose $A_0 \ge 64$ with $|B_J(Ax)| \ge |c_0|/2$ there for $A \ge A_0$. For such A, by Lemma 8.18,

$$e^{iA|x|}\eta_\nu(x) = A^\mu u_A(x) K(Ax) - e^{iA|x|}\eta_\nu(x)\frac{E_J(Ax)}{B_J(Ax)}, \qquad u_A(x) = \frac{\eta_\nu(x)|x|^\mu}{B_J(Ax)}.$$

The functions u_A are bounded in C^{n+1} with supports in a fixed compact set, so $\|\hat u_A\|_1 \le C$. The convolution operator with kernel $u_A(z/A)K(z) = (2\pi)^{-n}\int \hat u_A(\zeta) e^{i\langle z, \zeta/A\rangle} K(z)\,d\zeta$ is an average of the operators with kernels $e^{i\langle z,\zeta'\rangle}K(z)$, which are conjugates of $w(D)R^\lambda$ by modulations; so its $L^p \to L^r$ norm is at most $C\|w(D)R^\lambda\|_{L^p\to L^r}$, which is finite by hypothesis. Rescaling,

the kernel $A^\mu u_A(x)K(Ax) = A^{\mu-n}\,A^n[u_A(\cdot/A)K](Ax)$ gives an operator of norm at most $CA^{\mu-n+\frac{n}{p}-\frac{n}{r}} = CA^{\alpha-\frac{n}{p'}}$. The second kernel is bounded by CA^{-n} and supported in $|x|\le 4$, so by Young's inequality its operator has norm at most $CA^{-n} \le CA^{\alpha-n/p'}$. □

Proposition 8.20 ([BRS23, Proposition 6.1]). *Let $1 \le p \le r < \infty$ and $\alpha > 0$, and suppose that $R^{\lambda(r)+\alpha}$ maps $L^p(\mathbb{R}^n)$ to $L^r(\mathbb{R}^n)$. Then for $R \ge 2$ and every $f \in L^p(\mathbb{R}^n)$ supported in a cube of side R,*

$$\Big(\int_{S^{n-1}} |\hat f(\omega)|^r d\sigma(\omega)\Big)^{1/r} \le C\,R^{2\alpha}\|f\|_p.$$

Proof. A translation of f does not change $|\hat f|$, so let $\operatorname{supp} f \subset [-\frac{R}{2},\frac{R}{2}]^n$. Let $R_0 = 4\pi\sqrt{n}$; a cube of side $R < R_0$ lies in one of side R_0, so it suffices to take $R \ge R_0$. By homogeneity $\int_{1\le|x|\le 2} |\hat f(x/|x|)|^r dx = \frac{2^n-1}{n}\int_{S^{n-1}} |\hat f|^r d\sigma$. Let $g(y) = R^{2n} f(R^2 y)$, which is supported in $|y|_\infty \le \frac{1}{2R}$, so that $\hat f(x/|x|) = \int e^{-iR^2\langle y, x/|x|\rangle} g(y)\,dy$ and $\|g\|_p = R^{2n/p'}\|f\|_p$.

For $|x| \ge \frac12$ and $|y| \le \frac14$, Taylor's formula for $s \mapsto |x - sy|$ gives

$$\langle y, \tfrac{x}{|x|}\rangle = |x| - |x-y| + \langle y, H(x,y)y\rangle, \qquad H(x,y) = \int_0^1 (1-s)\nabla^2|\cdot|(x-sy)\,ds.$$

Fix $\zeta_1 \in C_c^\infty(\{\frac12 < |x| < 3\})$ with $\zeta_1 = 1$ on $1 \le |x| \le 2$, and $\zeta_2 \in C_c^\infty((-\pi,\pi)^n)$ with $\zeta_2 = 1$ on $[-1,1]^n$, and let $\Psi_R(x,w) = \zeta_1(x)\zeta_2(w)e^{-i\langle w, H(x,w/R)w\rangle}$. For $R \ge R_0$ and (x,w) in the support, $|w/R| \le \frac14$ and $|x - sw/R| \ge \frac14$, so Ψ_R is smooth, supported in $(-\pi,\pi)^{2n}$, with derivatives bounded uniformly in R. Its Fourier series $\Psi_R(x,w) = \sum_{k_1,k_2\in\mathbb{Z}^n} c_{k_1,k_2}(R)e^{i\langle k_1,x\rangle + i\langle k_2,w\rangle}$ on $[-\pi,\pi]^{2n}$ therefore satisfies $|c_{k_1,k_2}(R)| \le C(1+|k_1|+|k_2|)^{-2n-1}$. For $1 \le |x| \le 2$ and $y \in \operatorname{supp} g$, with $w = Ry$ we have $|w|_\infty \le \frac12$, $\Psi_R(x,w) = e^{-iR^2\langle y, H(x,y)y\rangle}$, and

$$\hat f(\tfrac{x}{|x|}) = e^{-iR^2|x|}\sum_{k_1,k_2} c_{k_1,k_2}(R)e^{i\langle k_1,x\rangle}\int e^{iR^2|x-y|}e^{iR\langle k_2,y\rangle}g(y)\,dy.$$

Let $\eta \in C_c^\infty(\{\frac14 < |x| < 4\})$ be equal to one on $\frac12 \le |x| \le 3$; then $\eta(x-y) = 1$ in the last integral, which is therefore $T_{R^2}[e^{iR\langle k_2,\cdot\rangle}g](x)$. By Minkowski's inequality and Proposition 8.19 with $A = R^2$,

$$\Big(\int_{1\le|x|\le2} |\hat f(x/|x|)|^r dx\Big)^{1/r} \le C\sum_{k_1,k_2} |c_{k_1,k_2}(R)|\,R^{2\alpha-\frac{2n}{p'}}\|g\|_p \le CR^{2\alpha}\|f\|_p.$$ □

8.7. **Large scales.** The next lemma is the multi-scale estimate that the separation of cubes produces.

Lemma 8.21 ([BRS23, Lemma 7.1]). *Let $L \ge 0$ be an integer. For finitely many $j \ge \max\{2L,3\}$ let $\mathcal{Q}_j$ be a finite set of cubes of $\mathcal{D}_j$, and for $Q \in \mathcal{Q}_j$ let $\mathcal{B}_Q$ be a family of cubes of $\mathcal{D}_L$ contained in Q which is S_Q-separated for some $S_Q \ge \max\{3\cdot 2^L, 2^{2L}(\#\mathcal{B}_Q)^{\frac{2}{n-1}}\}$. Then for $1 \le q_2 \le q_1 \le 2$,*

$$\Big\|\sum_j \sum_{Q\in\mathcal{Q}_j}\sum_{B\in\mathcal{B}_Q} 2^{j\frac{n+1}{2}}\mathcal{F}^{-1}[m_j(\widehat{\eta_B} * G_{Q,B})]\Big\|_{q_2} \le C\,2^{L/q_1}\Big(\sum_j\sum_{Q\in\mathcal{Q}_j}|Q|\Big(\sum_{B\in\mathcal{D}_L}\|G_{Q,B}\|_{q_1}^{q_1}\Big)^{q_2/q_1}\Big)^{1/q_2} \tag{8.25}$$

for all $G_{Q,B} \in L^1\cap L^2(\mathbb{R}^n)$, where $G_{Q,B} = 0$ for $B \notin \mathcal{B}_Q$.

Proof. Since $\mathcal{F}[\eta_B f] = (2\pi)^{-n}\widehat{\eta_B} * \hat f$, we have $\mathcal{F}^{-1}[m_j(\widehat{\eta_B} * G)] = (2\pi)^n m_j(D)[\eta_B \mathcal{F}^{-1}G]$. In Proposition 8.14, with $R = 2^L \le 2^j$, the hypothesis on S_Q gives $S_Q^{-\frac{n-1}{2}} R^{n-1}\#\mathcal{B}_Q \le 1$, so $\Lambda \le \sqrt{2}$.

(i) $q_1 = q_2 = 2$. By (8.19) and Plancherel's theorem, the j-th term of the left-hand side has L^2 norm at most $C2^{j\frac{n+1}{2}}(2^{L-j})^{1/2}(\sum_Q \sum_B \|G_{Q,B}\|_2^2)^{1/2} = C2^{L/2}(\sum_{Q\in\mathcal{Q}_j}|Q|\sum_B\|G_{Q,B}\|_2^2)^{1/2}$. The Fourier transform of the j-th term is supported in $\operatorname{supp} m_j$, and a point lies in at most two of these sets, so the squares of the norms add up to within a factor 2.

(ii) $q_1 = 2$, $q_2 = 1$. By (8.20), the j-th term has L^1 norm at most

$$C2^{j\frac{n+1}{2}}2^{j\frac{n-1}{2}}2^{L/2}\sum_Q\Big(\sum_B\|G_{Q,B}\|_2^2\Big)^{1/2} = C2^{L/2}\sum_Q|Q|\Big(\sum_B\|G_{Q,B}\|_2^2\Big)^{1/2};$$

sum over j.

(iii) $q_1 = q_2 = 1$. We show that for $B\in\mathcal{D}_L$ and $j\ge\max\{2L,3\}$

$$\big\|\mathcal{F}^{-1}[m_j(\widehat{\eta_B}*G)]\big\|_1 \le C\,2^L 2^{j\frac{n-1}{2}}\|G\|_1, \tag{8.26}$$

which gives (8.25) after multiplying by $2^{j\frac{n+1}{2}}$, since $|Q| = 2^{jn}$. We use the decomposition of Córdoba [Cór79]. Let Ω_j be a maximal $2^{-j/2}$-separated subset of S^{n-1}, so $\#\Omega_j \le C2^{j\frac{n-1}{2}}$. With ϕ_0 as in (8.11), the functions $\tilde\psi_\tau(\omega) = \phi_0(2^{j/2-1}(\omega-\tau))$, $\tau\in\Omega_j$, have bounded overlap and their sum is at least one on S^{n-1}; so $\psi_\tau = \tilde\psi_\tau/\sum_{\tau'}\tilde\psi_{\tau'}$ is a partition of unity on S^{n-1} with $\operatorname{supp}\psi_\tau\subset\{|\omega-\tau|\le 2^{1-j/2}\}$ and $|\nabla_\omega^k\psi_\tau|\le C_k2^{jk/2}$. Let $m_j^\tau(\xi) = m_j(\xi)\psi_\tau(\xi/|\xi|)$.

We claim $\|\mathcal{F}^{-1}m_j^\tau\|_1\le C$. Rotate so that $\tau = e_n$. On $\operatorname{supp} m_j^\tau$ we have $\frac34<|\xi|<1$, $|\xi'|\le 2^{1-j/2}$ and $0\le 1-\xi_n\le(1-|\xi|)+|\xi'|^2/(|\xi|+\xi_n)\le 2^{3-j}$. Let $\tilde m(t,s) = m_j^\tau(2^{-j/2}t, 1-2^{-j}s)$ for $(t,s)\in\mathbb{R}^{n-1}\times\mathbb{R}$; it is supported in $|t|\le 2$, $0\le s\le 8$. We check that $|\partial_\xi^\gamma m_j^\tau|\le C2^{j(\gamma_n+|\gamma'|/2)}$ for $|\gamma|\le n+1$, which gives $|\partial^\gamma\tilde m|\le C$. For the factor $\psi_\tau(\xi/|\xi|)$ this holds because $\xi\mapsto\xi/|\xi|$ has bounded derivatives. For $\chi(g(\xi))$ with $g(\xi) = 2^j(1-|\xi|)$, the chain rule writes $\partial^\gamma[\chi(g)]$ as a sum of terms $\chi^{(k)}(g)\prod_{i\le k}\partial^{\gamma_i}g$ with $\sum\gamma_i=\gamma$. On the support, $\partial_{\xi_l}g = -2^j\xi_l/|\xi|$ is $O(2^{j/2})$ for $l<n$ and $O(2^j)$ for $l=n$, and $\partial^{\gamma_i}g = O(2^j)$ when $|\gamma_i|\ge 2$; in each case $|\partial^{\gamma_i}g|\le C2^{j((\gamma_i)_n+|\gamma_i'|/2)}$. The product rule and $\chi\in C^{M_0}$ with $M_0\ge n+1$ give the claim, hence $|\mathcal{F}^{-1}\tilde m(y)|\le C(1+|y|)^{-n-1}$. The kernel $\mathcal{F}^{-1}m_j^\tau$ arises from $\mathcal{F}^{-1}\tilde m$ by an affine change of variables and a modulation, which preserve the L^1 norm.

Now $\widehat{\eta_B}$ is supported in $|\xi|\le\frac{1}{2R}$, $R = 2^L\le 2^{j/2}$. For $\omega\in\mathbb{R}^n$ let $\mathcal{A}_j(\omega)$ be the set of τ with $\operatorname{supp} m_j^\tau\cap B(\omega,\frac{1}{2R})\ne\emptyset$. If $R<8$ then $\#\mathcal{A}_j(\omega)\le C2^{j\frac{n-1}{2}}\le C'(2^{j/2}/R)^{n-1}$. If $R\ge 8$, two points ξ_1,ξ_2 of $B(\omega,\frac{1}{2R})$ with $|\xi_i|\ge\frac34$ satisfy $|\xi_1/|\xi_1|-\xi_2/|\xi_2||\le 2|\xi_1-\xi_2|/|\xi_1|\le 3/R$, so every $\tau\in\mathcal{A}_j(\omega)$ lies within $3/R+2^{1-j/2}\le 5/R$ of a fixed point of S^{n-1}; as the points of Ω_j are $2^{-j/2}$-separated, again $\#\mathcal{A}_j(\omega)\le C(2^{j/2}/R)^{n-1}$. Since $\mathcal{F}^{-1}[\widehat{\eta_B}(\cdot-\omega)] = e^{i\langle\omega,\cdot\rangle}\eta_B$ has L^1 norm $R^n\|\eta\|_1$,

$$\big\|\mathcal{F}^{-1}[m_j\,\widehat{\eta_B}(\cdot-\omega)]\big\|_1\le\sum_{\tau\in\mathcal{A}_j(\omega)}\|\mathcal{F}^{-1}m_j^\tau\|_1\,R^n\|\eta\|_1\le CR\,2^{j\frac{n-1}{2}}.$$

Writing $\mathcal{F}^{-1}[m_j(\widehat{\eta_B}*G)] = \int G(\omega)\mathcal{F}^{-1}[m_j\widehat{\eta_B}(\cdot-\omega)]\,d\omega$ proves (8.26).

Interpolation. The left-hand side of (8.25) is a linear function of $G = (G_{Q,B})$, and the right-hand side is $2^{L/q_1}$ times the norm of G in the mixed-norm space $L^{q_2}(X;L^{q_1}(Y))$, where X is the set of pairs (j,Q) with the measure giving mass $|Q|$ to (j,Q), and $Y = \mathcal{D}_L\times\mathbb{R}^n$ with counting measure times Lebesgue measure. These spaces form a complex interpolation scale [BP61]. Cases (i)–(iii) are the vertices $(\frac12,\frac12)$, $(\frac12,1)$, $(1,1)$ of the triangle $\{(\frac{1}{q_1},\frac{1}{q_2}):\frac12\le\frac{1}{q_1}\le\frac{1}{q_2}\le 1\}$, and $2^{L/q_1}$ is the exponential of an affine function of $\frac{1}{q_1}$; so (8.25) holds on the whole triangle. □

8.8. **Proof of Proposition 8.16.** Fix p_1, r, δ and M as in Proposition 8.16, let $r_1 = \max\{r, p_1\}$, so that $p_1 \le r_1 \le r_*(p_1) \le 2$ by (8.3), and let

$$V = \lceil 2/\delta \rceil, \qquad H = V^{2V+1}.$$

Then $V \ge 2$. By Corollary 8.9 with (p_1, r_1): for every $\varepsilon > 0$,

$$\|m_{j,\ell}(D)\|_{L^{p_1}\to L^{r_1}} \le C_\varepsilon 2^{-\ell(N-n)} 2^{j(\lambda(r_1)+\varepsilon)} \qquad (j \ge 3,\ \ell \ge 0), \tag{8.27}$$

and R^λ maps L^{p_1} to L^{r_1} for every $\lambda > \lambda(r_1)$. Write $g_Q = \beta_Q \sum_{z\in E_Q} F_{Q,z}$ and $\mathbf{N} = (\sum_Q |Q|\,|\beta_Q|^r (\#E_Q)^{r/p_1})^{1/r}$; only cubes with $1 \le \#E_Q \le M$ occur. The support of g_Q lies in $Q^+ = Q + [-\frac12, \frac12]^n$, and $|g_Q| \le |\beta_Q| \sum_{z\in E_Q} 1_{2q_z}$, where at most 3^n of the cubes $2q_z$ contain a given point; hence $\|g_Q\|_{p_1} \le C|\beta_Q|(\#E_Q)^{1/p_1}$.

Small scales, $2^j \le (2M)^H$. Fix such a $j \ge 3$ and let $\ell \ge 0$. For $Q' \in \mathcal{D}_{j+\ell}$ let $h_{Q'} = \sum_{Q\in\mathcal{D}_j,\, Q\subset Q'} g_Q$. The kernel of $m_{j,\ell}(D)$ is supported in $|x| < 2^{j+\ell}$, so $m_{j,\ell}(D)h_{Q'}$ is supported in the concentric cube $5Q'$, and these cubes overlap at most 5^n times. The sets Q^+, $Q \in \mathcal{D}_j$, overlap at most 2^n times. By Hölder's inequality on $5Q'$ and (8.27),

$$\|m_{j,\ell}(D)h_{Q'}\|_r \le C_\varepsilon 2^{(j+\ell)n(\frac1r - \frac{1}{r_1})} 2^{-\ell(N-n)} 2^{j(\lambda(r_1)+\varepsilon)} \Big(\sum_{Q\subset Q'} |\beta_Q|^{p_1} \#E_Q \Big)^{1/p_1}.$$

A cube Q' contains $2^{\ell n}$ cubes of $\mathcal{D}_j$, so by Hölder's inequality if $r \ge p_1$, and by $\ell^r \subset \ell^{p_1}$ if $r < p_1$, $(\sum_{Q\subset Q'} |\beta_Q|^{p_1}\#E_Q)^{1/p_1} \le 2^{\ell n \max\{0, \frac{1}{p_1}-\frac1r\}} (\sum_{Q\subset Q'} |\beta_Q|^r (\#E_Q)^{r/p_1})^{1/r}$. Since $\frac{n+1}{2} + \frac{n}{r} - \frac{n}{r_1} + \lambda(r_1) = \frac{n}{r}$ and $|Q| = 2^{jn}$, summing the r-th powers over Q' and then over ℓ gives

$$\Big\| 2^{j\frac{n+1}{2}} m_j(D) \sum_{Q\in\mathcal{D}_j} g_Q \Big\|_r \le C_\varepsilon 2^{j\varepsilon} \sum_{\ell\ge 0} 2^{-\ell(N-3n)}\, \mathbf{N} \le C_\varepsilon 2^{j\varepsilon} \mathbf{N},$$

because $N \ge 3n+1$. Take $\varepsilon = \delta/(2H)$, so that $2^{j\varepsilon} \le (2M)^{\delta/2}$. There are at most $H\log_2(2M)$ such j, and $H\log_2(2M)(2M)^{\delta/2} \le C_\delta M^\delta$.

Large scales, $2^j > (2M)^H$. Let $a = \frac{2}{n-1}\log_2 M$, $L_0 = 0$ and $L_{k+1} = \lfloor 2L_k + a \rfloor + 1$. Then $L_{k+1} > 2L_k + a \ge L_k$, and by induction $L_k \le (2^k - 1)(1+a)$, so, as $n \ge 2$,

$$2^{L_k} \le (2M^{\frac{2}{n-1}})^{2^k-1} \le (2M)^{2^{k+1}} \le (2M)^{2^V} \qquad (k \le V-1). \tag{8.28}$$

Moreover $2^V \le \frac12 V^{2V}$ for $V \ge 2$ and $\frac1V \le \frac{\delta}{2}$, so $2^V/H \le \frac{1}{2V} \le \frac{\delta}{4}$, and for these j, $2^{L_k} \le (2M)^{2^V} < 2^{j2^V/H} \le 2^{j\delta/4}$; in particular $L_k \le j/4$.

For each Q with $E_Q \ne \emptyset$ apply Lemma 8.17 to E_Q with this V and these L_k, obtaining $\mathcal{A}_Q^k$, $\mathcal{B}_Q^{k,\alpha}$ and $E_{Q,B}^k$, with $\#\mathcal{A}_Q^k \le 2^n M^{1/V} \le 2^n M^{\delta/2}$. A cube $B \in \mathcal{B}_Q^{k,\alpha}$ belongs to $\mathcal{D}_{L_k}$ and contains a point of Q, and $L_k \le j$, so $B \subset Q$. Two different cubes of $\mathcal{B}_Q^{k,\alpha}$ lie in different cubes of some $\mathcal{D}_{L_{k+1},\nu}$, so they are at distance at least $2^{L_{k+1}}$ and their centres differ by at least $2^{L_{k+1}} + 2^{L_k} \ge 3\cdot 2^{L_k}$ in one coordinate. Since the sets $E_{Q,B}^k$ are nonempty and disjoint, $\#\mathcal{B}_Q^{k,\alpha} \le M$, so $2^{L_{k+1}} > 2^{2L_k} M^{\frac{2}{n-1}} \ge 2^{2L_k}(\#\mathcal{B}_Q^{k,\alpha})^{\frac{2}{n-1}}$. Hence $\mathcal{B}_Q^{k,\alpha}$ is S_Q-separated with $S_Q = \max\{3\cdot 2^{L_k}, 2^{2L_k}(\#\mathcal{B}_Q^{k,\alpha})^{\frac{2}{n-1}}\}$.

Enumerate $\mathcal{A}_Q^k = \{\alpha_1, \dots, \alpha_{n(k,Q)}\}$. The decomposition of Lemma 8.17 and the triangle inequality bound the left-hand side of (8.23), restricted to these j, by

$$\sum_{k<V} \sum_{i\le 2^n M^{\delta/2}} \Big\| \sum_j \sum_{Q:\, n(k,Q)\ge i} \sum_{B\in\mathcal{B}_Q^{k,\alpha_i}} \sum_{z\in E_{Q,B}^k} 2^{j\frac{n+1}{2}} \beta_Q m_j(D) F_{Q,z} \Big\|_r.$$

Since $V2^nM^{\delta/2} \le C_\delta M^{\delta/2}$, it suffices to show that for each $k < V$, each subcollection of cubes Q and each choice $Q \mapsto \mathcal{B}_Q = \mathcal{B}_Q^{k,\alpha(Q)}$ with $\alpha(Q) \in \mathcal{A}_Q^k$,

$$\Big\| \sum_j \sum_Q \sum_{B\in\mathcal{B}_Q} \sum_{z\in E_{Q,B}^k} 2^{j\frac{n+1}{2}} \beta_Q m_j(D) F_{Q,z} \Big\|_r \le C_\delta M^{\delta/2}\, \mathbf{N}. \tag{8.29}$$

Let $L = L_k$. For $B \in \mathcal{B}_Q$ let $f_{Q,B} = \beta_Q \eta_B^{-1} \sum_{z\in E_{Q,B}^k} F_{Q,z}$ on the union of the cubes $2q_z$, $z \in E_{Q,B}^k$, and $f_{Q,B} = 0$ elsewhere. These cubes lie in $B^+ = B + [-\frac12, \frac12]^n$, a cube of side $2^L + 1$ on which $\eta_B \ge 1$, since $|x - x_B|_\infty \le 2^{L-1} + \frac12 \le 2^{L+1}$ there. Hence $\eta_B f_{Q,B} = \beta_Q \sum_z F_{Q,z}$ and, as above, $\|f_{Q,B}\|_{p_1} \le C|\beta_Q|(\#E_{Q,B}^k)^{1/p_1}$. Let $A_L = \{\xi : ||\xi| - 1| \le 2^{2-L}\}$ and $G_{Q,B} = 1_{A_L}\widehat{f_{Q,B}}$. If $\zeta \in \operatorname{supp} m_j$ and $|\zeta - \xi| \le 2^{-L}$, then $||\xi| - 1| \le ||\zeta| - 1| + |\zeta - \xi| < 2^{1-j} + 2^{-L} \le 2^{2-L}$ because $j \ge L$; so $m_j(\widehat{\eta_B} * \widehat{f_{Q,B}}) = m_j(\widehat{\eta_B} * G_{Q,B})$ and

$$\sum_{z\in E_{Q,B}^k} \beta_Q m_j(D) F_{Q,z} = m_j(D)[\eta_B f_{Q,B}] = (2\pi)^{-n}\mathcal{F}^{-1}\big[m_j(\widehat{\eta_B} * G_{Q,B})\big].$$

Since $j \ge 4L$ and $j \ge 3$, Lemma 8.21 with $(q_1, q_2) = (r_1, r)$ bounds the left-hand side of (8.29) by

$$C2^{L/r_1}\Big(\sum_j \sum_Q |Q|\Big(\sum_{B\in\mathcal{B}_Q} \int_{A_L} |\widehat{f_{Q,B}}|^{r_1}\Big)^{r/r_1}\Big)^{1/r}. \tag{8.30}$$

Let $\alpha = \delta 2^{-V-2}$. We claim that $\int_{A_L} |\hat f|^{r_1} \le C_\delta 2^{-L} 2^{2\alpha r_1 L} \|f\|_{p_1}^{r_1}$ for f supported in a cube of side $2^L + 1$. If $L \le 2$, then $|A_L| \le C$ and $|\hat f| \le \|f\|_1 \le C\|f\|_{p_1}$. If $L \ge 3$, then $A_L \subset \{\frac12 \le |\xi| \le \frac32\}$, and for $\frac12 \le t \le \frac32$ the function $\hat f(t\,\cdot)$ is the Fourier transform of $f_t = t^{-n} f(\cdot/t)$, which is supported in a cube of side between 2 and 2^{L+1} and has $\|f_t\|_{p_1} \le 2^n \|f\|_{p_1}$. Proposition 8.20 with $(p, r) = (p_1, r_1)$, whose hypothesis holds by Corollary 8.9, and integration in polar coordinates over $|t - 1| \le 2^{2-L}$ prove the claim.

By the claim, (8.30) is at most $C_\delta 2^{2\alpha L}(\sum_j \sum_Q |Q|\,|\beta_Q|^r (\sum_B (\#E_{Q,B}^k)^{r_1/p_1})^{r/r_1})^{1/r}$. By (8.28), $2^{2\alpha L} \le (2M)^{2^{V+1}\alpha} = (2M)^{\delta/2}$. Since $r_1 \ge p_1$, $(\sum_B (\#E_{Q,B}^k)^{r_1/p_1})^{p_1/r_1} \le \sum_B \#E_{Q,B}^k \le \#E_Q$. This proves (8.29), and with the small scales, Proposition 8.16. This completes the proof of Theorem 8.2. □

Part III. Endpoint bounds

9. Endpoint bounds

This section proves Theorems A, B and C. The two results we cite, the sparse domination theorem of Beltran, Roos and Seeger and the weighted theorem of Frey and Nieraeth, are stated first in the form in which they are used. Throughout, $R_a^\lambda = R_{a,1}^\lambda$.

9.1. **Sparse forms.** A *dyadic lattice* in the sense of [LN19, §2] is a family $\mathcal{D}$ of cubes in $\mathbb{R}^n$ such that the 2^n halves of every cube of $\mathcal{D}$ belong to $\mathcal{D}$, every two cubes of $\mathcal{D}$ are contained in a common cube of $\mathcal{D}$, and every compact set is contained in a cube of $\mathcal{D}$. For a cube Q, a locally integrable f and $1 \le p < \infty$ put $\langle f\rangle_{Q,p} = \big(|Q|^{-1}\int_Q |f|^p\big)^{1/p}$.

Definition 9.1. Let $0 < \gamma < 1$. A family $\mathcal{S}$ of cubes, each belonging to some dyadic lattice, is *γ-sparse* if there are pairwise disjoint measurable sets $E_Q \subset Q$, $Q \in \mathcal{S}$, with $|E_Q| \ge \gamma|Q|$. For $1 \le p, q < \infty$ put

$$\Lambda_{p,q}^{\mathcal{S}}(f_1, f_2) = \sum_{Q\in\mathcal{S}} |Q|\,\langle f_1\rangle_{Q,p}\langle f_2\rangle_{Q,q}, \qquad \Lambda_{p,q}^{*}(f_1, f_2) = \sup_{\mathcal{S}} \Lambda_{p,q}^{\mathcal{S}}(f_1, f_2),$$

the supremum over all γ-sparse families. A linear operator T from $C_c^\infty(\mathbb{R}^n)$ to distributions belongs to $\mathrm{Sp}(p,q)$ if there is C with

$$|\langle Tf_1, f_2\rangle| \le C\,\Lambda^*_{p,q}(f_1,f_2) \qquad (f_1,f_2\in C_c^\infty(\mathbb{R}^n)), \tag{9.1}$$

and $\|T\|_{\mathrm{Sp}(p,q)}$ is the least such C.

Here $\langle f,g\rangle = \int fg$. This is [BRS25, (1.2)–(1.4)]. The class $\mathrm{Sp}(p,q)$ does not depend on γ [LN19], and we fix $\gamma=\frac12$. Since $\langle f\rangle_{Q,p}$ increases with p, $\mathrm{Sp}(p,q)\subset\mathrm{Sp}(\tilde p,\tilde q)$ when $p\le\tilde p$ and $q\le\tilde q$.

Lemma 9.2. *Let T be a Fourier multiplier operator $m(D)$ with $m\in L^\infty$, and $T_t=m(D/t)$ for $t>0$. Then $\|T_t\|_{\mathrm{Sp}(p,q)}=\|T\|_{\mathrm{Sp}(p,q)}$.*

Proof. For $f\in C_c^\infty$ put $f^{(t)}(y)=f(y/t)$. Its Fourier transform is $t^n\hat f(t\xi)$, so

$$T(f^{(t)})(y)=\frac{1}{(2\pi)^n}\int m(\xi)\,t^n\hat f(t\xi)e^{i\langle y,\xi\rangle}\,d\xi=\frac{1}{(2\pi)^n}\int m(\eta/t)\hat f(\eta)e^{i\langle y/t,\eta\rangle}\,d\eta=T_tf(y/t).$$

Hence $\langle T_tf_1,f_2\rangle=\int T(f_1^{(t)})(tx)f_2(x)\,dx=t^{-n}\langle Tf_1^{(t)},f_2^{(t)}\rangle$. For a cube Q, $\langle f^{(t)}\rangle_{Q,p}=\langle f\rangle_{t^{-1}Q,p}$ and $|Q|=t^n|t^{-1}Q|$. The image of a dyadic lattice under $x\mapsto t^{-1}x$ is a dyadic lattice, and the image of a γ-sparse family is γ-sparse, with the sets $t^{-1}E_Q$. Therefore $t^{-n}\Lambda^{\mathcal S}_{p,q}(f_1^{(t)},f_2^{(t)})=\Lambda^{t^{-1}\mathcal S}_{p,q}(f_1,f_2)\le\Lambda^*_{p,q}(f_1,f_2)$, and $\|T_t\|_{\mathrm{Sp}(p,q)}\le\|T\|_{\mathrm{Sp}(p,q)}$. Exchanging the roles of T and T_t gives equality. □

9.2. **The sparse domination theorem.** For $1\le p<p_\circ<\frac{2n}{n+1}$ with $p_{\mathrm{ST}}\le p_\circ$ define $q_*(p,p_\circ)$ by

$$1-\frac{1}{q_*(p,p_\circ)}=\begin{cases}\dfrac{1}{r_*(p,p_\circ)} & \text{if } p_{\mathrm{ST}}\le p<p_\circ,\\[2mm] \dfrac{n+1}{n-1}\Big(1-\dfrac1p\Big) & \text{if } 1\le p\le p_{\mathrm{ST}},\end{cases} \tag{9.2}$$

with r_* from (8.2). Both lines give $\frac12$ at $p=p_{\mathrm{ST}}$, since $\frac{n+1}{n-1}(1-\frac{n+3}{2(n+1)})=\frac12$. This is [BRS25, (2.1)] with $r_\circ=p_\circ$: the first line of [BRS25, (2.1)] has weights $\frac{1}{r_\circ}$ and $\frac12$ on the same two differences as (8.2). For $p\le p_{\mathrm{ST}}$ the second line gives

$$q_*(p,p_\circ)=q_{\mathrm{opt}}(p)=\frac{(n-1)p}{n+1-2p}, \tag{9.3}$$

because $1-\frac{n+1-2p}{(n-1)p}=\frac{(n+1)(p-1)}{(n-1)p}$. The same computation and (8.4) show that for $p_{\mathrm{ST}}\le p<\frac{2n}{n+1}$,

$$\lim_{p_\circ\to\frac{2n}{n+1}}\Big(1-\frac{1}{q_*(p,p_\circ)}\Big)=\frac{n+1}{n-1}\Big(1-\frac1p\Big)=1-\frac{1}{q_{\mathrm{opt}}(p)}. \tag{9.4}$$

Theorem 9.3 ([BRS25, Theorem 2.3])**.** *Let $n\ge2$, $a>0$, $p_{\mathrm{ST}}\le p_\circ<\frac{2n}{n+1}$, and $p_\circ\le r_\circ\le\frac{n-1}{n+1}p_\circ'$. Suppose that* $\mathrm{VBR}(p,r)$ *holds for all $p\in[p_{\mathrm{ST}},p_\circ)$ and $r\in[p,r_*(p,p_\circ,r_\circ))$. Then $R_a^{\lambda(p)}\in\mathrm{Sp}(p,q)$ for $1\le p<p_\circ$ and $q>q_*(p,p_\circ,r_\circ)$.*

In [BRS25] the exponents $r_*(p,p_\circ,r_\circ)$, $q_*(p,p_\circ,r_\circ)$ are defined for every admissible $r_\circ$, and $\mathrm{VBR}(p,r)$ is [BRS25, Definition 2.2], which is Definition 8.1 for the Euclidean norm. We use only $r_\circ=p_\circ$, for which these exponents are $r_*(p,p_\circ)$ and $q_*(p,p_\circ)$ above. The constraint $r_\circ\le\frac{n-1}{n+1}p_\circ'$ then reads $p_\circ-1\le\frac{n-1}{n+1}$, that is $p_\circ\le\frac{2n}{n+1}$, which holds.

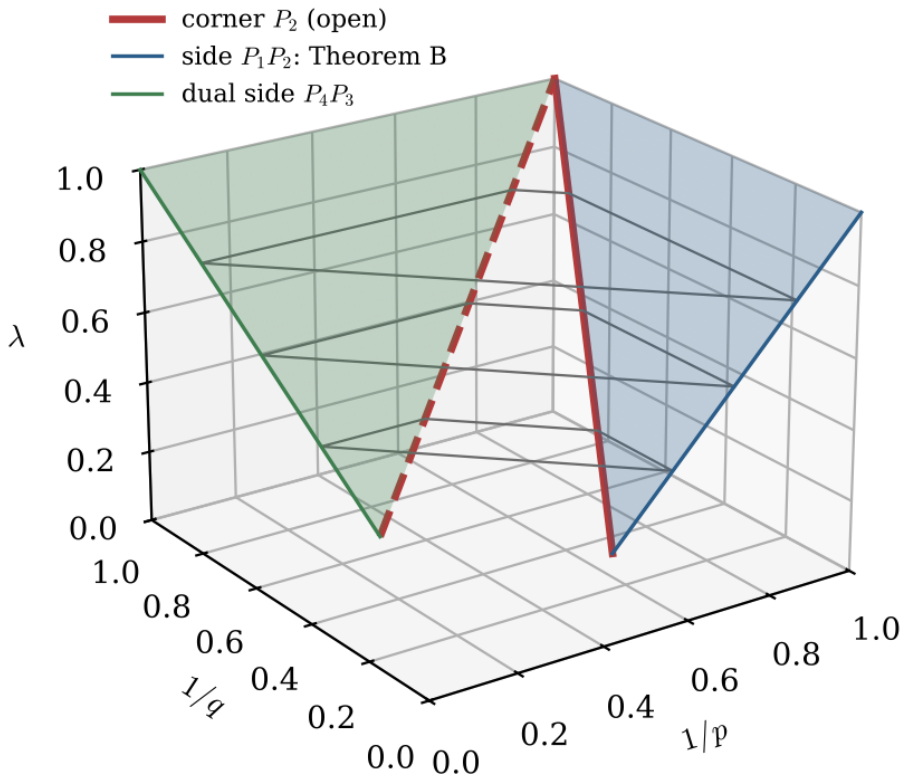


FIGURE 3. Sparse exponents $(1/p, 1/q)$ for $n = 3$ at height $\lambda = \lambda(p)$. The grey quadrilaterals are the trapezoids $P_1P_2P_3P_4$ of [BRS25, (1.5)] at $\lambda = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$. As λ runs over $(0, \frac{n-1}{2})$ the critical side P_1P_2 sweeps the blue surface, which Theorem B covers except for the red edge traced by the corner P_2, where $q = q_{\mathrm{opt}}(p)$; the green surface and the dashed edge are their duals P_4P_3 and P_3.

9.3. **Proof of Theorem B.** Figure 3 shows the exponents reached for $n = 3$. Let $1 \le p < \frac{2n}{n+1}$ and $q > q_{\mathrm{opt}}(p)$. By (9.3) and (9.4) there is $p_\circ$ with

$$\max\{p, p_{\mathrm{ST}}\} < p_\circ < \tfrac{2n}{n+1}, \qquad q > q_*(p, p_\circ)\,:$$

when $p \le p_{\mathrm{ST}}$ every $p_\circ \in (p_{\mathrm{ST}}, \frac{2n}{n+1})$ will do, since then $q_*(p, p_\circ) = q_{\mathrm{opt}}(p)$; when $p > p_{\mathrm{ST}}$, $1 - \frac{1}{q_*}$ tends to $1 - \frac{1}{q_{\mathrm{opt}}} < 1 - \frac{1}{q}$ as $p_\circ \to \frac{2n}{n+1}$. By Corollary 8.4, $\mathrm{VBR}(\tilde p, r)$ holds for $\tilde p \in [p_{\mathrm{ST}}, p_\circ)$ and $r \in [\tilde p, r_*(\tilde p, p_\circ))$, which is the hypothesis of Theorem 9.3 with $r_\circ = p_\circ$. Hence $R_a^{\lambda(p)} \in \mathrm{Sp}(p, q)$. Since $R_{a,t}^{\lambda(p)} = m(D/t)$ with $m(\xi) = (1 - |\xi|^a)_+^{\lambda(p)}$, Lemma 9.2 gives $\|R_{a,t}^{\lambda(p)}\|_{\mathrm{Sp}(p,q)} = \|R_a^{\lambda(p)}\|_{\mathrm{Sp}(p,q)}$ for every $t > 0$. □

9.4. **Weights.** A *weight* is a locally integrable function $w > 0$ almost everywhere. For a ball B write $⨍_B w = |B|^{-1} \int_B w$ and $w(E) = \int_E w$. The weight belongs to A_1 if

$$[w]_{A_1} = \sup_B \Big(⨍_B w\Big)\Big(\operatorname*{ess\,inf}_B w\Big)^{-1} < \infty,$$

and, for $1<\sigma<\infty$, to RH_σ if

$$[w]_{RH_\sigma}=\sup_B\Big(⨍_B w^\sigma\Big)^{1/\sigma}\Big(⨍_B w\Big)^{-1}<\infty,$$

the suprema over all balls. By Hölder's inequality $RH_\sigma\subset RH_{\tilde\sigma}$ when $\tilde\sigma<\sigma$. Every A_1 weight belongs to RH_σ for some $\sigma>1$, by the reverse Hölder inequality of Coifman and Fefferman [CF74], and $A_1\subset A_p$ for $p>1$, so the Hardy–Littlewood maximal operator M is bounded on $L^p(w)$ for $w\in A_1$ and $1<p<\infty$ [Gra14, Ch. 7]. The weak space $L^{p,\infty}(w)$ has the quasinorm $\|f\|_{L^{p,\infty}(w)}=\sup_{s>0}s\,w(\{|f|>s\})^{1/p}$, which satisfies $\|f+g\|_{L^{p,\infty}(w)}\le 2(\|f\|_{L^{p,\infty}(w)}+\|g\|_{L^{p,\infty}(w)})$ and $\|f\|_{L^{p,\infty}(w)}\le\|f\|_{L^p(w)}$, and $L^{p,\infty}(w)$ is complete.

Theorem 9.4 ([FN19, Theorem 1.4])**.** *Let $1\le p<q'$, so that $\frac1p+\frac1q>1$, and put $\sigma=(q'/p)'=\frac{q}{q+p-pq}$. There is a function Φ, increasing in each variable, such that every $T\in\mathrm{Sp}(p,q)$ satisfies*

$$\|Tf\|_{L^{p,\infty}(w)}\le\Phi\big([w]_{A_1},[w]_{RH_\sigma}\big)\,\|T\|_{\mathrm{Sp}(p,q)}\|f\|_{L^p(w)}\qquad(f\in C_c^\infty(\mathbb{R}^n))$$

for every $w\in A_1\cap RH_\sigma$.

This is [FN19, Theorem 1.4] in the notation of [BRS25, (9.2)], where the constant is, up to a factor depending on n,p,q,

$$\|T\|_{\mathrm{Sp}(p,q)}[w^\sigma]_{A_\infty}^{1/p}[w]_{A_1}^{1+1/p}[w]_{RH_\sigma}^{1/p},$$

and $[v]_{A_\infty}=\sup_B v(B)^{-1}\int_B M(v\mathbf{1}_B)$ is Wilson's constant. It is bounded by a function of $[w]_{A_1}$ and $[w]_{RH_\sigma}$: for every ball,

$$⨍_B w^\sigma\le[w]_{RH_\sigma}^\sigma\Big(⨍_B w\Big)^\sigma\le([w]_{RH_\sigma}[w]_{A_1})^\sigma\operatorname*{ess\,inf}_B w^\sigma,$$

so $v=w^\sigma\in A_1$ with $[v]_{A_1}\le([w]_{RH_\sigma}[w]_{A_1})^\sigma$; then $Mv\le[v]_{A_1}v$ almost everywhere, for M the maximal operator over balls containing the point, and $[v]_{A_\infty}\le[v]_{A_1}$.

9.5. **Proof of Theorem C.** Let $w\in A_1\cap RH_\sigma$ with $\sigma>1$, and $1<p<p_1(w)=1+\frac{n-1}{n+1}(1-\frac1\sigma)$. Then $p<\frac{2n}{n+1}$. For $q\in(q_{\mathrm{opt}}(p),p')$ put $\sigma(q)=\frac{q}{q+p-pq}$, the exponent of Theorem 9.4. The interval is not empty: $q_{\mathrm{opt}}(p)<p'$ is equivalent to $(n-1)(p-1)<n+1-2p$, that is to $p<\frac{2n}{n+1}$. On it the denominator $q+p-pq=q(1-p(1-\frac1q))$ is positive, since $1-\frac1q<\frac1p$, and $\sigma(q)=\big(1+\frac pq-p\big)^{-1}$ is continuous and increasing in q. At $q=q_{\mathrm{opt}}(p)$,

$$q+p-pq=\frac{(n-1)p(1-p)+p(n+1-2p)}{n+1-2p}=\frac{p\,(2n-(n+1)p)}{n+1-2p},$$

so that

$$\sigma(q_{\mathrm{opt}}(p))=\frac{n-1}{2n-(n+1)p}.$$

The hypothesis $p<p_1(w)$ is equivalent to $(n+1)p<2n-\frac{n-1}{\sigma}$, that is to $\sigma(q_{\mathrm{opt}}(p))<\sigma$. By continuity there is $q\in(q_{\mathrm{opt}}(p),p')$ with $\sigma(q)<\sigma$, and then $w\in RH_{\sigma(q)}$.

For this q, Theorem B gives $\|R_{a,t}^{\lambda(p)}\|_{\mathrm{Sp}(p,q)}\le C$ with C independent of t, and Theorem 9.4 gives

$$\|R_{a,t}^{\lambda(p)}f\|_{L^{p,\infty}(w)}\le C\,\Phi\big([w]_{A_1},[w]_{RH_{\sigma(q)}}\big)\|f\|_{L^p(w)}\qquad(f\in C_c^\infty(\mathbb{R}^n),\ t>0).\tag{9.5}$$

The functions C_c^∞ are dense in $L^p(w)$: continuous functions with compact support are dense since $w\,dx$ is a Radon measure [Rud87, Theorem 3.14], and a mollification of such a function converges uniformly with supports in a fixed compact set, hence in $L^p(w)$ because $w\in L^1_{\mathrm{loc}}$. As $L^{p,\infty}(w)$ is complete, $R_{a,t}^{\lambda(p)}$ extends uniquely to a bounded operator $L^p(w)\to L^{p,\infty}(w)$ satisfying (9.5), uniformly in t.

Convergence. Let $\mathcal{G}$ be the set of $g \in \mathcal{S}(\mathbb{R}^n)$ with $\hat g \in C_c^\infty(\mathbb{R}^n \setminus \{0\})$. We first show that $\mathcal{G}$ is dense in $L^p(w)$. Fix $\varphi_0 \in C_c^\infty(\mathbb{R}^n)$ with $\varphi_0 = 1$ on $\{|\xi| \le 1\}$, let $\phi = \mathcal{F}^{-1}\varphi_0 \in \mathcal{S}$, so that $\int \phi = 1$, and for $g \in C_c^\infty$ and $k \ge 1$ let $\hat g_k(\xi) = (\varphi_0(\xi/k) - \varphi_0(k\xi))\hat g(\xi)$. Then $\hat g_k$ is smooth with compact support and vanishes on $\{|\xi| < 1/k\}$, so $g_k \in \mathcal{G}$, and $g - g_k = g - \phi_{1/k} * g + \phi_k * g$ with $\phi_\varepsilon(x) = \varepsilon^{-n}\phi(x/\varepsilon)$. Both $|\phi_{1/k} * g|$ and $|\phi_k * g|$ are at most $C_\phi Mg$, with $C_\phi = \int \sup_{|y|\ge|x|} |\phi(y)|\,dx$ [Gra14, Theorem 2.1.10], and $Mg \in L^p(w)$. Moreover $\phi_{1/k} * g \to g$ and $|\phi_k * g| \le k^{-n}\|\phi\|_\infty \|g\|_1 \to 0$ pointwise. Dominated convergence gives $g_k \to g$ in $L^p(w)$, and density of C_c^∞ gives density of $\mathcal{G}$.

For $g \in \mathcal{G}$ choose $0 < \rho_1 < \rho_2$ with $\operatorname{supp} \hat g \subset \{\rho_1 \le |\xi| \le \rho_2\}$. For $t \ge 2\rho_2$ the function $h_t = R_{a,t}^{\lambda(p)} g - g$ has $\hat h_t(\xi) = ((1 - |\xi/t|^a)^{\lambda(p)} - 1)\hat g(\xi)$, and on $\{\rho_1 \le |\xi| \le \rho_2\}$ we have $|\xi/t|^a \le 2^{-a}$. There the map $s \mapsto (1-s)^{\lambda(p)} - 1$ is smooth on $[0, 2^{-a}]$ and vanishes at $s = 0$, and every derivative of $\xi \mapsto |\xi/t|^a = t^{-a}|\xi|^a$ is $O(t^{-a})$ uniformly on the annulus. Hence $\|\partial^\beta \hat h_t\|_\infty \le C_\beta t^{-a}$ for every multi-index β, with $\hat h_t$ supported in a fixed compact set, and integration by parts gives $|h_t(x)| \le C_N t^{-a}(1 + |x|)^{-N}$ for every N. Now let $c_1 = \operatorname{ess\,inf}_{B(0,1)} w$. For $\varrho \ge 1$ the A_1 condition on $B(0,\varrho) \supset B(0,1)$ gives $w(B(0,\varrho)) \le [w]_{A_1} c_1 |B(0,\varrho)|$, so with $Np > n$,

$$\int (1 + |x|)^{-Np} w(x)\,dx \le w(B(0,1)) + \sum_{k\ge 0} 2^{-kNp} w(B(0, 2^{k+1})) \le C \sum_{k \ge 0} 2^{-kNp + kn} < \infty.$$

Therefore $\|h_t\|_{L^p(w)} \le Ct^{-a} \to 0$ as $t \to \infty$.

Finally let $f \in L^p(w)$ and $\varepsilon > 0$. Choose $g \in \mathcal{G}$ with $\|f - g\|_{L^p(w)} \le \varepsilon$. With C_w the constant of (9.5),

$$\begin{aligned} \|R_{a,t}^{\lambda(p)} f - f\|_{L^{p,\infty}(w)} &\le 4\|R_{a,t}^{\lambda(p)}(f - g)\|_{L^{p,\infty}(w)} + 4\|h_t\|_{L^{p,\infty}(w)} + 2\|g - f\|_{L^{p,\infty}(w)} \\ &\le (4C_w + 2)\varepsilon + 4\|h_t\|_{L^p(w)}, \end{aligned}$$

and the last term tends to zero. Hence $R_{a,t}^{\lambda(p)} f \to f$ in $L^{p,\infty}(w)$. □

9.6. **Proof of Theorem A.** The weight $w \equiv 1$ has $[w]_{A_1} = 1$ and $[w]_{RH_\sigma} = 1$ for every $\sigma > 1$. Given $1 < p < \frac{2n}{n+1}$, choose $\sigma > 1$ so large that $p < 1 + \frac{n-1}{n+1}(1 - \frac{1}{\sigma})$; this is possible because the right side tends to $\frac{2n}{n+1}$ as $\sigma \to \infty$. Theorem C with this weight is Theorem A. □

Remark 9.5. For $n = 2$ the inputs of Theorems A–C were known before [OAI26]: the Bochner–Riesz conjecture in the plane [CS72] gives the hypothesis of Theorem 8.2 for every $p_\circ < \frac{4}{3}$. For $n \ge 3$, [BRS25] proves these conclusions in partial ranges, unconditionally where the Stein–Tomas theorem suffices; for the weights this is $\sigma \le \frac{n+3}{2}$, which gives $p < p_{\mathrm{ST}}$. The full ranges rest on Theorem 7.1 for every $p_\circ < \frac{2n}{n+1}$.

10. What remains open

10.1. **The corner of the trapezoid.** Theorem B covers the critical side P_1P_2 of the trapezoid of [BRS25, (1.5)] except the corner P_2, where $q = q_{\mathrm{opt}}(p)$. At P_2 the scale-by-scale estimates behind Theorem 9.3 sum only with a loss: the exponent $r_*(p, p_\circ)$ reaches the Knapp line $r = \frac{n-1}{n+1}p'$ only in the limit (8.4), and the hypothesis of Theorem 9.3 asks for $r < r_*$. The method would need $\mathrm{VBR}(p, \frac{n-1}{n+1}p')$, the endpoint of the conjectured range of these inequalities, which is established in [BRS25] only at $p = p_{\mathrm{ST}}$. For $p = p_{\mathrm{ST}}$ the corner is reached in [BRS25, Theorem 1.5], through the L^2 restriction theorem; for $n = 2$ the corner is also open except at $\lambda = \frac{1}{6}$ [BRS25, §1.2].

10.2. **The other sides.** For $0 < \lambda < \frac{n-1}{2}$, sparse bounds on the top side P_2P_3 of the trapezoid, away from the critical line, are open for $n \geq 3$, and in the plane except at $\lambda = \frac{1}{6}$. They are a strengthening of off-diagonal $L^p \to L^r$ bounds for Bochner–Riesz means below the critical index, and Theorem 7.1 does not reach them. By duality the side P_3P_4 is equivalent to the side P_1P_2, so Theorem B also gives it, except for the corner P_3.

10.3. **The range of the weights.** The exponent $p_1(w) = 1 + \frac{n-1}{n+1}(1 - \frac{1}{\sigma})$ of Theorem C is the one produced by Theorem 9.4 from the sparse bounds of Theorem B. Whether it is the sharp range in terms of the reverse Hölder exponent is not known, as already noted in [BRS25, §1.1]; weighted weak-type bounds fail for $p > 1$ for the power weights $|x|^{\varepsilon}$, $\varepsilon > 0$ [BRS25, §1.1], so some restriction of this kind is necessary.

10.4. **Other domains.** Parts I and II are written for the round sphere. Beltran, Roos and Seeger work with the Minkowski functional ρ of a convex domain with smooth boundary of positive curvature, and their Theorem 9.3 holds in that generality. The weak-type endpoint for such ρ in dimensions $n \geq 3$ would follow in the same way from the Bochner–Riesz conjecture for $(1 - \rho(\xi))_+^{\lambda}$, which the local smoothing argument of Part I does not address.

10.5. **Maximal functions.** The maximal operator $\sup_{t>0} |R_{a,t}^{\lambda(p)} f|$ is not treated here. Already for $\lambda > \lambda(p)$ and $p < 2$ its L^p boundedness requires conditions beyond those of the Bochner–Riesz conjecture [Tao02], and nothing in this paper bears on it.

Appendix A. The Dirichlet problem on a periodic fibre

The energy argument of Section 4 uses a potential that solves a quaternionic Monge–Ampère equation on each spatial fibre, is smooth up to a boundary that moves with the parameters, and is compared with explicit functions on the fibre and on small balls inside it. This appendix proves the existence, comparison and smooth dependence that are needed; the quantitative bounds in the wavelength come from the comparison functions of Section 5. The argument is that of [OAI26, Appendix A], written out in full. Alesker solved the Dirichlet problem on quaternionic balls [Ale03b, Theorem 1.4] and Zhu on bounded domains of $\mathbb{H}^m$ carrying a smooth subsolution [Zhu16, Theorem 1.1]. Our fibres have periodic real directions and only three real coordinates per variable; a fourth, periodic coordinate is adjoined to provide the quaternionic rotations used for the mixed derivatives at the boundary.

A.1. **Setting and statement.** Let $m \geq 1$, let $\Lambda \subset \mathbb{R}^m$ be a lattice of rank m, let $V \subset \mathbb{C}_{\mathbf{k}}^m \cong \mathbb{R}^{2m}$ be a bounded domain with smooth boundary, and put $\Omega = (\mathbb{R}^m/\Lambda) \times V$. In the notation of Section 3.1 with three slots, $z_a = x_a + P_a\mathbf{i}$, the periodic variables x_a occupy the real slot and $P_a = p_a + \mathbf{k}q_a \in \mathbb{C}_{\mathbf{k}}$ the slots $\mathbf{i}, \mathbf{j}$; H is the three-slot Hessian (3.1). A function on V is regarded as a function on Ω independent of x.

Theorem A.1 ([OAI26, Theorem A.1])**.** *Let r be smooth on a neighbourhood of $\overline{V}$, with $r < 0$ in V, $r = 0$ and $dr \neq 0$ on ∂V, and $H(r) > 0$ on $\overline{\Omega}$. Let $B_0, g \in C^{\infty}(\overline{\Omega})$ be real with*

$$H(B_0) \geq 0, \qquad g > \det H(B_0) \qquad \text{on } \overline{\Omega}.$$

Then there is a unique $u \in C^{\infty}(\overline{\Omega})$ with

$$\det H(B_0 + u) = g, \qquad H(B_0 + u) > 0 \quad \text{on } \overline{\Omega}, \qquad u|_{\partial\Omega} = 0, \tag{A.1}$$

and there are constants $b \geq b' > 0$ with

$$b\, r \leq u \leq b'\, r. \tag{A.2}$$

If r, B_0, g and the domain depend smoothly on finitely many parameters, the constants b, b' can be chosen locally uniformly in the parameters, and the solution depends smoothly on the parameters up to the boundary after any smooth local identification of the domains with a fixed domain.

Here $C^\infty(\overline{\Omega})$ denotes functions all of whose derivatives extend continuously to $\overline{\Omega}$, and similarly for the Hölder spaces $C^{k,\alpha}(\overline{\Omega})$; $C_0^{k,\alpha}$ is the subspace of functions vanishing on $\partial\Omega$. The *smooth dependence* means: if $\Theta_\tau : \overline{V}_{\tau_0} \to \overline{V}_\tau$ are diffeomorphisms depending smoothly on (τ, y), then $(\tau, x, y) \mapsto u_\tau(x, \Theta_\tau(y))$ is smooth up to the boundary.

A.2. **Matrix facts.** We use the following consequences of Lemma 3.2. Let Q, Q_1, Q_2 be positive definite Hermitian quaternionic $m \times m$ matrices and B Hermitian.

(M1) If $A \geq 0$ and $B \geq 0$ then $A : B \geq 0$. Indeed $A = UDU^*$ with U unitary and $D \geq 0$ diagonal, and $A : B = \operatorname{Re}\operatorname{tr}(D^{1/2}U^*BUD^{1/2})$ by cyclicity of $\operatorname{Re}\operatorname{tr}$, a sum of the nonnegative numbers $e_i^* D^{1/2}U^*BUD^{1/2}e_i$.

(M2) $\frac{d}{ds}\log\det(Q + sB)|_{s=0} = Q^{-1} : B$, by (3.6) and (3.7). Since $Q_\theta = (1-\theta)Q_2 + \theta Q_1 > 0$ for $0 \leq \theta \leq 1$,

$$\log\det Q_1 - \log\det Q_2 = \Big(\int_0^1 Q_\theta^{-1}\, d\theta\Big) : (Q_1 - Q_2). \tag{A.3}$$

(M3) Put $S = \operatorname{Re}\operatorname{tr} Q^{-1}$, the sum of the reciprocals of the eigenvalues of Q. By the arithmetic–geometric mean inequality $S \geq m(\det Q)^{-1/m}$. If $-\beta I \leq B \leq \beta I$ then $|Q^{-1} : B| \leq \beta S$, by (M1) applied to $\beta I \pm B$.

(M4) If $A \geq 0$ and $P \geq cI$ with $c > 0$, then $\det(A+P) > \det A$. Indeed the real representation (3.5) turns A and $A + P$ into symmetric real matrices with $(A+P)_{\mathbb{R}} \geq A_{\mathbb{R}} + cI$, so by the minimax characterisation each eigenvalue of $(A+P)_{\mathbb{R}}$ exceeds the corresponding eigenvalue of $A_{\mathbb{R}}$ by at least c, and (3.5) gives $\det(A+P)^4 > \det(A)^4 \geq 0$.

A.3. **The linearised operator and comparison.** Work with $k \in \{3, 4\}$ slots and m coordinate indices; the real dimension is $N = km$, and H is the k-slot Hessian. For a real covector $\xi \in \mathbb{R}^N$ write $\hat{\xi} = (\sum_\alpha e_\alpha \xi_{a\alpha})_a \in \mathbb{H}^m$; then $|\hat{\xi}| = |\xi|$. For a real symmetric $N \times N$ matrix M let $H(M)$ be the Hermitian matrix with $H(D^2 v) = H(v)$; it is linear in M, and $H(\xi\xi^{\mathrm{t}}) = \hat{\xi}\hat{\xi}^*$ by Lemma 3.1. If $Q > 0$ is a Hermitian matrix field, the operator

$$L_Q w = Q^{-1} : H(w)$$

is a real linear operator $\sum_{i,j} a^{ij}\partial_i\partial_j w$ with no lower-order terms, whose coefficient matrix is defined by $a : M = Q^{-1} : H(M)$. For $M = \xi\xi^{\mathrm{t}}$ this gives $a(\xi,\xi) = \hat{\xi}^* Q^{-1}\hat{\xi}$, so

$$\lambda_{\max}(Q)^{-1}|\xi|^2 \leq a(\xi,\xi) \leq \lambda_{\min}(Q)^{-1}|\xi|^2. \tag{A.4}$$

If ψ is a C^2 function with $Q = H(\psi) > 0$ we write $L_\psi = L_{H(\psi)}$, the derivative of $\psi \mapsto \log\det H(\psi)$ by (M2).

In what follows a *periodic domain* is an open set $U \subset (\mathbb{R}^m/\Lambda) \times W$, where W is bounded in the remaining real coordinates and Λ is a lattice (possibly of lower rank, possibly acting on more coordinates), so that $\overline{U}$ is compact and at least one real coordinate y is a bounded function on U. The domain Ω, its four-slot extension below, and balls in a fibre are periodic domains.

Lemma A.2 (Maximum principle)**.** *Let U be a periodic domain and $L = \sum a^{ij}\partial_i\partial_j$ with a continuous and positive definite in U. If $f \in C^2(U) \cap C(\overline{U})$ satisfies $Lf \geq 0$ in U, then $\max_{\overline{U}} f = \max_{\partial U} f$.*

Proof. At an interior maximum $D^2 f \le 0$, so $Lf = a : D^2 f \le 0$; hence a function with $Lf > 0$ has no interior maximum. Let y be a bounded real coordinate and $h = e^y$. Then $Lh = a^{yy} e^y > 0$, so for $\varepsilon > 0$ the function $f + \varepsilon h$ satisfies $L(f + \varepsilon h) > 0$ and attains its maximum over the compact set $\overline{U}$ on ∂U. Letting $\varepsilon \to 0$ gives the claim. □

Lemma A.3 (Comparison)**.** *Let U be a periodic domain in k slots, and $\psi_1, \psi_2 \in C^2(U) \cap C(\overline{U})$ with $H(\psi_i) > 0$ in U. If $\det H(\psi_1) \ge \det H(\psi_2)$ in U and $\psi_1 \le \psi_2$ on ∂U, then $\psi_1 \le \psi_2$ in U. In particular (A.1) has at most one solution in $C^2(\Omega) \cap C(\overline{\Omega})$.*

Proof. By (A.3), $0 \le \log\det H(\psi_1) - \log\det H(\psi_2) = L_{\bar{Q}}(\psi_1 - \psi_2)$ with $\bar{Q}^{-1} = \int_0^1 Q_\theta^{-1} d\theta$, $Q_\theta = (1-\theta)H(\psi_2) + \theta H(\psi_1)$. The coefficients are continuous and positive definite in U by (A.4), and Lemma A.2 gives $\psi_1 - \psi_2 \le \max_{\partial U}(\psi_1 - \psi_2) \le 0$. For uniqueness apply this twice with $\psi_i = B_0 + u_i$. □

A.4. **The four-slot extension.** Let $\tilde{\Omega} = \Omega \times (\mathbb{R}/\mathbb{Z})^m$, with the new coordinates z_{a3} in the **k** slots, and extend r, B_0, g to $\tilde{\Omega}$ as functions independent of them. For a function independent of the **k** slots the four-slot G and H of (3.1) agree with the three-slot ones, since every term containing ∂_{a3} vanishes. Hence the hypotheses of Theorem A.1 hold on $\tilde{\Omega}$, and a solution on Ω extends to a solution on $\tilde{\Omega}$. Conversely, suppose $\tilde{u}$ solves the four-slot problem on $\tilde{\Omega}$. For $c \in (\mathbb{R}/\mathbb{Z})^m$ the translate $\tilde{u}(\cdot + c)$ in the **k** slots solves the same problem, since the data are invariant and H commutes with translations. By Lemma A.3 (with $k = 4$), $\tilde{u}(\cdot + c) = \tilde{u}$, so $\tilde{u}$ is independent of the **k** slots and restricts to a solution on Ω. Therefore it suffices to prove Theorem A.1 on $\tilde{\Omega}$, and the rest of the proof works there, with four slots and $N = 4m$.

The manifold $\overline{\tilde{\Omega}}$ is compact with smooth boundary $\partial\tilde{\Omega} = (\mathbb{R}^m/\Lambda) \times \partial V \times (\mathbb{R}/\mathbb{Z})^m$ and a flat metric, and each point has a neighbourhood isometric to an open subset of $\mathbb{R}^{4m}$ in which the boundary is a smooth hypersurface. We call such a neighbourhood a *flat chart*. By compactness $\overline{\tilde{\Omega}}$ is covered by finitely many flat charts of a fixed size. The local Schauder estimates [GT01, Corollary 6.3 and Lemma 6.5], their higher-order forms, and the local L^2 regularity [GT01, Theorems 8.8 and 8.12] are local statements and apply in such charts; summing over a finite cover gives the global estimates of [GT01, Theorem 6.6] and their higher-order analogues on $\overline{\tilde{\Omega}}$, exactly as in the proofs in [GT01, Ch. 6].

A.5. **The linear Dirichlet problem.**

Lemma A.4. *Let $Q > 0$ be a smooth Hermitian field on $\overline{\tilde{\Omega}}$, $k \ge 0$ and $0 < \alpha < 1$. Then $L_Q : C_0^{k+2,\alpha}(\overline{\tilde{\Omega}}) \to C^{k,\alpha}(\overline{\tilde{\Omega}})$ is an isomorphism.*

Proof. For the Laplacian: by Poincaré's inequality along a bounded coordinate of V, the form $\int \nabla v \cdot \nabla w$ is coercive on $H_0^1(\tilde{\Omega})$, so for $f \in C^\infty(\overline{\tilde{\Omega}})$ the Lax–Milgram theorem gives a weak solution $v \in H_0^1$ of $\Delta v = f$. The local regularity theorems, interior and at the flat boundary pieces, give $v \in H^j$ for every j, hence $v \in C^\infty(\overline{\tilde{\Omega}})$ by Sobolev embedding. For $f \in C^{k,\alpha}$ mollify in charts to get smooth f_ℓ bounded in $C^{k,\alpha}$ and converging in $C^{k,\beta}$, $\beta < \alpha$. The estimate below bounds the solutions v_ℓ in $C^{k+2,\alpha}$; they converge in $C^{k+2,\beta}$, and the limit lies in $C_0^{k+2,\alpha}$ and solves $\Delta v = f$. Uniqueness follows from Lemma A.2.

For $L_s = (1-s)\Delta + sL_Q$, $0 \le s \le 1$, the coefficients are smooth and uniformly elliptic, uniformly in s, since $\Delta = L_I$ by Lemma 3.1. Since $H(r) \ge c_r I$ on $\overline{\tilde{\Omega}}$ for some $c_r > 0$, (M1) gives $L_s r \ge c' > 0$ uniformly in s. If $v \in C_0^2$ and $|L_s v| \le K$, then $L_s(\pm v + \frac{K}{c'} r) \ge 0$ and Lemma A.2 gives $|v| \le \frac{K}{c'}|r|$. With the global Schauder estimate $\|v\|_{C^{k+2,\alpha}} \le C(\|v\|_{C^0} + \|L_s v\|_{C^{k,\alpha}})$ this gives $\|v\|_{C^{k+2,\alpha}} \le C\|L_s v\|_{C^{k,\alpha}}$ on $C_0^{k+2,\alpha}$, uniformly in s, and the method of continuity [GT01, Theorem 5.2] transfers invertibility from $L_0 = \Delta$ to $L_1 = L_Q$. □

A.6. **The continuity path.** Write $\psi = B_0 + u$, so that (A.1) reads $\det H(\psi) = g$, $H(\psi) > 0$, $\psi = B_0$ on $\partial\tilde\Omega$. For $0 \le t \le 1$ put

$$g_t = (1-t)\det H(B_0 + r) + tg.$$

At $t = 0$ the function $u = r$ is a solution: $H(B_0 + r) \ge H(r) > 0$ and $r = 0$ on the boundary. By (M4), $\det H(B_0 + r) > \det H(B_0)$, and $g > \det H(B_0)$ by hypothesis; by compactness there are $\delta, c_0, C_0 > 0$ with

$$g_t \ge \det H(B_0) + \delta, \qquad c_0 \le g_t \le C_0 \qquad \text{on } \overline{\tilde\Omega},\ 0 \le t \le 1. \tag{A.5}$$

Let T be the set of $t \in [0,1]$ for which (A.1) with g_t in place of g has a solution $u_t \in C^\infty(\overline{\tilde\Omega})$. Then $0 \in T$, and we show that T is open and closed.

Openness. Let $t_0 \in T$, fix $\alpha \in (0,1)$ and $k \ge 0$, and consider

$$\mathcal{F}(t,u) = \log\det H(B_0 + u) - \log g_t$$

on the open set of $u \in C_0^{k+2,\alpha}$ with $H(B_0 + u) > 0$ on $\overline{\tilde\Omega}$, with values in $C^{k,\alpha}$. It is a smooth map: $\log\det H(M)$ is a smooth function of the real Hessian M on the open set $H(M) > 0$, and composition with a smooth function is a smooth map of Hölder spaces. Its derivative in u at u_{t_0} is $L_{\psi_{t_0}}$, an isomorphism by Lemma A.4. The implicit function theorem gives, for t near t_0, a solution $u_t \in C_0^{k+2,\alpha}$ depending smoothly on t. By Lemma A.3 the solutions obtained for different k coincide, so $u_t \in C^\infty(\overline{\tilde\Omega})$ and T is open.

Closedness. By the a priori estimates below, every u_t, $t \in T$, satisfies $\|u_t\|_{C^j(\overline{\tilde\Omega})} \le C_j$ for every j, and $H(B_0 + u_t) \ge c\,I$, with C_j, c independent of t. If $t_\ell \in T$ and $t_\ell \to t$, the Arzelà–Ascoli theorem gives a subsequence of u_{t_ℓ} converging in every C^j to some $u \in C^\infty(\overline{\tilde\Omega})$, which vanishes on the boundary, satisfies $H(B_0 + u) \ge cI$ and $\det H(B_0 + u) = g_t$. So $t \in T$.

Hence $T = [0,1]$ and $u = u_1$ solves (A.1). It remains to prove the a priori estimates. For the rest of this appendix $u = u_t$ is a smooth solution for some $t \in T$, $\psi = B_0 + u$, $Q = H(\psi)$, $L = L_\psi$, $S = \operatorname{Re}\operatorname{tr} Q^{-1}$, and C denotes constants independent of t and u. By (M3) and (A.5),

$$S \ge m\, g_t^{-1/m} \ge m C_0^{-1/m} =: s_0 > 0, \qquad Lr = Q^{-1} : H(r) \ge c_r S. \tag{A.6}$$

A.7. **Barriers and the gradient.** *The barriers* (A.2). As $b' \downarrow 0$, $\det H(B_0 + b'r) \to \det H(B_0)$ uniformly, so by (A.5) we may fix $b' \in (0,1)$ with $\det H(B_0 + b'r) < g_t$ for all t. Since $\det H(B_0 + br) \ge \det(bH(r)) \ge (bc_r)^m$ by (M4) and monotonicity, we may fix $b > 1$ with $\det H(B_0 + br) > C_0 \ge g_t$. All three potentials $B_0 + b'r$, ψ, $B_0 + br$ equal B_0 on the boundary and have positive definite Hessians, so Lemma A.3 gives $B_0 + br \le \psi \le B_0 + b'r$, which is (A.2). Since r is a defining function, near $\partial\tilde\Omega$ we can write $u = \rho\, r$ with ρ smooth up to the boundary (Section 3.3). On the collar $r < 0$, so (A.2) gives $b' \le \rho \le b$, and by continuity also on the boundary. There $\nabla u = \rho\nabla r$, so

$$|\nabla u| \le b\,|\nabla r| \qquad \text{on } \partial\tilde\Omega. \tag{A.7}$$

First and second derivatives along constant directions. For real symmetric M with $H(M) > 0$ put $F(M) = \log\det H(M)$. It is concave, being the composition of the linear map $M \mapsto H(M)$ with the concave function $\log\det$ (Lemma 3.2), and $DF(M)[\dot M] = H(M)^{-1} : H(\dot M)$ by (M2). For a constant real unit vector e write ψ_e, ψ_{ee} for the directional derivatives. Differentiating $F(D^2\psi) = \log g_t$ once and twice along e gives

$$L\psi_e = (\log g_t)_e, \qquad L\psi_{ee} = (\log g_t)_{ee} - D^2F(D^2\psi)[D^2\psi_e, D^2\psi_e] \ge (\log g_t)_{ee}. \tag{A.8}$$

By (A.6), $L(\pm\psi_e + Kr) \geq -C + Kc_r s_0 > 0$ for a large constant K, and Lemma A.2 gives $\pm\psi_e \leq \max_{\partial\tilde{\Omega}} |\nabla\psi| + K|r|$. With (A.7) this bounds $|\nabla\psi|$, hence $|\nabla u|$, by a constant C. In the same way $L(\psi_{ee} + Kr) > 0$ for large K, so

$$\psi_{ee} \leq \max_{\partial\tilde{\Omega}} |D^2\psi| + C \qquad \text{in } \tilde{\Omega},\ |e| = 1. \tag{A.9}$$

It remains to bound $D^2\psi$ on the boundary; B_0 is smooth, so it suffices to bound D^2u there.

A.8. **Second derivatives at the boundary.** Fix a boundary point and a flat chart centred at it, with coordinates $x \in \mathbb{R}^{4m}$, identified with $\mathbb{H}^m$ through the slots, so that the real inner product is $\operatorname{Re}(v^*w)$. Let ν be the outward unit normal at 0; then $\nabla r(0) = |\nabla r(0)|\nu$, since $r < 0$ inside. All constants below are uniform over boundary points, by compactness.

Tangential directions. From $u = \rho r$, $D^2u = \rho D^2 r + \nabla\rho \otimes \nabla r + \nabla r \otimes \nabla\rho + r D^2\rho$, and at 0 the last three terms vanish on pairs of tangent vectors. Hence $|D^2u(0)(v, w)| = \rho|D^2r(0)(v, w)| \leq C$ for unit tangent v, w.

Mixed directions. Fix a unit tangent vector e at 0. Let $P_{\tan}$ be the real orthogonal projection onto the tangent space and put

$$v = \frac{P_{\tan} D^2 r(0) e}{|\nabla r(0)|}, \qquad A = v\nu^* - \nu v^* - \nu(\nu^* v)\nu^*,$$

with vectors read as quaternionic columns. Since v is tangent, $\operatorname{Re}(\nu^* v) = 0$, so the quaternion $\nu^* v$ satisfies $\overline{\nu^* v} = -\nu^* v$. Hence $A^* = \nu v^* - v\nu^* + \nu(\nu^* v)\nu^* = -A$, and, using $\nu^*\nu = 1$ and $v^*\nu + \nu^* v = 2\operatorname{Re}(\nu^* v) = 0$,

$$A\nu = v - \nu(v^*\nu) - \nu(\nu^* v) = v.$$

Consider the affine vector field $X(x) = e + Ax$. Its flow is $\phi_s(x) = e^{sA}x + c_s$ with $c_s = \int_0^s e^{\sigma A} e\, d\sigma$, and $U_s = e^{sA}$ is quaternionic unitary since $A^* = -A$.

First, Xr vanishes to second order along the boundary at 0. Indeed $Xr(0) = \nabla r(0) \cdot e = 0$, and for a tangent vector w,

$$\begin{aligned} \partial_w(Xr)(0) &= D^2 r(0)(w, e) + \nabla r(0) \cdot (Aw) = |\nabla r(0)|(v \cdot w + \operatorname{Re}(\nu^* A w)) \\ &= |\nabla r(0)|(v \cdot w - \operatorname{Re}((A\nu)^* w)) = 0, \end{aligned}$$

because $D^2r(0)(w, e) = w \cdot P_{\tan} D^2 r(0) e$ and $\operatorname{Re}(\nu^* A w) = \operatorname{Re}((A^*\nu)^* w) = -\operatorname{Re}((A\nu)^* w)$. So the restriction of Xr to the boundary hypersurface vanishes at 0 with its differential, and $|Xr(x)| \leq C|x|^2$ for boundary points x in the chart. On the boundary $Xu = \rho\, Xr + r\, X\rho = \rho\, Xr$, so

$$|Xu(x)| \leq C|x|^2 \qquad (x \in \partial\tilde{\Omega},\ |x| \leq \varrho_0), \tag{A.10}$$

for a fixed chart radius ϱ_0.

Secondly, $L(X\psi) = X \log g_t$. By the covariance in Lemma 3.1 and its invariance under translations, $H(\psi \circ \phi_s)(x) = U_s^* H(\psi)(\phi_s(x)) U_s$, and the Moore determinant is invariant under unitary conjugation, since the eigenvalues are. Hence $\log\det H(\psi \circ \phi_s)(x) = \log g_t(\phi_s(x))$ for x in the chart and small s, and differentiation at $s = 0$, using (M2) and $\frac{d}{ds}\psi \circ \phi_s|_{s=0} = X\psi$, gives $L(X\psi) = X \log g_t$. Since XB_0 is smooth with $|H(XB_0)| \leq C$, (M3) and (A.6) give

$$|L(Xu)| \leq |X \log g_t| + |L(XB_0)| \leq C + CS \leq CS. \tag{A.11}$$

Let $U = \{|x| < \varrho_0\} \cap \tilde{\Omega}$ and $w = K_1|x|^2 - K_2 r$. On the part of ∂U in the boundary of $\tilde{\Omega}$, $w = K_1|x|^2 \geq |Xu|$ by (A.10) once $K_1 \geq C$. On the part where $|x| = \varrho_0$, $w \geq K_1\varrho_0^2 \geq |X||\nabla u| \geq |Xu|$ once K_1 is large, by the gradient bound. In four slots $H(|x|^2) = 8I$, since $|x|^2 = \sum_{a,\alpha} z_{a\alpha}^2$ has real Hessian $2I$ and each diagonal entry collects $\sum_\alpha e_\alpha \overline{e_\alpha} \cdot 2 = 8$. So by (A.6) and (A.11),

$$L(w \pm Xu) \leq 8K_1 S - K_2 c_r S + CS \leq 0$$

once K_2 is large. Lemma A.2 applied to $-(w \pm Xu)$ gives $|Xu| \le w$ in U. At 0 both w and $Xu = \rho Xr$ vanish. Along the inward normal, $x = -s\nu$ with small $s > 0$, we get $|Xu(-s\nu)| \le K_1 s^2 - K_2 r(-s\nu) \le Cs$, so $|\partial_\nu(Xu)(0)| \le C$. Now

$$\partial_\nu(Xu)(0) = D^2u(0)(\nu, e) + \nabla u(0) \cdot (A\nu) = D^2u(0)(\nu, e),$$

because $\nabla u(0) = \rho \nabla r(0)$ is normal and $A\nu = v$ is tangent. Hence $|D^2u(0)(\nu, e)| \le C$.

The normal direction. By (3.15) with $u = \rho r$, at 0 the compression of $H(u)$ to the quaternionic orthogonal complement $\nu^\perp$ is $\rho H(r)|_{\nu^\perp} \ge b' c_r I$. Since $H(B_0) \ge 0$,

$$Q|_{\nu^\perp} \ge b' c_r I. \tag{A.12}$$

Choose a real orthonormal basis $\varepsilon_1 = \nu, \varepsilon_2, \dots, \varepsilon_{4m}$ of $\mathbb{R}^{4m}$ and write $M = D^2\psi(0) = \sum_{i,j} M_{ij} \varepsilon_i \varepsilon_j^{\mathrm{t}}$. By linearity and polarisation of $H(\xi\xi^{\mathrm{t}}) = \hat\xi\hat\xi^*$,

$$Q = H(M) = M_{11}\nu\nu^* + R, \qquad R = \sum_{(i,j)\neq(1,1)} M_{ij}\, \tfrac{1}{2}\big(\hat\varepsilon_i\hat\varepsilon_j^* + \hat\varepsilon_j\hat\varepsilon_i^*\big),$$

and the entries M_{ij} with $(i,j) \neq (1,1)$ are bounded by the two previous steps, so $|R| \le C$. Let $f_2, \dots, f_m$ be a quaternionic orthonormal basis of $\nu^\perp$. In the basis $\nu, f_2, \dots, f_m$,

$$Q = \begin{pmatrix} q_{11} & z^* \\ z & D \end{pmatrix}, \qquad q_{11} = M_{11} + \nu^* R \nu, \quad z_i = f_i^* R \nu, \quad D = Q|_{\nu^\perp} = R|_{\nu^\perp},$$

so z and D are bounded and $D \ge b' c_r I$ by (A.12). If $m > 1$, (3.9) gives $g_t = \det Q = (\det D)(q_{11} - z^* D^{-1} z)$, so

$$0 < q_{11} = \frac{g_t}{\det D} + z^* D^{-1} z \le \frac{C_0}{(b' c_r)^{m-1}} + \frac{|z|^2}{b' c_r} \le C;$$

if $m = 1$, $q_{11} = Q = g_t \le C_0$. In both cases $|M_{11}| \le |q_{11}| + |R| \le C$, so $|D^2\psi(0)| \le C$.

A.9. **Hölder estimates and higher derivatives.** By (A.9) and the boundary bound, $\psi_{ee} \le C$ for every unit e. Since $\Delta\psi = \operatorname{Re}\operatorname{tr} Q > 0$ by Lemma 3.1, every eigenvalue of $D^2\psi$ is at least $-(4m-1)C$. So $|D^2\psi| \le C$ on $\overline{\widetilde{\Omega}}$, hence $|Q| \le C$; with $\det Q = g_t \ge c_0$ every eigenvalue of Q lies in an interval $[c, C] \subset (0, \infty)$. Thus $D^2\psi$ takes values in the compact convex set

$$\mathcal{K} = \{M : |M| \le C,\ cI \le H(M) \le CI\},$$

on which F is smooth and concave, and by (A.4) $DF(M)[\xi\xi^{\mathrm{t}}] = \hat\xi^* H(M)^{-1} \hat\xi \in [C^{-1}|\xi|^2, c^{-1}|\xi|^2]$ for $M \in \mathcal{K}$. Multiply F by a smooth cut-off supported in $\{H(M) > 0\}$ and equal to one near $\mathcal{K}$, and extend by zero; the equation $F(D^2\psi) = \log g_t$ is then a fully nonlinear, uniformly elliptic, concave equation on the range of $D^2\psi$, with smooth dependence on x through g_t and boundary values B_0. These are the hypotheses (i)′, (ii)′ of the interior and boundary estimates [GT01, Theorems 17.14 and 17.26′], which are due to Evans [Eva82] and Krylov [Kry83, Kry84] and are local; in the flat charts of a finite cover they give $\|\psi\|_{C^{2,\alpha}(\overline{\widetilde{\Omega}})} \le C$ for some $\alpha \in (0,1)$.

For higher derivatives we argue by induction: suppose $\|u\|_{C^{j+2,\alpha}} \le C_j$. In the interior, for each constant direction e, ψ_e solves the linear equation (A.8) whose coefficients Q^{-1} are bounded in $C^{j,\alpha}$ and uniformly elliptic; the interior Schauder estimates bound ψ_e in $C^{j+2,\alpha}$ on compact subsets of the chart. Near the boundary, flatten it by a smooth change of coordinates $y = \Phi(x)$ with the boundary at $y_{4m} = 0$. The equation becomes $\mathcal{G}(y, D_y v, D_y^2 v) = 0$ for $v = u \circ \Phi^{-1}$, with $\mathcal{G}$ smooth and $\partial\mathcal{G}/\partial v_{ij}$ uniformly elliptic on the range of the solution. For a tangential coordinate direction y_ι, $\iota < 4m$, the derivative v_ι vanishes on $y_{4m} = 0$, because v does, and solves the linear equation obtained by differentiating, with coefficients $\partial\mathcal{G}/\partial v_{ij}, \partial\mathcal{G}/\partial v_i$ bounded in $C^{j,\alpha}$ and right-hand side $-\partial_{y_\iota}\mathcal{G}$ bounded in $C^{j,\alpha}$. The local boundary Schauder estimates bound v_ι in $C^{j+2,\alpha}$ near the boundary. This controls in $C^{j+1,\alpha}$ every second derivative v_{il} except

$v_{4m\,4m}$. Uniform ellipticity gives $\partial\mathcal{G}/\partial v_{4m\,4m} \geq c > 0$, so by the implicit function theorem $v_{4m\,4m} = \Psi(y, D_y v, (v_{il})_{(i,l)\neq(4m,4m)})$ with Ψ smooth, and the right side is bounded in $C^{j+1,\alpha}$. Hence $\|u\|_{C^{j+3,\alpha}} \leq C_{j+1}$. All constants are independent of t, which completes the a priori estimates and the existence proof.

A.10. **Parameters.** Let the data depend smoothly on a parameter τ in an open set of $\mathbb{R}^K$, and let Θ_τ identify $\overline{V}_{\tau_0}$ with $\overline{V}_\tau$ as above. The gap (A.5) at $t = 1$ and the bounds c_r, C_0 are locally uniform in τ by continuity, so b, b' can be fixed locally uniformly. Pulling the equation back by $\mathrm{id} \times \Theta_\tau$ gives on the fixed domain $\tilde{\Omega}_{\tau_0}$ an equation $\mathcal{F}_\tau(v) = 0$, $v \in C_0^{k+2,\alpha}$, depending smoothly on (τ, v). At τ_0 its linearisation is $L_{\psi_{\tau_0}}$ conjugated by the pull-back, an isomorphism by Lemma A.4. The implicit function theorem gives solutions v_τ depending smoothly on τ with values in $C_0^{k+2,\alpha}$, for every k; by uniqueness they are the pull-backs of the solutions u_τ. A map from parameters to $C^{k+2,\alpha}$ that is smooth for every k is jointly smooth in (τ, x), which is the asserted dependence. The descent from $\tilde{\Omega}$ to Ω is compatible with all of this.

Lemma A.3 also holds in three slots on subdomains of Ω, in particular on the balls of the fibres used in Section 5. □

Declarations

Data and code availability. The sources of this paper, with exact-arithmetic checks of its algebraic identities and exponent computations in Python and a Lean 4 file that verifies the quaternion identities of Part I and the exponent arithmetic of Sections 8 and 9, are available at `https://github.com/creelie/scaling-pancake`, release v1.0.0, archived on Zenodo under doi:10.5281/zenodo.23701803. No statement of the paper depends on them.

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